

SAP BTP Documentation - KYMA

BTP Documentation: kyma

URL: <https://help.sap.com/docs/btp/btp-admin-guide/468c2f3c3ca24c2c8497ef9f83154c44.html>

Title: Kyma Environment | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformBasic Platform

ConceptsEnvironmentsKyma Environment

SAP Business Technology PlatformEnglish

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Kyma Environment

SAP BTP, Kyma runtime provides a fully managed cloud-native Kubernetes application runtime based on the open-source project "Kyma". Based on modular building blocks, Kyma runtime includes all the necessary capabilities to simplify the development and to run enterprise-grade cloud-native applications.

Kyma as a Managed Serviceà

Kyma environment permits a native consumption of the Multi-Cloud Foundation Services (What Is the Multi-Cloud Foundation?) and smooth consumption of SAP and non-SAP applications. It also supports out-of-the-box CAP, SAP Cloud SDK, application router, and HTML5 deployer.

Kyma implements a dedicated application runtime to deploy highly scalable, robust, and secure containerized microservices.

Note

Kyma as a managed service automatically checks all Kyma-managed resources. Any unexpected modifications are discarded, and the resource is reverted to the original state.

Every Kyma environment consists of:

One or more Kubernetes clusters based on project "Gardener" on a cloud provider and region (data center) of your choice. To find out the available regions and providers, see Regions for the Kyma Environment.

A set of Kyma modules picked by a user in the release channel of their choice installed on the provisioned cluster.

Integrationà

Kyma environment runs on one or more Kubernetes clusters created in a specific subaccount. The configuration of the Kyma environment enables you to connect it to a multitude of SAP systems. This way, you can build various formations that aggregate the SAP systems and environment according to your business use cases.

SAP systems connected to a Kyma environment expose APIs and events. To extend the existing logic of these SAP services, you can build serverless applications called "Functions", and trigger them to react to particular events or calls to your application's API. You can also use the Kyma environment to deploy microservices or even build full-stack applications.

Related Informationà

Getting Started in the Kyma Environment

Development in the Kyma Environment

Administration and Operations in the Kyma Environment

Security in the Kyma Environment

Discovery Center

What's New for SAP Business Technology Platform

URL: <https://help.sap.com/docs/btp/btp-admin-guide/557ec3adc3174ed4914ec9d6d13487cf.html>

Title: Regions for the Kyma Environment | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformBasic Platform

ConceptsRegionsRegions for the Kyma Environment

SAP Business Technology PlatformEnglish

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Regions for the Kyma Environment

To work with the Kyma environment, you need to specify the region for both your subaccount and the cluster.

Note

In Kyma, the ingress and egress IP addresses are independent of the subaccount region. For more information, see [Ingress and Egress IP Addresses](#).

Subaccount Regions

The table lists the regions you can choose from when creating a subaccount.

Subaccount Regions for Kyma

Global Account Type

Region

IaaS Provider

Technical Key

Region Name

Plan ID

Enterprise account

Partner Test, Demo, and Development account

Trial account

ap21

Microsoft Azure

cf-ap21

Singapore

azure

azure_lite

trial

build-runtime-azure

Enterprise account

Partner Test, Demo, and Development account

us20

Microsoft Azure

cf-us20

US West (WA)

azure

azure_lite

build-runtime-azure

Enterprise account

Partner Test, Demo, and Development account

jp20

Microsoft Azure

cf-jp20

Japan (Tokyo)

azure

azure_lite

build-runtime-azure

Enterprise account

Partner Test, Demo, and Development account

us21

Microsoft Azure

cf-us21

US East (VA)

azure

azure_lite

build-runtime-azure

Enterprise account

Partner Test, Demo, and Development account

eu20

Microsoft Azure

cf-eu20
Europe (Netherlands)
azure
azure_lite
build-runtime-azure
Enterprise account
Partner Test, Demo, and Development account
eu22
Microsoft Azure
cf-eu22
Europe (Frankfurt)
azure
azure_lite
Enterprise account
Partner Test, Demo, and Development account
uk20
Microsoft Azure
cf-uk20
UK South (London)
azure
azure_lite
Enterprise account
Partner Test, Demo, and Development account
ap20
Microsoft Azure
cf-ap20
Australia (Sydney)
azure
azure_lite
build-runtime-azure
Enterprise account
Partner Test, Demo, and Development account
br20
Microsoft Azure
cf-br20
Brazil (São Paulo)
azure
azure_lite
build-runtime-azure
Enterprise account
Partner Test, Demo, and Development account
ca20
Microsoft Azure
cf-ca20
Canada (Toronto)
azure
azure_lite
Enterprise account
ch20
Microsoft Azure
cf-ch20
Switzerland (Zurich) EU Access
azure
Enterprise account

cn20
Microsoft Azure
cf-cn20
China (North 3) 1
azure
Enterprise account
Trial account
us10
Amazon Web Services
cf-us10
US East (VA)
aws
trial
build-runtime-aws
Enterprise account
Trial account
eu10
Amazon Web Services
cf-eu10
Europe (Frankfurt)
aws
trial
build-runtime-aws
Enterprise account
eu11
Amazon Web Services
cf-eu11
Europe (Frankfurt) EU Access
aws
build-runtime-aws
Enterprise account
eu13
Amazon Web Services
cf-eu13
Europe (Milan)
aws
Enterprise account
br10
Amazon Web Services
cf-br10
Brazil (São Paulo)
aws
build-runtime-aws
Enterprise account
jp10
Amazon Web Services
cf-jp10
Japan (Tokyo)
aws
build-runtime-aws
Enterprise account
ca10
Amazon Web Services
cf-ca10

Canada (Montreal)
aws
build-runtime-aws
Enterprise account
ap12
Amazon Web Services
cf-ap12
South Korea (Seoul)
aws
Enterprise account
ap10
Amazon Web Services
cf-ap10
Australia (Sydney)
aws
build-runtime-aws
Enterprise account
ap11
Amazon Web Services
cf-ap11
Singapore
aws
build-runtime-aws
Enterprise account
us11
Amazon Web Services
cf-us11
US West (Oregon)
aws
Enterprise account
us30
Google Cloud
cf-us30
US Central (IA)
gcp
build-runtime-gcp
Enterprise account
eu30
Google Cloud
cf-eu30
Europe (Frankfurt)
gcp
build-runtime-gcp
Enterprise account
eu31
Google Cloud
cf-eu31
Europe (Netherlands)
gcp
Enterprise account
in30
Google Cloud
cf-in30
India (Mumbai)

gcp
build-runtime-gcp
Enterprise account
jp30
Google Cloud
cf-jp30
Japan (Osaka)
gcp
Enterprise account
jp31
Google Cloud
cf-jp31
Japan (Tokyo)
gcp
Enterprise account
sa30
Google Cloud
cf-sa30
KSA (Dammam) GCP public sector
gcp
Enterprise account
sa31
Google Cloud
cf-sa31
KSA (Dammam)
gcp
Enterprise account
il30
Google Cloud
cf-il30
Israel (Tel Aviv)
gcp
Enterprise account
br30
Google Cloud
cf-br30
Brazil (São Paulo)
gcp
Enterprise account
ap30
Google Cloud
cf-ap30
Australia (Sydney)
gcp
build-runtime-gcp
Enterprise account
ap31
Google Cloud
cf-ap31
Singapore (Jurong West)
gcp
Enterprise account
eu01
SAP Cloud Infrastructure

cf-eu01

Europe (Frankfurt) EU Access

sap-converged-cloud available in the Germany (Frankfurt) (eu-de-2) region

Enterprise account

eu02

SAP Cloud Infrastructure

cf-eu02

Europe (Rot) EU Access

sap-converged-cloud available in the Germany (Rot) (eu-de-1) region

Enterprise account

ap01

SAP Cloud Infrastructure

cf-ap01

Australia (Sydney)

sap-converged-cloud available in the Australia (Sydney) (ap-au-1) region

Enterprise account

ae01

SAP Cloud Infrastructure

cf-ae01

UAE (Dubai)

sap-converged-cloud available in the UAE (Dubai) (ap-ae-1) region

Enterprise account

us01

SAP Cloud Infrastructure

cf-us01

US East (Sterling)

sap-converged-cloud available in the US East (Sterling) (na-us-1) region

Enterprise account

us02

SAP Cloud Infrastructure

cf-us02

US West (Colorado)

sap-converged-cloud available in the US West (Colorado) (na-us-2) region

Enterprise account

jp01

SAP Cloud Infrastructure

cf-jp01

Japan (Tokyo)

sap-converged-cloud available in the Japan (Tokyo) (ap-jp-1) region

Enterprise account

cn40

Alibaba Cloud

cf-cn40

China (Shanghai)

alicloud

Note

1 This region is available only in the BTP region cf-cn20 and is the sole region available within the Microsoft Azure (azure) plan in China.

Cluster Regions¹

When you enable a Kyma environment for a given subaccount, you must select a plan and region where the cluster is going to be created. Note that there is a number of regions available within each plan. They are all listed in the table:

Cluster Regions

IaaS Provider

Plan ID
Region
Region Name
Microsoft Azure
azure
azure_lite
build-runtime-azure
centralus
US Central (IA)
eastus
US East (VA)
westus2
US West (WA)
northeurope
North EU (Ireland)
uksouth
UK South (London)
germanywestcentral
Europe (Frankfurt)
switzerlandnorth
Switzerland (Zurich)
westeurope
Europe (Netherlands)
southeastasia
Singapore
australiaeast
Australia (Sydney)
japaneast
Japan (Tokyo)
brazilsouth
Brazil (São Paulo)
canadacentral
Canada (Toronto)
chinanorth3 1
China (North 3) 1
Amazon Web Services
aws
build-runtime-aws
eu-central-1
Europe (Frankfurt)
eu-west-2
Europe (London)
eu-south-1
Europe (Milan)
ca-central-1
Canada (Montreal)
sa-east-1
Brazil (São Paulo)
us-east-1
US East (VA)
ap-northeast-1
Japan (Tokyo)
ap-northeast-2
South Korea (Seoul)

ap-south-1
India (Mumbai)
ap-southeast-1
Singapore
ap-southeast-2
Australia (Sydney)
us-west-2
US West (Oregon)
trial
eu-central-1
Europe (Frankfurt)
us-east-1
US East (VA)
ap-southeast-1
Singapore
Google Cloud
gcp
build-runtime-gcp
europa-west3
Europe (Frankfurt)
europa-west4
Europe (Netherlands)
us-central1
US Central (IA)
asia-south1
India (Mumbai)
asia-northeast2
Japan (Osaka)
me-central2
KSA (Dammam)
me-west1
Israel (Tel Aviv)
australia-southeast1
Australia (Sydney)
southamerica-east1
Brazil (São Paulo)
asia-northeast1
Japan (Tokyo)
asia-southeast1
Singapore (Jurong West)
us-west1
North America (Oregon)
us-east4
North America (Virginia)
SAP Cloud Infrastructure
sap-converged-cloud
eu-de-1
Germany (Rot)
eu-de-2
Germany (Frankfurt)
na-us-1
US East (Sterling)
na-us-2
US West (Colorado)

ap-au-1

Australia (Sydney)

ap-jp-1

Japan (Tokyo)

ap-ae-1

UAE (Dubai)

Alibaba Cloud

alicloud

cn-shanghai

China (Shanghai)

Load Balancersà^

Depending on the IaaS Provider, the following Load Balancers are provisioned by default:

Default Load Balancers

IaaS Provider

Default Load Balancer

Microsoft Azure

Standard

Amazon Web Services

Classic Network Load Balancer

Google Cloud

External passthrough Network Load Balancer

For more details on the Load Balancers and their features, check out the official documentation of the respective IaaS provider.

Related Informationà^

Creating Kyma Instances

Available Plans in the Kyma Environment

Provisioning and Updating Parameters in the Kyma Environment

URL: <https://help.sap.com/docs/btp/btp-admin-guide/sharing-clusters-in-kyma>

Title: *Sharing Clusters in Kyma | SAP Help Portal*

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Administrator's Guide...Subaccounts or Spaces/
Namespaces?Sharing Clusters in Kyma

SAP BTP Administrator's GuideEnglish

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Sharing Clusters in Kyma

Sharing a Kyma cluster among multiple teams, applications, or in partner scenarios, can improve efficiency with regard to cost, administration, and rule enforcement. However, before you set up cluster sharing, carefully consider your specific use case to ensure optimal security, isolation, and resource allocation.

The following recommendations help you decide whether sharing a cluster is a good option for you and provide best practices and effective mitigation actions, based on Kubernetes: Multi-tenancy.

Use Casesà^

You must understand and define the objectives and expected outcomes for your specific use case. Typical reasons for cluster sharing include cost optimization, resource utilization, and collaboration between teams. The following use cases are common:

Multiple Customers

This use case is typical if you want to serve several external customers (for example, in a partner scenario).

The customers don't have access to the cluster; Kubernetes is invisible from their perspective and is only used by the vendor to manage the customer workloads. You must ensure that the workloads are strictly isolated from each other.

Caution

It's not recommended to share the cluster among multiple customers. Instead, use separate clusters to ensure strict isolation of the users' workloads, guarantee absolute fairness on all levels, and prevent manipulations.

Multiple Teams

Members of the teams often have direct access to Kubernetes resources using tools. Typically, there is some

level of trust between members of different teams, but you must apply Kubernetes policies such as role-based access control (RBAC), quotas, and network policies to safely and fairly share clusters.

For multiple teams, cluster sharing (with the respective mitigations) is useful in the following scenarios:

Small teams or applications: If you have a small number of teams or applications with low resource requirements, sharing a cluster can maximize resource utilization and reduce infrastructure costs.

Development and testing environments: Sharing a cluster is often acceptable for development and testing purposes. These environments typically have lower security and isolation requirements, making it easier to share resources. For more information, see [The Basic Account Model](#).

Non-critical workloads: If the workloads running on the cluster are non-critical or have a low impact on business operations, sharing a cluster can be a viable option. This is especially true for non-production environments or temporary projects.

Checklist for Shared Clusters[^]

Consider the following mitigative actions for the control plane and data plane:

Control Plane Isolation

Control plane isolation ensures that different tenants cannot access or affect each others' Kubernetes API resources. The control plane manages the Kubernetes resources, enforces access to those, and prevents tenants from viewing, creating, editing, or deleting objects in another tenant. The control plane also manages global resources that affect the entire cluster. To isolate tenants from the impact of changes to global resources, consider the following mitigative actions:

Use namespaces for isolation.

Kubernetes provides namespaces as a logical grouping mechanism. Each team or user can have their own namespace, which isolates the groups of API resources within a single cluster.

Use RBAC to control their access to resources and scope them to their respective namespaces.

Implement RBAC to manage user access and permissions within the cluster. Assign appropriate roles and permissions to ensure that users have the necessary access without compromising security. See [Role-Based Access Control \(RBAC\) in Kyma](#).

Cluster admins manage global resources and Custom Resource Definitions (CRDs).

Global resources are outside of the namespace concept and must be managed by administrators. They are applied to all "tenants" in the same way and updates affect everyone at the same time. While this is limiting, there are several benefits: It incentivizes teams to use the same tools without needing to operate them, keeps the versions consistent across teams, and encourages that global best practices are provided or even enforced.

Use policies for compliance.

Use policy engines, like Gatekeeper or Kyverno, to enforce policies on what "tenants" can and cannot do. In addition, these policy engines can add metadata, like labels, to every workload tagging it with the "tenant" name for easier filtering of logs.

Use quotas, limits, and priorities for controlling resource allocation.

To control how many resources a tenant can occupy, especially in clusters that have a limited ability to scale, it's recommended to set resource quotas for tenants. Also, use resource quotas to limit the maximum resources of Pods, to assure that they fit on nodes besides the global workload.

Example

If a node has 64GB of RAM and the "global tooling" needs 4GB, the Pod size must be limited to 60GB of RAM; otherwise, the Pods cannot be scheduled reliably or run the risk of being evicted frequently.

If multiple namespaces are supported, you must establish a way to limit the consumption across multiple namespaces using limit ranges. Similarly, restrict the Pod priorities that can be set in a specific tenant, for example, prioritizing production workloads over background batch jobs.

Data Plane Isolation

Data plane isolation ensures that workloads from one tenant cannot interfere with workloads from another tenant. Two primary domains of isolation are networking and storage. However, you can use additional measures to prevent breached workloads from affecting other tenants of the same cluster.

Define network policies to control traffic flow.

Implement network policies to control traffic flow between different namespaces or applications. This helps enforce security and isolation and prevents unauthorized access or interference.

Use centralized observability.

To track cluster-wide resource usage, performance metrics, and health, implement a centralized observability

solution that the tenants can access and filter for their data. Ideally, access can be restricted to the tenants' data only.

Use Istio with dedicated ingress gateways.

Istio, an open source service mesh, improves traffic shaping and supports mTLS between the Pods, securing the traffic even more against other tenants sniffing traffic. Use dedicated ingress gateways per tenant to configure the domains individually. For details, see Istio Module.

URL: <https://help.sap.com/docs/btp/btp-developers-guide/7c9496c88a294b7f9ccc69a7e0998817.html>

Title: Develop Resilient Applications in the Kyma Runtime | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology

PlatformDevelopmentDevelopment in the Kyma EnvironmentDevelop Resilient Applications in the Kyma

Runtime

SAP Business Technology PlatformEnglish

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Develop Resilient Applications in the Kyma Runtime

All SAP BTP, Kyma runtime production plans ensure high availability. To benefit from the high-availability setup, make sure the architecture and deployment of your application comply with resiliency best practices. Use the following guidelines for Kubernetes and microservice-based applications to develop a stable application. Read further to find some Istio-related resiliency tips.

Resilient Applications Deployed on Kubernetesà^

To develop resilient applications deployed on Kubernetes, use:

Kubernetes built-in health checks. For more information, read how to Configure Liveness, Readiness and Startup Probes.

Deployment across multiple availability zones for high availability. For more details, see the blog post.

The Keda Module or Kubernetes autoscaling to automatically adjust the number of replicas based on resource usage.

Rolling updates to minimize downtime during application updates. If you don't provide any configuration, rolling updates are by default enabled in your SAP BTP, Kyma runtime.

BTP backing services, such as PostgreSQL or Object Store on SAP BTP. If you have to store some application data on SAP BTP, Kyma runtime, use persistent storage to ensure data is not lost in the event of a pod failure.

Resource quotas to prevent resource starvation and ensure fair resource allocation.

The Telemetry Module to configure a logging and monitoring solution to identify and troubleshoot issues in real time.

Resilient Microservice-Based Applicationsà^

The Istio Module and Istio traffic management features help you achieve inbuilt microservices resiliency.

Moreover, to develop resilient microservice-based applications, use:

Istio's circuit breaker and retry features to prevent service overloads and recover from service failures.

Distributed tracing with Istio's observability features to monitor the flow of requests across microservices and identify performance and error issues by shipping logs to SAP Cloud Logging (for details, see Integrate With SAP Cloud Logging).

The Eventing module for asynchronous communication patterns, such as messaging queues, to decouple services and reduce dependencies. See how to Configure SAP Event Mesh for Kyma Eventing.

Health check endpoints as a part of your microservice. For more details, read Kubernetes API health endpoints.

A centralized logging and monitoring solution to identify issues across multiple microservices and quickly troubleshoot them.

A clear and well-defined boundary in your microservice's design to minimize the impact of failures and enable easy replacement or scaling of individual services.

Istio's service mesh to implement mutual TLS and authorization policies to secure communication between microservices. For more details, read the blog post.

Network resiliency using Istio's traffic management features to test how your microservices respond to various failure scenarios, such as slow or failing services.

The Telemetry Module to ship your Istio's access logs and metrics to your logging backend to monitor traffic patterns and quickly troubleshoot issues.

Related Informationà^

Improve resiliency of your applications deployed on Kyma runtime by using multiple availability zones
Developing enterprise-grade applications with mutual TLS easy with SAP BTP, Kyma runtime and BTP destinations
SAP BTP Kyma API Rules with destinations and a managed approuter
Protect your Kyma workloads from eavesdropping with JWT access strategy and BTP destinations
Protect your Kyma workloads from eavesdropping with introspection and BTP destinations

URL: <https://help.sap.com/docs/btp/btp-developers-guide/cloud-application-programming-model>

Title: *Developing with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Kyma Runtimes* | SAP Help Portal

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SAP BTP Developer's GuideEnglish

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Developing with CAP in the SAP BTP, Cloud Foundry and SAP BTP, Kyma Runtimes

Overview^

The SAP Cloud Application Programming Model (CAP) is a framework of languages, libraries, and tools for building enterprise-grade services and applications. It guides developers along a path of proven best practices and a great wealth of out-of-the-box solutions to recurring tasks.

CAP-based projects benefit from a primary focus on domain. Instead of delving into overly technical disciplines, CAP focuses on accelerated development and safeguarding investments in a world of rapidly changing cloud technologies.

The following graphic shows that the CAP framework features a mix of proven and broadly adopted open-source and SAP technologies:

On top of open-source technologies, CAP mainly adds:

Core Data Services (CDS) as its universal modeling language for both domain models and service definitions.

Service SDKs and runtimes for Node.js and Java, offering libraries to implement and consume services as well as generic provider implementations serving many requests automatically.

These are the CAP development profiles you can use:

Development profile: use mocked services for local testing and development.

You can begin building SAP Cloud Application Programming (CAP) applications locally without needing a subscription or initial setup in SAP BTP. This approach offers several advantages during the early stages of development:

Rapid Prototyping

You can quickly design and test domain models and services locally, enabling faster iteration and feedback cycles.

Deferred Integration

You can postpone the integration with SAP BTP services until later stages, allowing you to focus on core functionality first.

Mocked Platform Services

The CAP development profile includes mocked variants of some SAP BTP services dealing with authentication, database, messaging, and application gateway. These mocks allow applications to run locally without requiring the application to be deployed to SAP BTP.

Hybrid profile: local development and testing using selected real services

Over time, you can add things gradually, only when they're needed. For example, you can move ahead to running your applications in close-to-productive setups for integration tests and delivery, without any change in models or code:

Run locally, but bind to real services (for example, SAP HANA Cloud, and the Destination service) using cds bind or a local default-env.json that simulates Cloud Foundry or Kyma service bindings.

Great for integration tests (for example, real SAP HANA Cloud schema, real authorization flows) before you deploy.

Production profile: use SAP BTP services and deploy the application to SAP BTP.

Run in SAP BTP, Cloud Foundry or SAP BTP, Kyma runtime with platform-managed service bindings and production-grade settings.

Real service instances (SAP HANA Cloud, Destination services, and others) are bound by the platform.

Production configuration for logging, security, scaling, and multitenancy.
Database migrations done via cds deploy (for example, to SAP HANA Cloud).
See Grow as You Go in the CAP documentation.

Related Informationà

Strengths and Benefits of the Runtimes and Their Programming Models

Explore

Discover

Design

Deliver

Run and Scale

Tutorials and Reference Sample for Full-Stack Multitenant SaaS Applications for Partners

Tutorials for SAP Cloud Application Programming Model

URL: <https://help.sap.com/docs/btp/comparison-btp-runtimes/archetypes>

Title: Archetypes | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry Runt...Archetypes

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Archetypes

Compare the archetypes supported by SAP BTP, Kyma runtime and SAP BTP, Cloud Foundry runtime.

Archetypes

Category

SAP BTP, Kyma runtime

SAP BTP, Cloud Foundry runtime

RESTful Web services

Yes

Yes

SOAP Web Services

Yes

Yes

Micro Services

Yes

Yes

CAP Applications

Yes

Yes

Webhooks

Yes

Yes

Custom Docker Container

Yes

Yes

See Deploy Docker Images in the Cloud Foundry Runtime and Understand the Cloud Foundry Application Lifecycle for Docker Applications

URL: <https://help.sap.com/docs/btp/comparison-btp-runtimes/definition-of-runtime>

Title: Definition of the Runtime | SAP Help Portal

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Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry RuntimeEnglish

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Definition of the Runtime

Introduces the SAP BTP, Kyma runtime and SAP BTP, Cloud Foundry runtime, together with aspects such as who interacts with the platform and the provided tenancy model.

Definition of the Runtime

Category

SAP BTP, Kyma runtime

SAP BTP, Cloud Foundry runtime

Definition

SAP BTP, Kyma runtime is a fully managed Kubernetes runtime based on the open-source project "Kyma". This cloud-native solution allows the developers to extend SAP solutions with serverless Functions and combine them with containerized microservices. The offered functionality ensures smooth consumption of SAP and non-SAP applications, running workloads in a highly scalable environment, and building event- and API-based extensions.

The SAP BTP, Cloud Foundry runtime lets you develop polyglot cloud-native applications and run them on the SAP BTP Cloud Foundry environment.

Consumption and interaction with the runtime

Provisioned in SAP BTP subaccount via SAP BTP cockpit or btp CLI

Access to Technical UI

Access to Kubernetes API

Access control is managed and handled by the runtime by using Kubernetes RBAC (Role Based Access Control)

Commercial integration into SAP BTP subaccount and global account for quota- and entitlement management and cost control

Managed and embedded in SAP BTP subaccount via SAP BTP cockpit or btp CLI. See Getting Started with an Enterprise Account in the Cloud Foundry Environment

Access to Cloud Foundry API

Access control is configurable for Orgs and Spaces

Commercial integration into SAP BTP subaccount and global account for quota- and entitlement management and cost control

Sharing mode

Non-shared cluster based on individual cloud provider subscription.

The cluster is provisioned using the requested virtual machine types from the selected cloud provider in the requested region.

Shared access of cloud provider virtual machines and infrastructure amongst other consumers of the platform in the landscape.

URL: <https://help.sap.com/docs/btp/comparison-btp-runtimes/observability>

Title: *Observability | SAP Help Portal*

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Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry RuntimeEnglish

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Observability

Compare the observability capabilities of SAP BTP, Kyma runtime and SAP BTP, Cloud Foundry runtime.

Observability

Category

SAP BTP, Kyma runtime

SAP BTP, Cloud Foundry runtime

Custom metric telemetry

Yes

Yes

Monitoring

Limited support for metrics and dashboarding with SAP Cloud Logging integration; see Ship Metrics to SAP Cloud Logging

Integration with any vendor-specific Kubernetes APM solution possible

Yes

Tracing

Support for distributed tracing with SAP Cloud Logging integration; see Ship Distributed Traces to SAP Cloud Logging

Integration with any vendor-specific tracing solution possible based on OTLP; see Traces

Zipkin tracing support, see Supported and Unsupported Cloud Foundry Features

Logging

Supported using SAP Cloud Logging integration; see Ship Logs to SAP Cloud Logging

Integration with any vendor-specific logging solution possible

Yes

Alerting

Supported using SAP Cloud Logging integration; see Use SAP Cloud Logging Alerts

Yes

URL: <https://help.sap.com/docs/btp/comparison-btp-runtimes/offerings-by-service>

Title: Offerings by the Runtime Service | SAP Help Portal

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Offerings by the Runtime Service

This section provides general information about the runtime offerings. Both runtimes in the focus of this comparison provide the same level of service.

Offering

Category

SAP BTP, Kyma runtime

SAP BTP, Cloud Foundry runtime

SLA

Standard SLA for cloud services, see SAP Trust Center and the Service Level Agreement for SAP Cloud Services.

Standard SLA for cloud services, see SAP Trust Center and the Service Level Agreement for SAP Cloud Services.

Release frequency

Continuous delivery; ad-hoc patches and hotfixes

Bi-weekly updates; ad-hoc patches and hotfixes

Upgrades

Upgrades and updates are included. Rollout depending on change type.

The change type defines which of the following update/upgrade windows are being used:

Major Upgrades (rarely happening, notification 4 weeks ahead of time)

Continuous delivery, see Kyma Release Channels

Immediate updates (urgent patches, not affecting productive applications)

For more information, see Platform Updates and Notifications.

Upgrades and updates are included. Rollout depending on change type.

The change type defines which of the following update/upgrade windows are being used:

Major Upgrades (rarely happening, notification 4 weeks ahead of time)

Monthly updates

Immediate updates (urgent patches, not affecting productive applications)

For more information, see Platform Updates and Notifications.

Operating model

See Operating Model in the Kyma Environment.

See Operating Model in the Cloud Foundry Environment.

Incident handling

See Operating Model in the Kyma Environment.

See Operating Model in the Cloud Foundry Environment.

Support levels

All support levels

All support levels

Regional availability

Available for Amazon Web Services, Google Cloud, and Microsoft Azure.

Regions for the Kyma Environment.

Datacenter Roadmap in Roadmap Explorer

Region information in SAP Discovery Center

Available for Amazon Web Services, Google Cloud, Microsoft Azure, and Alibaba Cloud.

Regions and API Endpoints Available for the Cloud Foundry Environment

Datacenter Roadmap in Roadmap Explorer

Region information in SAP Discovery Center

Contractually planned downtimes

Only when upgrades are planned for quarterly major upgrade window (MUW). See Platform Updates and Notifications

Only when upgrades are planned for quarterly major upgrade window (MUW). See Platform Updates and Notifications

URL: <https://help.sap.com/docs/btp/comparison-btp-runtimes/other-technical-capabilities>

Title: Other Information | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry Runt...Other Information

Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry RuntimeEnglish

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Other Information

Compare other information for SAP BTP, Kyma runtime and SAP BTP, Cloud Foundry runtime.

Other Information

Category

SAP BTP, Kyma runtime

SAP BTP, Cloud Foundry runtime

Docker container support

Yes

Yes, see Supported and Unsupported Cloud Foundry Features

SSH access into workload

Yes

Yes, see Supported and Unsupported Cloud Foundry Features

URL: <https://help.sap.com/docs/btp/comparison-btp-runtimes/protocols>

Title: Protocols | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry Runt...Protocols

Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry RuntimeEnglish

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Protocols

Compare protocols supported by SAP BTP, Kyma runtime and SAP BTP, Cloud Foundry runtime.

Protocols

Category

SAP BTP, Kyma runtime

SAP BTP, Cloud Foundry runtime

AMQP

Yes

Yes

MQTT

Yes
Yes
gRPC
Yes
Yes
HTTP/1.1
Yes
Yes
HTTP/2
Yes
Yes, see Supported and Unsupported Cloud Foundry Features
OData
Yes
Yes
SOAP
Yes
Yes
WebSockets
Yes
Yes, see Supported and Unsupported Cloud Foundry Features
Streaming endpoints (RTMP, RTSP, WebRTC)
Yes
No

URL: <https://help.sap.com/docs/btp/comparison-btp-runtimes/runtime-comparison>

Title: Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry Runtime | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry Runt...

Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry RuntimeEnglish

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Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry Runtime

In SAP BTP, you have the choice between several runtimes.

If you want to run ABAP, you'll choose the SAP BTP ABAP environment. Hence, this document only compares the non-ABAP runtimes SAP BTP, Cloud Foundry runtime and SAP BTP, Kyma runtime.

Both runtimes can be used to host and run your custom code, while at the same time being connected to SAP BTP Multi-Cloud Services and your on-premise systems. Depending on your use case and the skill set of your developers, choose the runtime that fits your needs.

Note

The following pages present features of SAP's non-ABAP runtimes grouped by category. Compare the features that are relevant for your runtime decision and follow the links to the respective detail documentation.

Last update: July 2025

Business Aspects Drive the Decisionà^

The difference and the main decision point is your application architecture, driven by business aspects. Usually, scaling patterns and economies of scale drive certain requirements towards the implementation of the business logic. That logic can vary from very simplistic pieces down to very complex chains of logic.

Some architectural patterns need more control over the infrastructure. Simple implementations of business logic need less infrastructure control.

No Difference in Non-Functionalsà^

Both non-ABAP runtimes offer parity with regards to non-functionals. Both bring multi-availability-zone infrastructure support and the same SLA levels. Applications do benefit from high availability of the runtime and can run in high availability mode themselves. Both runtimes are managed for you by SAP.

Decide Only When You Have Toà^

Because the runtimes don't differ from a non-functional point of view, you don't have to take the decision up front. If your business grows and your scaling requirements or use cases grow or change, then you can

augment your current architecture with more infrastructure by adding certain use cases on either runtime of choice. The decision which runtime to use doesn't need to be binary. It can very well be a combination of the two.

Related Information^

Definition of the Runtime

Runtime Strategy and Roadmap

Offerings by the Runtime Service

Development Skills

Security and Resilience

Archetypes

Protocols

Observability

Technical Details

Sizing

Other Information

URL: <https://help.sap.com/docs/btp/comparison-btp-runtimes/runtime-strategy>

Title: Runtime Strategy and Roadmap | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry Runt...Runtime Strategy and Roadmap

Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry RuntimeEnglish

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Runtime Strategy and Roadmap

Both runtime offerings are a fundamental part of SAP BTP's runtime service offerings and are part of SAP's long term strategy and commitment. This section shows where you find the relevant material.

Runtime Strategy and Roadmap

Category

SAP BTP, Kyma runtime

SAP BTP, Cloud Foundry runtime

Strategic embedding

Part of SAP BTP's App Development Strategy

Part of SAP BTP's App Development Strategy

Support for SAP Cloud Application Programming Model (CAP) as the strategic programming model.

Yes

For more information, see Deploy to Kyma Runtime

Yes

For more information, see Deploy to Cloud Foundry

Roadmap

SAP Road Map Explorer for SAP BTP, Kyma runtime

SAP Road Map Explorer for SAP BTP, Cloud Foundry runtime

Commercial Information

Commercial Information for Kyma Runtime

Commercial Information for Cloud Foundry Runtime

URL: <https://help.sap.com/docs/btp/comparison-btp-runtimes/sizing>

Title: Sizing | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry Runt...Sizing

Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry RuntimeEnglish

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Sizing

Compare the sizing for SAP BTP, Kyma runtime and SAP BTP, Cloud Foundry runtime.

Sizing

Category

SAP BTP, Kyma runtime

SAP BTP, Cloud Foundry runtime

Supported CPU architecture

AMD64

AMD64

Max. assignable CPU

Most of the available vCPUs can be assigned to workloads.

CPUs and memory can be defined individually. They are not connected.

Guaranteed CPU share of ¼ core per GB instance memory.

Because the maximum instance memory per application is 16 GB, vertical scaling can go up to 4 CPUs.

See SAP BTP-Specific Configurations.

Max. assignable memory

Up to 90% of the VM size.

For a list of available options, see the Kyma Price Calculator.

Up to 16 GB per single app instance.

See SAP BTP-Specific Configurations.

Minimum infrastructure size

3 Kubernetes Nodes, each 2 CPU and 8 GB memory. Roughly 40% reserved to run the infrastructure.

1 GB Memory with ¼ CPU core

URL: <https://help.sap.com/docs/btp/comparison-btp-runtimes/technical-details>

Title: *Technical Details | SAP Help Portal*

HomeSAP Business Technology Platform (SAP BTP)Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry Runt...Technical Details

Comparison: SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry RuntimeEnglish

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Technical Details

Compare the technical details for SAP BTP, Kyma runtime and SAP BTP, Cloud Foundry runtime.

Technical Details

Category

SAP BTP, Kyma runtime

SAP BTP, Cloud Foundry runtime

Based on open source and open standards

Based on Kyma open source, augmented by Istio, Open Telemetry, and SAP proprietary extensions.

Based on Cloud Foundry open source, augmented by HAProxy, BOSH, and SAP proprietary extensions.

Supported buildpacks

Cloud Native Buildpacks

SAP-managed and Cloud Foundry community buildpacks for several development languages

SAP BTP service support

Yes

Yes

Internal network support

Yes

No

See Supported and Unsupported Cloud Foundry Features

CI/CD support

Yes, see Roadmap

Yes

Service mesh support

Yes

No

Elasticity

Yes

Yes

Autoscaling support

Yes

Yes, with the Application Autoscaler service.

See Application Autoscaler and SAP Discovery Center: Application Autoscaler.

Speed of autoscaling

App scaling speed depends on startup time of the workload.

Vertical scale-up (by adding more Kubernetes Nodes to the cluster): < 1 min

App scaling speed depends on startup time of the workload

Partial zone outage detection

Yes

Yes

In-system build & deploy using buildpacks

No

Yes

Platform-as-a-service: MTA support

No

Yes

Identity provisioning support

No

No

Custom domains

Yes

See Set up a custom domain for a workload

Yes

See SAP Custom Domain Service

Transport services

No

For Kyma, use Helm charts and a CI/CD pipeline. See Delivery Options.

Yes

See SAP Cloud Transport Management

Connectivity

Yes

Yes

Task and Jobs support

Yes

Yes

See Supported and Unsupported Cloud Foundry Features

RFC connection support

Yes

Yes

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform-internal/regions-for-kyma-environment>

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/06b6fb3d45d040429e36f0359d2fe1f2.html>

Title: [What Is the Multi-Cloud Foundation?](#) | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)Migrating from the Neo Environment to the Multi-Cloud Foundation ...What Is the Multi-Cloud Foundation?

Migrating from the Neo Environment to the Multi-Cloud Foundation (Cloud Foundry and Kyma)English

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What Is the Multi-Cloud Foundation?

Get to know the multi-cloud foundation and its environments.

The multi-cloud foundation is the basis of SAP BTP, which through its multi-cloud compatibility, lets you choose from different infrastructures, environments, and runtimes.

It comprises three different environments: The Cloud Foundry environment, the Kyma environment, and the ABAP environment.

Environments within the Multi-Cloud Foundation

For more information about the different environments within the multi-cloud foundation, have a look at the following sections.

Note

The ABAP environment isn't part of this guide as it is not possible to migrate from the Neo environment to the ABAP environment.

For more information about the ABAP environment, see [ABAP Environment](#), instead.

Cloud Foundry Environmentà^

The Cloud Foundry environment allows you to create polyglot cloud applications in Cloud Foundry and is based on the Cloud Foundry Open Source Cloud Application Platform.

Cloud Foundry is an open source, industry standard Platform-as-a-Service (PaaS) technology for developing and deploying cloud applications in both private and public cloud environments. It's designed to run on a variety of Infrastructure-as-a-Service (IaaS) providers, such as Amazon Web Services, Google Cloud Platform, Alibaba Cloud, and Microsoft Azure. It uses a container-based architecture that executes applications in any programming language and lets you choose from different runtimes and backing services. For more information, see [Cloud Foundry - Open Source Cloud Application Platform](#).

SAP is a platinum-level founding member of the Cloud Foundry Foundation, which oversees the development of Cloud Foundry software and its ecosystem. To provide an open and portable platform to run and manage web-based applications and services in a cloud environment, it has adopted the open source Cloud Foundry technology as basis for its SAP BTP Cloud Foundry environment. You can use it to enhance SAP products and integrate business applications, as well as to develop entirely new enterprise applications based on business APIs.

Caution

Not all features supported by Cloud Foundry open source are automatically also supported by the SAP BTP Cloud Foundry environment. For a list of supported and unsupported features, see [Cloud Foundry Environment](#).

Kyma Environmentà^

SAP BTP, Kyma runtime provides a fully managed cloud-native Kubernetes application runtime based on the open-source project "Kyma". Based on modular building blocks, Kyma runtime includes all the necessary

capabilities to simplify the development and to run enterprise-grade cloud-native applications.

Kyma environment permits a native consumption of the Multi-Cloud Foundation Services and smooth consumption of SAP and non-SAP applications. It also supports out-of-the-box CAP, SAP Cloud SDK, application router, and HTML5 deployer.

Kyma implements a dedicated application runtime to deploy highly scalable, robust, and secure containerized microservices.

Every Kyma environment consists of:

A Kubernetes cluster based on project "Gardener" on a cloud provider and region (data center) of your choice.

To find out the available regions and providers, see [Regions for the Kyma Environment](#).

The open-source project "Kyma" installed in its latest version on the provisioned cluster.

Every Kyma environment runs on a single Kubernetes cluster created for a specific subaccount. However, the configuration of the Kyma environment allows you to connect it to a multitude of SAP systems.

For all functionalities of the Kyma environment, see the project "Kyma" documentation.

Become Familiar with the Multi-Cloud Foundation

Before migrating your existing scenarios, take your time to get to know the multi-cloud foundation and its components.

Multi-Cloud Foundation

The following content helps you get a basic understanding of the multi-cloud foundation:

[Why You Should Migrate from the Neo Environment to the Multi-Cloud Foundation](#)

Learn how the multi-cloud foundation differs from the Neo environment and why you should migrate to it.

Cloud Foundry Environment

The following links help you become familiar with the Cloud Foundry environment:

[Cloud Foundry Environment](#)

Get more detailed information about the Cloud Foundry environment.

[Getting Started with a Trial Account in the Cloud Foundry Environment](#)

Use an SAP BTP trial account to get a first practical impression of the Cloud Foundry environment.

[Create Your First App on Cloud Foundry](#)

Gain first hands-on experience in the Cloud Foundry environment.

Kyma Environment

The following links help you become familiar with the Kyma environment:

[Kyma Environment](#)

Get more detailed information about the Kyma environment.

[Getting Started with a Trial Account in the Kyma Environment](#)

Use an SAP BTP trial account to get a first practical impression of the Kyma environment.

[Build an Application in the Kyma Runtime](#)

Explore SAP BTP, Kyma runtime by deploying a containerized microservice onto Kubernetes, exposing it via API, and making it consumable with a UI5 frontend.

Related Information

Environments

[Cloud Foundry Environment](#)

[Kyma Environment](#)

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/aae4e0ae1cdf434b908c3c8cf3ea942a.html>

Title: [Migrating from the Neo Environment to the Multi-Cloud Foundation for SAP BTP \(Cloud Foundry and Kyma\) | SAP Help Portal](#)

HomeSAP Business Technology Platform (SAP BTP)Migrating from the Neo Environment to the Multi-Cloud Foundation ...

Migrating from the Neo Environment to the Multi-Cloud Foundation (Cloud Foundry and Kyma)English

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Migrating from the Neo Environment to the Multi-Cloud Foundation for SAP BTP (Cloud Foundry and Kyma)

Learn why and how to migrate scenarios from the Neo environment to the multi-cloud foundation for SAP BTP.

This guide is for SAP Business Technology Platform (SAP BTP) customers and partners with scenarios in the Neo environment that need to move to the multi-cloud foundation, including the Cloud Foundry environment or the Kyma environment.

What Is This Guide About?

The multi-cloud foundation has become the strategic focus for SAP BTP because of its significant benefits as compared with the Neo environment, including its principles of openness and freedom of choice in terms of support for multiple infrastructure providers, data centers, and a wide range of runtimes and programming languages. Unlike for Neo where you can use selected services only, you can choose from the full service portfolio on SAP Discovery Center.

Therefore, the Neo environment will be sunset and scenarios on Neo must be migrated to the multi-cloud foundation (that is, the Cloud Foundry environment or the Kyma environment).

Remember

SAP Business Technology Platform, Neo environment will sunset on December 31, 2028, subject to terms of customer or partner contracts.

For more information, see SAP Note 3351844 .

Learn about the characteristics of the multi-cloud foundation and its differences and advantages compared to the Neo environment.

Use this guide to plan and prepare for the migration to the multi-cloud foundation, to get help in deciding on your migration path and to learn where you find detailed information about the steps involved in migrating your scenarios from the Neo environment to the multi-cloud foundation. It presents a strategic approach to handling your migration, helps you to get started, and during the process, improves your expertise in working with the new environment.

Is This Guide for You?

This guide addresses SAP BTP customers and partners who have at least one existing scenario in SAP BTP, Neo environment that they must move to the multi-cloud foundation (that is, the Cloud Foundry environment or the Kyma environment).

A scenario on SAP BTP pursues a specific business objective (that is, a use case) and most likely, comprises several different applications, services, and integrations to achieve it. Therefore, migrating a scenario from the Neo environment to the multi-cloud foundation implies migrating all components involved.

Documentation Feedback Survey

Give Feedback

Help us improve our documentation. Please take this two-minute survey and let us know what you think.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/about-trial-account>

Title: About the Trial Account | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Getting Started with a Trial Account in the Kyma EnvironmentAbout the Trial Account

SAP Business Technology PlatformEnglish

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About the Trial Account

A trial account lets you try out the Kyma environment for free with a restricted use of the platform resources and services. Access is open to everyone. Trial accounts are intended for personal exploration, and not for production use or team development.

Trial Period

The trial account is valid for 90 days. During this time, you can explore and use the basic functionality of SAP BTP, Kyma runtime for 14 days. After 90 days, your account is automatically deleted, and you'll no longer be able to access your data. See Trial Accounts and Free Tier to learn more about the trial periods.

Note that while your trial account is valid for 90 days, the Kyma environment that you create with it expires 14 days after its creation and is then deleted permanently. To create a new Kyma environment, you must first disable the expired Kyma in your subaccount. To learn how to do that, see Disable Expired Trial Kyma Environment.

Caution

Once your account has expired, your cluster is deleted and all the cluster resources are removed. We recommend that you back up your cluster configuration to restore it quickly if you choose to create a new Kyma environment.

Scope and Limitations

It's possible to provision only one Kyma environment per global account.

You can only create a single Kyma cluster per subaccount.

You can manage members in your trial account.

Kyma trial accounts are available in all SAP BTP regions.

A trial account provides you with a 1-node cluster with 4 vCPU and 16 GB of memory. An idle cluster without any customer workload uses around 33% of vCPU and 33% of memory.

A trial account comes with a cluster with built-in eventing.

Events are processed in-memory and are not persisted to disk. Not delivered events will be lost on restart of the eventing infrastructure.

There's no service level agreement with regards to the availability of the platform.

Related Information^

Trial Accounts and Free Tier

Assign Roles in the Kyma Environment

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/access-kyma-application-logs>

Title: Access Kyma Application Logs | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformAdministration and OperationsAdministration and Operations in the Kyma EnvironmentAccess Kyma Application Logs
SAP Business Technology PlatformEnglish

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Access Kyma Application Logs

Get insights into your applications, microservices, and Functions by viewing the respective logs. To check out real-time logs immediately, use the Kubernetes functionalities - either in Kyma dashboard, or with kubectl.

Contextà^

If you prefer to see the logs in your terminal, replace the placeholders in the following command and run it:

âE

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```
kubectl logs <{POD_NAME}> --namespace <{NAMESPACE_NAME}> --container <{CONTAINER_NAME}>
```

àf

To see real-time logs in Kyma dashboard, follow these steps:

Procedureà^

Open Kyma dashboard and select the namespace.

Access the Pod.

Find the container and select View Logs.

Resultsà^

You have access to short-term logs of every container in your Kyma runtime.

Next Stepsà^

Recommendation

If you want long-term storage of logs with advanced querying, dashboarding, and alerting capabilities, set up SAP Cloud Logging to ingest observability data with Kyma runtime.

Related Informationà^

Auditing and Logging Information in Kyma

Create an SAP Cloud Logging Instance through SAP BTP Service Operator

SAP Cloud Logging: Ingest via Kyma Runtime

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/accessing-your-audit-log-data>

Title: Accessing Your Audit Log Data | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Provisioning and Updating Parameters in the Kyma EnvironmentAccessing Your Audit Log Data
SAP Business Technology PlatformEnglish

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Accessing Your Audit Log Data

Use the credentials stored in your Kyma cluster to query your audit log data using the SAP Audit Log Retrieval API v2.

Prerequisites

You have a Kyma instance created. See [Creating Kyma Instances](#).

You have the Audit Log Access (auditLogAccess) parameter enabled. See [Audit Log Access](#).

Context

When you enable the Audit Log Access parameter, Kyma stores the audit log read credentials in a Kubernetes Secret named auditlog-read-credentials in the kyma-system namespace of your cluster. The Secret contains the following fields:

Field

Description

url

Base URL of the SAP Audit Log Retrieval API

clientID

OAuth 2.0 client ID

clientSecret

OAuth 2.0 client Secret

tokenUrl

OAuth 2.0 token endpoint

Caution

Enabling Audit Log Access is irreversible. Once you enable the feature, you cannot disable it.

Procedure

Connect to your Kyma cluster using kubectl. See [Access a Kyma Instance Using kubectl](#).

To retrieve the Secret, run:

```
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```

```
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```

```
kubectl get secret -n kyma-system auditlog-read-credentials -o yaml
```

```
àf
```

Use the credentials from the Secret to authenticate against the SAP Audit Log Retrieval API v2 using OAuth 2.0 client credentials. See [Create an OAuth Access Token](#).

Retrieve your audit log records. See [Audit Log Retrieval](#).

Results

You can read your audit log data from the SAP Audit Log Retrieval API v2 using the credentials stored in the auditlog-read-credentials Secret.

Related Information

[Audit Log Retrieval API for Global Accounts in the Cloud Foundry Environment](#)

[Retrieve Audit Log Data Using Audit Log CLI](#)

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/administration-and-operations-in-kyma-environment>

Title: *Administration and Operations in the Kyma Environment* | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformAdministration and

OperationsAdministration and Operations in the Kyma Environment

SAP Business Technology PlatformEnglish

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Administration and Operations in the Kyma Environment

This is the managed offering of SAP BTP, Kyma runtime (based on the open-source project "Kyma"). The administrators of the Kyma environment take care of setting it up and make sure it is ready for developers to work with. Create your Kyma instance to build applications and extensions to SAP and third-party solutions, manage roles, have your Kubernetes objects backed up, and view metrics and logs.

Caution

Do not make changes in any Kyma-managed namespaces, the kube-system namespace, or to any resources marked with Do Not Edit. Changes to those resources could lead to exclusion or disruptions from the agreed SLA. If you decide to make changes despite this, consult support in advance, and proceed with caution. To identify a Kyma-managed namespace, check if it has the operator.kyma-project.io/managed-by: kyma label.

Creating Kyma Instances

Set up a Kubernetes cluster with SAP BTP, Kyma runtime and use it to build applications and extensions to

your SAP and third-party solutions. You can create one or multiple Kyma clusters in a single SAP BTP subaccount.

Tracking Kubeconfig and Cluster Associations in Kyma

You can identify which kubeconfig is associated with a particular Kyma cluster by the value of the Cluster Name parameter.

Managing Kyma Runtime Using the Provisioning Service API

The SAP Cloud Management service for SAP BTP (technical name: cis) provides the Provisioning Service API to create and manage available environments. Use the Provisioning Service API to automatically manage and access SAP BTP, Kyma runtime.

Available Plans in the Kyma Environment

Depending on your global account type, you have access to a different plan that specifies the cluster parameters for the Kyma environment.

Provisioning and Updating Parameters in the Kyma Environment

When creating a Kyma cluster, you can configure various parameters to adjust it to your specific needs.

Adding and Deleting a Kyma Module

To use a Kyma module, you must add it first. Use Kyma dashboard or Kyma CLI to do that. If you don't need the module anymore, delete it to save resources.

Setting Your Module to the Unmanaged and Managed State

In some cases, for example, for testing, you may need to modify your module beyond what is supported by its configuration. By default, when a module is in the managed state, Kyma Control Plane governs its Kubernetes resources, reverting any manual changes during the next reconciliation loop. To modify Kubernetes objects directly without them being reverted, you must set the module to the unmanaged state. In this state, reconciliation is disabled, ensuring your manual changes are preserved.

Access Kyma Application Logs

Get insights into your applications, microservices, and Functions by viewing the respective logs. To check out real-time logs immediately, use the Kubernetes functionalities - either in Kyma dashboard, or with kubectl.

Kyma Environment Backup

Kyma Modules' Sizing

The following tables show the resource consumption, such as memory, CPU, and storage, of Kyma modules.

Change Storage Size in Kyma

If the amount of data for the applications in your Kyma environment grows, you can expand the storage size for your customer data by resizing the respective Persistent Volume Claim (PVC).

Assigning Workloads to Worker Node Pools

In addition to the mandatory worker node pool existing in each Kyma runtime, you can add and configure custom worker node pools. These additional worker nodes are reserved for running your workloads with special computing requirements on optimized machines. To ensure your workload runs on the right worker node pool, assign workloads to particular worker nodes.

Admission Policies

Learn about the validating admission policies deployed in SAP BTP, Kyma runtime clusters, why they block specific actions, and what you can do if you need an exception. These policies are enforced at admission time and protect cluster stability, security, and operational correctness.

Kyma Runtime with Dual-Stack Support

Learn how to apply dual-stack support in Kyma instances.

Ingress and Egress IP Addresses

Learn how to check ingress and egress IP addresses for your Kyma runtime cluster. These addresses are needed to integrate with partner systems, enforce IP-based access controls, connect to external databases, and more.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/api-gateway-module>

Title: API Gateway Module | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Kyma ModulesAPI Gateway Module

SAP Business Technology PlatformEnglish

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API Gateway Module

API Gateway is a Kyma module with which you can expose and secure APIs.

What Is API Gateway?

To use the API Gateway module, you must also add the Istio module. Moreover, to expose a workload using the APIRule custom resource (CR), the workload must be part of the Istio service mesh.

By default, both the API Gateway and Istio modules are automatically added when you create a SAP BTP, Kyma runtime instance.

Features

The API Gateway module offers the following features:

API exposure: The module uses Istio features to help you easily and securely expose your workloads by creating APIRule CRs. With an APIRule, you can perform the following actions:

- Group multiple workloads and expose them under a single host.

- Use a short host name to simplify the migration of resources to a new cluster.

- Configure the noAuth access strategy, which offers a simple configuration to allow access to specific HTTP methods.

- Secure your workloads by configuring jwt or extAuth access strategies. The jwt access strategy enables you to use Istio's JWT configuration to protect your exposed services and interact with them using JSON Web Tokens. The extAuth access strategy allows you to implement custom authentication and authorization logic.

Default Kyma gateway configuration: The module creates and manages the default Istio TLS Gateway that receives incoming traffic in your Kyma cluster. The gateway uses the cluster's default domain and a self-signed certificate.

Rate limiting: The module simplifies local rate limiting on the Istio service mesh layer. By configuring a RateLimit CR, you can limit the number of requests targeting an exposed application in a unit of time, based on specific paths and headers.

External gateway integration: Configure Istio mTLS Gateway and integrate it with an external gateway deployed outside of the Kyma cluster.

Architecture

Within the API Gateway module, API Gateway Operator manages the application of API Gateway's configuration and handles resource reconciliation. It contains the following controllers: APIGateway Controller, APIRule Controller, RateLimit Controller, and ExternalGateway Controller.

API/Custom Resource Definitions

The `apigateways.operator.kyma-project.io CustomResourceDefinition (CRD)` describes the APIGateway CR that APIGateway Controller uses to manage the module and its resources. See [APIGateway Custom Resource](#). The `apirules.operator.kyma-project.io CRD` describes the APIRule CR that APIRule Controller uses to expose and secure APIs. See [APIRule Custom Resource](#).

The `externalgateways.gateway.kyma-project.io CRD` describes the kind and the format of data that ExternalGateway Controller uses to configure external gateway integration. See [ExternalGateway Custom Resource](#).

The `ratelimits.gateway.kyma-project.io CRD` describes the kind and the format of data that RateLimit Controller uses to configure request rate limits for applications. See [RateLimit Custom Resource](#).

Authorization

To assign access permissions to the API Gateway module resources, use the following aggregated ClusterRoles:

- `kyma-api-gateway-view` - Grants read-only access to all API Gateway resources.

- `kyma-api-gateway-edit` - Grants full access to `gateway.kyma-project.io` resources and read-only access to `operator.kyma-project.io` resources.

- `kyma-api-gateway-admin` - Grants full access to all API Gateway resources.

Resource Consumption

To learn more about the resources used by the Istio module, see [Kyma Modules' Sizing](#).

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/application-connector-module>

Title: *Application Connector Module | SAP Help Portal*

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Kyma
ModulesApplication Connector Module
SAP Business Technology PlatformEnglish

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Application Connector Module

With the Application Connector module, you can connect your workflows deployed on Kyma with external solutions. No matter if you want to integrate an on-premise or a cloud system, the integration process does not change, which prevents any configuration or network-related problems.

What is Application Connectivity in Kyma?à^

Application Connectivity in Kyma simplifies the interaction between external systems and your Kyma workloads.

The main benefits are:

Smooth and loosely coupled integration of external systems with Kyma workloads using Kyma Eventing

Easy consumption of SAP BTP services by supporting the SAP BTP Extensibility approach

Establishing high-security standards for any interaction between systems by using trusted communication channels and authentication methods

Reducing configuration changes for Kyma workloads through encapsulating configuration details of external API endpoints

Monitoring and tracing capabilities to facilitate operational aspects

The Application Connector module bundles all features of Application Connectivity in Kyma. You can install and manage the module using Kyma dashboard.

The module includes Kubernetes operators and is fully configurable over its own Kubernetes custom resources (CRs). For each external system, a dedicated configuration is used. This allows for individual configuration of security and aspects (like encryption and authentication) per system.

Besides proxying any ingress and egress requests to external systems and dealing with security concerns, it also includes full integration with SAP BTP Unified Customer Landscape (UCL) to simplify the consumption of SAP BTP services.

Featuresà^

The Application Connector module provides the following features:

Easy installation of Kyma's Application Connectivity capabilities by enabling the Application Connector module in your SAP BTP, Kyma runtime.

Simple configuration using Kubernetes CRs and easy management with Kyma dashboard.

Full integration of BTP's UCL service, which implements the SAP BTP Extensibility concept. This allows for the automated integration of external systems registered in the UCL service.

Dispatching of incoming requests from external systems to Kyma workloads (for example, a Kyma Serverless Function) by using an Istio Gateway with mTLS and the Kyma Eventing module.

Proxying outgoing requests to external APIs and transparently covering security requirements like encryption and authentication (like OAuth 2.0 + mTLS, Basic Auth, and Client Certificates).

Metering of throughput and exposing monitoring metrics.

Architectureà^

Application Connector Manager

When you add the Application Connector module, Application Connector Manager takes care of installation and configuration of all the Application Connector module components in your cluster. It manages the lifecycle of the Application Connector module based on the dedicated ApplicationConnector custom resource (CR).

API/Custom Resource Definitionsà^

The API of the Application Connector module is based on Kubernetes Custom Resource Definitions (CRD), which extend the Kubernetes API with custom additions. To inspect the specification of the Application Connector module API, see:

Application CRD

CompassConnection CRD

ApplicationConnector CRD

Resource Consumptionà^

To learn more about the resources used by the Application Connector module, see Application Connector.

Useful Linksà^

To learn more how to work with the Application Connector module, see the Application Connectivity tutorials.

To learn more about the architecture, the configuration parameters, and any other references, see Technical Reference.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/assigning-workloads-to-worker-node-pools>

Title: Assigning Workloads to Worker Node Pools | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformAdministration and OperationsAdministration and Operations in the Kyma EnvironmentAssigning Workloads to Worker Node Pools
SAP Business Technology PlatformEnglish

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à

Assigning Workloads to Worker Node Pools

In addition to the mandatory worker node pool existing in each Kyma runtime, you can add and configure custom worker node pools. These additional worker nodes are reserved for running your workloads with special computing requirements on optimized machines. To ensure your workload runs on the right worker node pool, assign workloads to particular worker nodes.

Prerequisitesà

You have at least one additional worker node pool in your Kyma runtime. To learn how to add and configure a custom worker node pool, see [Additional Worker Node Pools](#).

Contextà

Worker node pools are collections of one or more worker nodes. To assign your workloads (Pods) to a particular worker node pool, use the following worker node's label: `worker.gardener.cloud/pool=<worker-pool-name>`. The label specifies the worker pool to which the node belongs.

You can assign your workloads to worker node pools using one of the following recommended methods:

Set a node selector.

Configure node affinity.

Node affinity offers more flexibility for assigning workloads than node selector. You can define multiple affinity rules and configure them as hard or soft requirements by using one of the following strategies:

`requiredDuringSchedulingIgnoredDuringExecution` or `preferredDuringSchedulingIgnoredDuringExecution`. You can also invert affinity by setting it as anti-affinity.

You can also assign your workloads to worker node pools using taints with tolerations. This method is useful for strict control over where workloads are placed, ensuring isolation or prioritization of specific workloads. For more information, see [Taints and Tolerations](#).

Procedureà

Use a Node Selector.

Use Pod Affinity.

Resultsà

With a node selector or affinity constraint, Pods are only scheduled on nodes that match the defined label conditions.

Troubleshootingà

Pod stuck in the Pending state

Symptom: Your Pod stays in the Pending state.

Cause: There's no suitable node.

Solution: Check the labels used to find matching nodes or use a soft node assignment by applying the node affinity type `preferredDuringSchedulingIgnoredDuringExecution`.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/availability-zones-in-kyma-environment>

Title: Availability Zones in the Kyma Environment | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Kyma
EnvironmentAvailability Zones in the Kyma Environment

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Availability Zones in the Kyma Environment

The Kyma environment uses availability zones to provide high availability and fault tolerance for your applications and workloads.

Availability zones are isolated failure domains within a single geographical region, connected through low-latency networks, and operating independently with their own power, network, and cooling infrastructure. This

separation minimizes the risk that a failure in one zone affects the others, and a localized issue causes widespread disruption to applications and services.

High Availability

The Kubernetes and SAP BTP, Kyma runtime configurations are optimized for production use cases. By operating on a three-availability-zone architecture, Kyma offers the high availability feature within the following enterprise plans:

Standard: Amazon Web Services, Google Cloud, Microsoft Azure, and Alibaba Cloud

Build Runtime: Amazon Web Services, Google Cloud, and Microsoft Azure

This means that the default worker nodes are deployed in three availability zones of the respective cloud region, and thus can provide zone-level failure tolerance for Kyma and applications deployed on Kyma runtime. The Kubernetes control plane is also hosted in three availability zones. Thus, the runtime automatically manages node and zone failures for all managed components, including the Kubernetes API server.

High Availability in Custom Worker Node Pools

We recommend enabling high availability for additional worker node pools in production environments to enhance resilience. To ensure optimized deployment, Kyma runtime automatically selects specific availability zones during provisioning, so you cannot choose them yourself.

If you disable high availability, your additional worker node pool lacks zone-level failure tolerance because it's deployed in a single, randomly selected availability zone.

Caution

You cannot change the high availability setting of an existing additional worker node pool. To alter this setting, you must delete and recreate the pool with the desired configuration.

High availability for additional worker node pools depends on your choice of virtual machine types:

For general-purpose machines, three availability zones are always available in all supported regions.

For other machine types, for example, compute-intensive machines, the number of availability zones varies by region. For more information on machine types and regional availability, see [Machine Type: Machine Type in Additional Worker Node Pools](#).

High Availability in User Applications

While high availability is guaranteed for Kubernetes and native Kyma components, it is not automatically provided for your own applications deployed on Kyma. To ensure your applications are resilient to zone failures, you must manually configure high availability by following these guidelines:

Deploy multiple replicas across different availability zones.

Use appropriate resource requests and limits.

Configure health checks and readiness probes.

Implement proper service discovery and load balancing.

If you deploy with topology spread constraints to distribute the replicas across multiple availability zones, the Kubernetes scheduler distributes them across zones, ensuring that if one zone becomes unavailable, your applications continue to run on replicas in the remaining zones.

For more information on building resilient applications, see [Develop Resilient Applications in the Kyma Runtime](#).

Updates and Upgrades

To maintain high availability and reliability of the Kyma environment during updates and upgrades, the following processes are performed:

Regular updates: Kyma modules are updated regularly across all availability zones to ensure no downtime. By distributing updates across zones, Kyma maintains application and workload availability during routine maintenance.

Major upgrades and maintenance schedules: For critical upgrades, such as Kubernetes version changes, Kyma manages the process within the multi-zone architecture. Updates occur one node at a time across different availability zones to minimize the risk of simultaneous failures and ensure that essential services remain operational within other zones. In Kyma, necessary downtimes are planned and managed during designated periods, following the BTP maintenance window schedule. The sequential update process ensures continuous operation within unaffected zones, maintaining service integrity and platform availability.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/available-plans-in-kyma-environment>

Title: *Available Plans in the Kyma Environment* | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformAdministration and OperationsAdministration and Operations in the Kyma EnvironmentAvailable Plans in the Kyma Environment

Available Plans in the Kyma Environment

Depending on your global account type, you have access to a different plan that specifies the cluster parameters for the Kyma environment.

Trialà^

The technical name of the plan is trial. For details on the trial cluster specification, see [Scope and Limitations](#).

Remember

The parameters marked with an asterisk "*" are mandatory.

Trial Plan Specification

Parameter

Description

Supported Operation

More Information

Cluster Name*

btp CLI parameter: name

Defines the name of your cluster.

Provisioning

Cluster Name*

Modules

btp CLI parameter: modules

Defines which Kyma modules are provisioned in your cluster.

Provisioning

Modules

OpenID Connect

btp CLI parameter: oidc

Provides a custom OpenID Connect (OIDC) configuration.

Provisioning

Updating

OpenID Connect (OIDC)

Administrators

btp CLI parameter: administrators

Specifies the list of runtime administrators.

Provisioning

Updating

Administrators

Freeà^

The technical name of the plan is free. With the free service plans for Kyma, you can try out services in global accounts without any additional cost for 30 days. However, the free model account has certain limitations.

The free plan offers you a one-node cluster and is only available on AWS.

Only community support is available for free tier service plans and these are not subject to SLAs.

Before you enable the Kyma environment, you must first assign it as an entitlement to your subaccount.

You can use the free plan only once in a global account for up to 30 days.

Remember

Once you have started the free plan in a subaccount, you cannot use it in another subaccount within the same global account even though the 30-day period has not ended.

The services you plan to use must be available in the same region as the subaccount for the Kyma runtime. If necessary, change the default subaccount region.

The upgrade to the paid plan is not yet supported.

For more information, read [Using Free Service Plans](#).

Remember

The parameters marked with an asterisk "*" are mandatory.

Free Plan Specification

Parameter

Description

Supported Operation

More Information

Cluster Name*

btp CLI parameter: name

Defines the name of your cluster.

Provisioning

Cluster Name*

Region*

btp CLI parameter: region

Defines a region (set of datacenters) where your cluster runs.

Provisioning

Region*

Modules

btp CLI parameter: modules

Defines which Kyma modules are provisioned in your cluster.

Provisioning

Modules

Networking

btp CLI parameter: networking

Provides a custom IP range for worker nodes.

Provisioning

Networking

OpenID Connect

btp CLI parameter: oidc

Provides a custom Open ID Connect (OIDC) configuration.

Provisioning

Updating

OpenID Connect (OIDC)

Administrators

btp CLI parameter: administrators

Specifies the list of runtime administrators.

Provisioning

Updating

Administrators

Standard: Amazon Web Services, Google Cloud, Microsoft Azure, and Alibaba Cloud[^]

The technical names of the standard enterprise plans are aws, gcp, azure, and alicloud.

Within these plans, Kyma offers the high availability feature. For more information, see [Availability Zones in the Kyma Environment](#).

Note

To indicate that your Kyma runtime is used for production, select **Used for production** in your subaccount details. This setting allows SAP BTP, Kyma runtime operators to prioritize incidents and support cases affecting production subaccounts over subaccounts used for non-production purposes. See [Change Subaccount Details](#). After your Kyma instance is created, you can switch from standard enterprise plans to build runtime plans within the same cloud provider. However, you can't switch from build runtime plans to the standard plans. For more information, see [Updating Kyma Environment Plans](#).

Remember

The parameters marked with an asterisk "*" are mandatory.

Standard Plan Specification

Parameter

Description

Supported Operation

More Information

Cluster Name*

btpt CLI parameter: name

Defines the name of your cluster.

Provisioning

Cluster Name*

Region*

btpt CLI parameter: region

Defines a region (set of datacenters) where your cluster runs.

Provisioning

Region*

Colocate Control Plane

btpt CLI parameter: colocateControlPlane

Allows you to have both the control plane and worker nodes in the same region.

Provisioning

Colocate Control Plane

Machine Type

btpt CLI parameter: machineType

Specifies the provider-specific virtual machine type.

Provisioning

Updating

Machine Type

Auto Scaler Min

btpt CLI parameter: autoScalerMin

Specifies the minimum number of virtual machines to create.

Provisioning

Updating

Auto Scaler Min

Auto Scaler Max

btpt CLI parameter: autoScalerMax

Specifies the maximum number of virtual machines to create.

Provisioning

Updating

Auto Scaler Max

Modules

btpt CLI parameter: modules

Defines which Kyma modules are provisioned in your cluster.

Provisioning

Modules

Networking

btpt CLI parameter: networking

Provides a custom IP range for worker nodes.

Provisioning

Networking

OpenID Connect

btpt CLI parameter: oidc

Provides a custom Open ID Connect (OIDC) configuration.

Provisioning

Updating

OpenID Connect (OIDC)

Administrators

btpt CLI parameter: administrators

Specifies the list of runtime administrators.

Provisioning

Updating

Administrators

Additional Worker Node Pools

btp CLI parameter: additionalWorkerNodePools

Defines a custom list of additional worker node pools.

Provisioning

Updating

Additional Worker Node Pools

Access Control List 1, 2

accessControlList

Specifies the IP ranges that can access the Kubernetes API.

Provisioning

Updating

Access Control List

Additional Volume Size 2

btp CLI parameter: additionalVolumeSizeGi

Specifies extra disk space on top of the default volume size for a worker node pool.

Provisioning

Updating

Additional Volume Size

Audit Log Access 2

btp CLI parameter: auditLogAccess

Enables direct read access to your own audit log data using the SAP Audit Log Retrieval API v2.

Provisioning

Updating

Audit Log Access

Note

1 Not available in the standard Google Cloud (gcp) plan.

Note

2 Not available in the standard Alibaba Cloud (alicloud) plan or in the China (North 3) (cf-cn20) region in the Microsoft Azure (azure) plan.

SAP Cloud Infrastructure[^]

The technical name of this enterprise plan is sap-converged-cloud.

Remember

The parameters marked with an asterisk "*" are mandatory.

SAP Cloud Infrastructure Plan Specification

Parameter

Description

Supported Operation

More Information

Cluster Name*

btp CLI parameter: name

Defines the name of your cluster.

Provisioning

Cluster Name*

Region*

btp CLI parameter: region

Defines a region (set of datacenters) where your cluster runs.

Provisioning

Region*

Colocate Control Plane

btp CLI parameter: colocateControlPlane

Allows you to have both the control plane and worker nodes in the same region.

Provisioning

Colocate Control Plane

Machine Type

btpt CLI parameter: machineType

Specifies the provider-specific virtual machine type.

Provisioning

Updating

Machine Type

Auto Scaler Min

btpt CLI parameter: autoScalerMin

Specifies the minimum number of virtual machines to create.

Provisioning

Updating

Auto Scaler Min

Auto Scaler Max

btpt CLI parameter: autoScalerMax

Specifies the maximum number of virtual machines to create.

Provisioning

Updating

Auto Scaler Max

Modules

btpt CLI parameter: modules

Defines which Kyma modules are provisioned in your cluster.

Provisioning

Modules

Networking

btpt CLI parameter: networking

Provides a custom IP range for worker nodes.

Provisioning

Networking

OpenID Connect

btpt CLI parameter: oidc

Provides a custom Open ID Connect (OIDC) configuration.

Provisioning

Updating

OpenID Connect (OIDC)

Administrators

btpt CLI parameter: administrators

Specifies the list of runtime administrators.

Provisioning

Updating

Administrators

Additional Worker Node Pools

btpt CLI parameter: additionalWorkerNodePools

Defines a custom list of additional worker node pools.

Provisioning

Updating

Additional Worker Node Pools

Additional Volume Size

btpt CLI parameter: additionalVolumeSizeGi

Specifies extra disk space on top of the default volume size for a worker node pool.

Provisioning

Updating

Additional Volume Size

Audit Log Access

btpt CLI parameter: auditLogAccess

Enables direct read access to your own audit log data using the SAP Audit Log Retrieval API v2.

Provisioning

Updating

Audit Log Access

Build Runtime: Amazon Web Services, Google Cloud, Microsoft Azure, and Alibaba Cloud[^]

The technical names of these enterprise plans are build-runtime-aws, build-runtime-gcp, build-runtime-azure, and build-runtime-alicloud. Use them to integrate Kyma's functionalities within SAP Build. See the SAP Build documentation.

You can switch from the standard enterprise plans to build runtime plans within the same cloud provider. For more information, see Updating Kyma Environment Plans.

You can also use the build runtime plans within the Pay-As-You-Go for SAP BTP for cloud test, demo, and development commercial model. For details, see Service Plans and Metering.

Remember

The parameters marked with an asterisk "*" are mandatory.

Build Runtime Plan Specification

Parameter

Description

Supported Operation

More Information

Cluster Name*

btp CLI parameter: name

Defines the name of your cluster.

Provisioning

Cluster Name*

Region*

btp CLI parameter: region

Defines a region (set of datacenters) where your cluster runs.

Provisioning

Region*

Colocate Control Plane

btp CLI parameter: colocateControlPlane

Allows you to have both the control plane and worker nodes in the same region.

Provisioning

Colocate Control Plane

Machine Type

btp CLI parameter: machineType

Specifies the provider-specific virtual machine type.

Provisioning

Updating

Machine Type

Auto Scaler Min

btp CLI parameter: autoScalerMin

Specifies the minimum number of virtual machines to create.

Provisioning

Updating

Auto Scaler Min

Auto Scaler Max

btp CLI parameter: autoScalerMax

Specifies the maximum number of virtual machines to create.

Provisioning

Updating

Auto Scaler Max

Modules

btp CLI parameter: modules

Defines which Kyma modules are provisioned in your cluster.

Provisioning

Modules

Networking

btp CLI parameter: networking

Provides a custom IP range for worker nodes.

Provisioning

Networking

OpenID Connect

btp CLI parameter: oidc

Provides a custom Open ID Connect (OIDC) configuration.

Provisioning

Updating

OpenID Connect (OIDC)

Administrators

btp CLI parameter: administrators

Specifies the list of runtime administrators.

Provisioning

Updating

Administrators

Additional Worker Node Pools

btp CLI parameter: additionalWorkerNodePools

Defines a custom list of additional worker node pools.

Provisioning

Updating

Additional Worker Node Pools

Access Control List 3, 4

accessControlList

Specifies the IP ranges that can access the Kubernetes API.

Provisioning

Updating

Access Control List

Additional Volume Size 4

btp CLI parameter: additionalVolumeSizeGi

Specifies extra disk space on top of the default volume size for a worker node pool.

Provisioning

Updating

Additional Volume Size

Audit Log Access 4

btp CLI parameter: auditLogAccess

Enables direct read access to your own audit log data using the SAP Audit Log Retrieval API v2.

Provisioning

Updating

Audit Log Access

Note

3 Not available in the Build Runtime: Google Cloud (build-runtime-gcp) plan.

Note

4 Not available in the Build Runtime: Alibaba Cloud (build-runtime-alicloud) plan.

Kyma Test Demo and Development (Azure Lite)â

The technical name of the plan is `azure_lite`. The Kyma Test Demo and Development plan is offered to Partners, within the Pay-As-You-Go for SAP BTP for cloud test, demo, and development commercial model. You can use this plan for testing, development, and demo purposes. In the SAP BTP cockpit, the plan is called "Kyma Runtime Partner TDD".

Remember

The parameters marked with an asterisk "*" are mandatory.

Kyma Test Demo and Development Plan Specification

Parameter

Description

Supported Operation

More Information

Cluster Name*

btp CLI parameter: name

Defines the name of your cluster.

Provisioning

Cluster Name*

Region*

btp CLI parameter: region

Defines a region (set of datacenters) where your cluster runs.

Provisioning

Region*

Machine Type

btp CLI parameter: machineType

Specifies the provider-specific virtual machine type.

Provisioning

Updating

Machine Type

Auto Scaler Min

btp CLI parameter: autoScalerMin

Specifies the minimum number of virtual machines to create.

Provisioning

Updating

Auto Scaler Min

Auto Scaler Max

btp CLI parameter: autoScalerMax

Specifies the maximum number of virtual machines to create.

Provisioning

Updating

Auto Scaler Max

Modules

btp CLI parameter: modules

Defines which Kyma modules are provisioned in your cluster.

Provisioning

Modules

Networking

btp CLI parameter: networking

Provides a custom IP range for worker nodes.

Provisioning

Networking

OpenID Connect

btp CLI parameter: oidc

Provides a custom Open ID Connect (OIDC) configuration.

Provisioning

Updating

OpenID Connect (OIDC)

Administrators

btp CLI parameter: administrators

Specifies the list of runtime administrators.

Provisioning

Updating

Administrators

Additional Worker Node Pools

btp CLI parameter: `additionalWorkerNodePools`

Defines a custom list of additional worker node pools.

Provisioning

Updating

Additional Worker Node Pools

Colocate Control Plane

btp CLI parameter: `colocateControlPlane`

Allows you to have both the control plane and worker nodes in the same region.

Provisioning

Colocate Control Plane

Audit Log Access

btp CLI parameter: `auditLogAccess`

Enables direct read access to your own audit log data using the SAP Audit Log Retrieval API v2.

Provisioning

Updating

Audit Log Access

Related Information

Provisioning and Updating Parameters in the Kyma Environment

Regions for the Kyma Environment

Kyma Modules

Account Administration Using the SAP BTP Command Line Interface (btp CLI)

Service Plans and Metering for Kyma Runtime

What Is the Consumption-Based Commercial Model?

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/change-storage-size-in-kyma>

Title: *Change Storage Size in Kyma | SAP Help Portal*

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformAdministration and OperationsAdministration and Operations in the Kyma EnvironmentChange Storage Size in Kyma

SAP Business Technology PlatformEnglish

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Change Storage Size in Kyma

If the amount of data for the applications in your Kyma environment grows, you can expand the storage size for your customer data by resizing the respective Persistent Volume Claim (PVC).

Prerequisites

Your user has the authorizations to change PVC resources.

Your workload uses a Persistent Volume, managed by a PVC.

Context

In a Kubernetes cluster, your customer data is stored in volumes. A “Persistent Volume Claim” (PVC) manages that volume for you and defines the size of storage. As the amount of customer data grows, you must adjust the PVC accordingly.

Procedure

Go to Kyma dashboard and select the correct namespace and workload.

Find the PVC you want to resize and enter the desired new size of the volume.

Save your changes.

The PVC shows the condition type Resizing. Wait until the PVC switches to condition type

FileSystemResizePending. This may take a few minutes.

Restart your workload by restarting all related Pods.

Results

The storage size for your customer data has been adjusted to your needs.

Next Steps

Remember to back up your customer data.

Related Information^

Kubernetes: Expanding Persistent Volumes Claims

Kyma Environment Backup

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/cloud-manager-module>

Title: *Cloud Manager Module | SAP Help Portal*

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ModulesCloud Manager Module

SAP Business Technology PlatformEnglish

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Cloud Manager Module

Use the Cloud Manager module to manage infrastructure providers' resources from SAP BTP, Kyma runtime.

What Is Cloud Manager?^

The Cloud Manager module manages access to cloud providers' chosen resources and products. Once you add Cloud Manager to your Kyma cluster, the module securely brings the offered resources. Cloud Manager is strictly coupled with the cloud provider where your Kyma cluster is deployed.

Remember

Using certain features of the Cloud Manager module introduces additional costs. For more information, see [Calculation with the Cloud Manager Module](#).

Scope^

The Cloud Manager module is available on Kyma clusters running on Amazon Web Services, Google Cloud, and Microsoft Azure.

Restriction

Cloud Manager supports the NFS feature of Amazon Web Services and Google Cloud only.

Features^

The Cloud Manager module provides the following features tailored for each of the cloud providers.

NFS: Network File System (NFS) server that can be used as a ReadWriteMany (RWX) volume in the Kyma cluster.

VPC Peering: Virtual Private Cloud (VPC) peering between your SAP BTP, Kyma runtime and remote cloud provider's project, account, or subscription.

Redis: cloud provider-flavored cache that can be used in your Kyma cluster.

Architecture^

Cloud Manager has read and write access to your IpRange, VpcPeering, NfsVolume, and Redis custom resources (CRs) in the Kyma cluster. The module also manages Kyma VPC networks, NFS Volume instances, and Redis instances in your cloud provider landscape in Kyma.

API/Custom Resource Definitions^

The `cloud-resources.kyma-project.io` Custom Resource Definition (CRD) describes the kind and the format of data that Cloud Manager uses to configure resources. For more information, see [Cloud Manager Resources](#).

Related Information^

[kyma-project.io: Cloud Manager module: Tutorials](#)

[Calculation with the Cloud Manager Module](#)

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/configure-entitlements-for-sap-s-4hana-cloud-extensibility-service>

Title: *Configure the Entitlements for the SAP S/4HANA Cloud Extensibility Service | SAP Help Portal*

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Extending SAP S/4HANA Cloud in the Cloud Foundry and Kyma Environ...Configure the Entitlements for the SAP S/4HANA Cloud Extensibilit...

SAP Business Technology PlatformEnglish

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Configure the Entitlements for the SAP S/4HANA Cloud Extensibility Service

Configure the required entitlements to make the APIs of the registered SAP S/4HANA Cloud system accessible in your subaccount in which your extension applications will reside.

Prerequisites^

You are an administrator of the global account in SAP BTP.

The subaccount is in the SAP BTP, Cloud Foundry environment with enabled Cloud Foundry, or Kyma, or both capabilities.

You have registered an SAP S/4HANA Cloud system. See [Register an SAP S/4HANA Cloud System in a Global Account in SAP BTP](#).

Context^

An entitlement is your right to provision and consume a resource. In other words, the entitlement is the respective service plan, api-access or messaging, that you're entitled to use.

Procedure^

In the SAP BTP cockpit, navigate to your global account.

In the navigation area, choose Entitlements Entity Assignments.

On the Entity Assignments screen, select your subaccount in the Subaccounts/Directories field.

Choose Edit, and then choose Add Service Plans.

Select the SAP S/4HANA Cloud Extensibility service.

Note

To have the SAP S/4HANA Cloud Extensibility service in the list, you need to have registered at least one SAP S/4HANA Cloud system.

In the Service Details area, select the system you have registered. In the Available Plans area, select the required service plans, and then choose Add Service Plan.

Note

For more information about the available service plans for SAP S/4HANA Cloud Extensibility service, see [Supported Service Plans for SAP S/4HANA Cloud](#).

Choose Save.

Related Information^

[Configure Entitlements and Quotas for Subaccounts](#)

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/configure-entitlements-for-sap-successfactors-extensibility-service>

Title: [Configure the Entitlements for the SAP SuccessFactors Extensibility Service | SAP Help Portal](#)

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Extending SAP SuccessFactors in the Cloud Foundry and Kyma Enviro...Configure the Entitlements for the SAP SuccessFactors Extensibili...

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Configure the Entitlements for the SAP SuccessFactors Extensibility Service

Configure the required entitlements to make the SAP SuccessFactors HXM Suite OData APIs of the registered SAP SuccessFactors system accessible in your subaccount in which your extension applications will reside.

Prerequisites^

You are an administrator of the global account in SAP BTP.

The subaccount is in the SAP BTP, Cloud Foundry environment with enabled Cloud Foundry, or Kyma, or both capabilities.

You have registered an SAP SuccessFactors system. See [Register an SAP SuccessFactors System in a Global Account in SAP BTP](#).

Context^

An entitlement is your right to provision and consume a resource. In other words, the entitlement is the api-access service plan that you're entitled to use, and the sso-configuration service plan that allows you to register an assertion consumer service in SAP SuccessFactors automatically.

Procedure^

In the SAP BTP cockpit, navigate to your global account.

In the navigation area, choose Entitlements Entity Assignments.

On the Entity Assignments screen, select your subaccount in the Subaccounts/Directories field.

Choose Edit, and then choose Add Service Plans.

Select the SAP SuccessFactors Extensibility service.

Note

To have the SAP SuccessFactors Extensibility service in the list, you need to have registered at least one SAP SuccessFactors system.

In the Service Details area, select the system you have registered. In the Available Plans area, select the api-access and sso-configuration service plans, and then choose Add 2 Service Plans.

Related Information

Configure Entitlements and Quotas for Subaccounts

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/configure-single-sign-on-between-subaccount-in-sap-btp-and-sap-successfactors>

Title: Single Sign-On Between a Subaccount in SAP BTP and SAP SuccessFactors | SAP Help Portal

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Single Sign-On Between a Subaccount in SAP BTP and SAP SuccessFactors

Use this procedure to configure the single sign-on (SSO) between SAP BTP and the SAP SuccessFactors system.

When a user attempts to access SAP SuccessFactors for the first time, the system redirects the user to the identity provider for authentication. From then on, the user session is kept active, and the user is no longer prompted for credentials when trying to use the SAP SuccessFactors application. This is called single sign-on (SSO).

Note

For SAML-based single sign-on integrations, we recommend using Identity Authentication as the primary identity provider instead of using SAP SuccessFactors as the identity provider.

Using Identity Authentication

You can use Identity Authentication to set up the single sign-on (SSO) between SAP SuccessFactors and SAP BTP. Identity Authentication provides you with controlled cloud-based access to business processes, applications, and data. It simplifies your user experience through authentication mechanisms, single sign-on, on-premise integration, and convenient self-service options. We recommend using Identity Authentication as a central point to integrate SAP applications for a seamless single sign-on experience.

Authentication: All SAP cloud applications can offer their users the same authentication mechanisms , as well as strong authentication with configurable multi-factor (MFA) enforcement; easy separation mechanism for multiple user stores and flexible configuration where to validate user's credentials.

Single Sign-On: Identity Authentication offers central SSO endpoint for all SAP cloud applications and pre-configured or semi-automated trust configuration .

Integrating SAP applications: Identity Authentication offers common identity for users, as well as a unified way for user management and security token service for protection of system-to-system communication. Data across applications can be correlated .

See SAP SuccessFactors Integration Scenario.

Using SAP SuccessFactors as a Trusted Identity Provider

Prerequisites

You have the UI and Role Administrator role assigned to your user. See Security Administration: Managing Authentication and Authorization.

Context

The authentication to SAP SuccessFactors applications is restricted to the authorized users. The identification of a user is verified by the identity provider, as specified by SAML 2.0. The identity provider stores a list of all users that are allowed to access the SAP SuccessFactors system along with their credentials. The integration between the SAP SuccessFactors and the identity provider is based on a trust configuration.

To ensure the required security for accessing the applications, you need to configure the single sign-on between the subaccount in SAP BTP and the SAP SuccessFactors system using a SAML identity provider. The single sign-on requires both solutions to be configured as trusted SAML service providers for the identity provider, and at the same time, the identity provider to be configured as trusted identity provider for the two

solutions.

Use a subaccount in SAP BTP with the SAP BTP, Cloud Foundry runtime enabled and follow these steps to set up the single sign-on (SSO):

Configure SAP SuccessFactors as a Trusted Identity Provider in SAP BTP

Configure the Subaccount as a Trusted Service Provider in SAP SuccessFactors

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/create-kyma-environment-instance>

Title: *Creating Kyma Instances | SAP Help Portal*

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformAdministration and OperationsAdministration and Operations in the Kyma EnvironmentCreating Kyma Instances

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Creating Kyma Instances

Set up a Kubernetes cluster with SAP BTP, Kyma runtime and use it to build applications and extensions to your SAP and third-party solutions. You can create one or multiple Kyma clusters in a single SAP BTP subaccount.

Prerequisitesà

Your subaccount has entitlements for Kyma runtime configured. See [Configure Entitlements and Quotas for Subaccounts](#).

Contextà

To set up Kyma environment in your subaccount, you must create an instance of it. You can create it from your subaccount Overview section by choosing **Enable Kyma** and following the wizard steps to configure the provisioning parameters. You can also create a Kyma environment instance from Service Marketplace in the same way as any other SAP BTP service or application.

If you prefer to work in a terminal or want to automate operations using scripts, you can create Kyma environment with the SAP BTP command line interface (btp CLI). See [Enable SAP BTP, Kyma Runtime Using the Command Line](#).

In the process of provisioning a Kyma cluster, an instance of SAP Service Manager is created (see [Credentials](#)). Therefore, each Kyma instance has an instance of Service Manager assigned to it. If you have multiple Kyma clusters in your subaccount, you can easily trace which Service Manager instance is assigned to which Kyma instance by the Service Manager instance's display name, which is the same as its matching Kyma instance's ID.

Note

If the creation of an SAP Service Manager instance is unsuccessful, Kyma provisioning cannot begin. If the situation persists, create a support ticket (see [Getting Support](#)).

Note

To indicate that your SAP BTP, Kyma runtime is used for production, select **Used for production** in your subaccount details. This setting allows Kyma runtime operators to prioritize incidents and support cases affecting production subaccounts over subaccounts used for non-production purposes. See [Change Subaccount Details](#).

Procedureà

In the SAP BTP cockpit, navigate to your subaccount Overview.

In the Kyma Environment section, choose **Enable Kyma** when creating your first Kyma instance. When provisioning the second and subsequent Kyma clusters, choose **Create**.

In the Basic Info view of the wizard window, perform the following actions:

Choose one of the plans assigned to your account.

You can keep the default instance and cluster names, but customizing them enables you to identify particular instances more easily.

Choose a region from the list.

To continue the configuration, choose **Next**.

In the Additional Parameters view, provide the required details in the Form. Alternatively, switch to the JSON tab and upload your configuration file or specify the parameters in JSON format.

For more information on the configurable parameters, see [Provisioning and Updating Parameters in the Kyma Environment](#).

To review your configuration, choose Next.

To confirm changes, choose Create.

Wait until the instance is created. The instance creation may take several minutes. In the Kyma Environment section of your subaccount Overview, you can monitor the instance creation status and view the operation details.

Note

If the Kyma instances quota assigned to your subaccount has been used up, you get a message informing you about it. To increase the quota, go to Entitlements and make the necessary changes or contact your administrator.

Results

You have created a Kyma environment instance. If needed, you can repeat the procedure to create more Kyma instances.

Next Steps

In the Kyma Environment section of your subaccount Overview, you can see all the Kyma instances created in your subaccount. If there is more than one, choose an instance from the dropdown list to view the following details:

Kyma instance ID

Kyma instance name

Plan

Kyma dashboard link

Kubeconfig link

Cluster name

You can also view the details of your Kyma instances under Instances and Subscriptions.

To manage your Kyma instances, go to Instances and Subscriptions. Choose the action you want to perform from an instance's actions menu.

To manage access to Kyma environment and Kyma dashboard, assign roles as needed. See Assign Roles in the Kyma Environment.

To use a variety of functionalities, such as telemetry and eventing, or to use SAP BTP services, add the respective Kyma modules. See Kyma Modules.

Tip

To manage a Kyma instance automatically, you can create a Kyma service binding. The binding enables getting a Kyma kubeconfig, which in turn allows for accessing a Kyma cluster, deploying applications, running tests, and deleting the resources in a fully automated way. See Managing Kyma Runtime Using the Provisioning Service API.

Related Information

Available Plans in the Kyma Environment

Configuring a Custom Identity Provider for Kyma

Adding and Deleting a Kyma Module

Account Administration Using the SAP BTP Command Line Interface (btp CLI)

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/create-service-instance-to-consume-sap-s-4hana-cloud-apis>

Title: Create a Service Instance to Consume the SAP S/4HANA Cloud APIs | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Extending SAP S/4HANA Cloud in the Cloud Foundry and Kyma Environ...Create a Service Instance to Consume the SAP S/4HANA Cloud APIs

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Create a Service Instance to Consume the SAP S/4HANA Cloud APIs

To enable the integration of your extension applications with the SAP S/4HANA Cloud system you have registered in the SAP BTP global account, and to configure the communication flow, you create a service instance of the SAP S/4HANA Cloud Extensibility service.

Prerequisites

You have configured the required entitlements to make the APIs of the registered SAP S/4HANA Cloud system

accessible in your subaccount in which your extension applications will reside. See [Configure the Entitlements for the SAP S/4HANA Cloud Extensibility Service](#).

In case of principal propagation, you first need to configure single-sign on (SSO) with the Identity Authentication service and protect your application. See [Integrating the Service with SAP S/4HANA Cloud, SAP Integrated Business Planning and SAP Analytics Cloud](#).

Contextà

In both Cloud Foundry and Kyma environments, you consume services by creating a service instance. Service instances are created using a specific service plan.

To allow applications running on SAP BTP to consume SAP S/4HANA Cloud APIs, you need to create a service instance of the SAP S/4HANA Cloud Extensibility service using the api-access service plan.

Note

These service plans have been deprecated:

sap_com_0109

sap_com_0009

sap_com_0008

However, you can still enable these communication scenarios using the api-access service plan:

Sales Order Integration (SAP_COM_0109): allows you to integrate your extension applications with sales order processing in SAP S/4HANA Cloud. For more information, see [https://api.sap.com/api/](https://api.sap.com/api/API_SALES_ORDER_SRV/overview)

API_SALES_ORDER_SRV/overview.

Product Integration (SAP_COM_0009): enables you to replicate product master data from client system to SAP S/4HANA system. For more information, see [https://api.sap.com/api/](https://api.sap.com/api/PRODUCTMDMBULKREPLICATEREQUEST/overview)

PRODUCTMDMBULKREPLICATEREQUEST/overview.

Business Partner, Customer and Supplier Integration (SAP_COM_0008): allows you to consume the Business Partner API which enables you to create, read, update, and delete master data related to Business Partners, Suppliers, and Customers in an SAP S/4HANA system. For more information, see [https://api.sap.com/api/](https://api.sap.com/api/API_BUSINESS_PARTNER/overview)

API_BUSINESS_PARTNER/overview.
For a sample JSON for these communication scenarios, see [Communication Arrangement JSON/YAML File - Properties](#).

The api-access service plan defines the access to the corresponding SAP S/4HANA Cloud APIs. It supports both predefined and custom communication scenarios for consuming the SAP S/4HANA Cloud APIs and integrating your extension applications. See:

[Supported Service Plans for SAP S/4HANA Cloud](#)

[Custom Communication Scenarios](#)

You create the service instance in your subaccount with the respective environment enabled. When creating the service instance, you configure the connectivity by specifying the required configurations in a JSON format. The following authentication scenarios are supported for the communication flow between the extension application and SAP S/4HANA Cloud:

Basic Authentication (inbound and outbound connections)

OAuth 2.0 Client Credentials (outbound connections)

OAuth 2.0 SAML Bearer Assertion (inbound connections)

To communicate with SAP S/4HANA Cloud the extension application can use principal propagation which is done using OAuth 2.0 SAML Bearer Assertion flows. Principal propagation means you forward the identity of the logged-in cloud users when accessing or updating data in the SAP S/4HANA Cloud system.

This is useful in scenarios where you need to have restricted data access based on the logged-in user from your extension. Or, you want to ensure only users with the right permissions are able to update the system via extensions deployed in SAP BTP.

To use this authentication scenario, you first need to configure single-sign on (SSO) with the Identity Authentication service and protect your application. See [Integrating the Service with SAP S/4HANA Cloud, SAP Integrated Business Planning and SAP Analytics Cloud](#).

Client Certificate Authentication (inbound and outbound connections)

OAuth2mTLS (outbound connections)

Depending on whether you are using Cloud Foundry or Kyma environment, you have to follow different steps to create an SAP S/4HANA Cloud Extensibility service instance:

[Create an SAP S/4HANA Cloud Extensibility Service Instance in the Cloud Foundry Environment](#)

Create an SAP S/4HANA Cloud Extensibility Service Instance in the Kyma Environment

Related Information

Communication Arrangement JSON/YAML File - Properties

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/create-service-instance-to-consume-sap-successfactors-hxm-suite-odata-api>

Title: Create a Service Instance to Consume the SAP SuccessFactors HXM Suite OData API | SAP Help Portal
HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Extending SAP SuccessFactors in the Cloud Foundry and Kyma Enviro...Create a Service Instance to Consume the SAP SuccessFactors HXM S...

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Create a Service Instance to Consume the SAP SuccessFactors HXM Suite OData API

To enable the integration of your extension applications with the SAP SuccessFactors system you have registered in the global account in SAP BTP, you first need to create a service instance of the corresponding service.

Context

In both Cloud Foundry and Kyma environment, you consume services by creating a service instance. Service instances are created using a specific service plan.

To allow applications running on SAP BTP to consume SAP SuccessFactors HXM Suite OData APIs, you need to create a service instance of the SAP SuccessFactors Extensibility service using the api-access service plan. You create the service instance in your subaccount with the respective environment enabled. When creating the service instance, you configure the authentication type for the communication by specifying the required configurations in a JSON format. SAP BTP supports the following authentication scenarios for SAP SuccessFactors:

OAuth access with OAuth 2.0 SAML bearer assertion

To use this authentication type, you must protect your application. See Adding Authentication and Authorization.

OAuth access with OAuth 2.0 SAML bearer assertion with technical user

The process for creating a service instance depends on the environment you are using:

For Cloud Foundry environment, see Create an SAP SuccessFactors Extensibility Service Instance in the Cloud Foundry Environment.

For Kyma environment, see Create an SAP SuccessFactors Extensibility Service Instance in the Kyma Environment.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/custom-openid-connect-configuration>

Title: Custom OpenID Connect Configuration | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Provisioning and Updating Parameters in the Kyma EnvironmentCustom OpenID Connect Configuration

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Custom OpenID Connect Configuration

During SAP BTP, Kyma runtime provisioning or update, you can provide your custom Open ID Connect (OIDC) configuration as a list of oidc objects or as a single oidc object. You can also have no OIDC configuration.

Configuring the OIDC Property as a List of oidc Objects

Configure the oidc property as a list of oidc objects. In this way, you can define zero, one, or multiple OIDC configurations. When using the list of oidc objects, you must provide values for each element except for requiredClaims. By configuring the OIDC details yourself, you gain full visibility of your settings within the SAP BTP cockpit. For your convenience, in the SAP BTP cockpit, one oidc object has the predefined values filled in, but you can change them. For each additional oidc object, you must provide your own settings.

Provisioning Kyma Runtime with the oidc List

When configuring your OIDC setup, you have the following options:

If you use an oidc list, you must explicitly define values for all its elements, because they do not inherit default values.

If you provide an empty oidc list with zero elements, no OIDC configuration is applied to the instance.

Example

Configuring the oidc list.

You can define one or multiple OIDC configurations, depending on the number of lists of oidc objects you add to your provisioning request. See an example of a request with one oidc object in the list:

```
âE
ã
{
  "service_id" : "47c9dcbf-ff30-448e-ab36-d3bad66ba281",
  "plan_id" : "4deee563-e5ec-4731-b9b1-53b42d855f0c",
  "context" : {
    "globalaccount_id" : {GLOBAL_ACCOUNT_ID}
  },
  "parameters" : {
    "region": {REGION},
    "name" : {CLUSTER_NAME},
    "oidc" : {
      "list": [
        {
          "clientId" : "custom-client-id",
          "issuerURL" : "https://custom.issuer.com",
          "groupsClaim" : "groups",
          "groupsPrefix" : "-",
          "signingAlgs" : ["RS256"],
          "usernamePrefix" : "-",
          "usernameClaim" : "sub",
          "requiredClaims" : ["first-claim=value", "second-claim=value"],
          "encodedJwksArray": ""
        }
      ]
    }
  }
}
```

àf

You see a configuration similar to this one:

```
âE
ã
{
  ...
  "oidc" : {
    "list": [
      {
        "clientId" : "custom-client-id",
        "issuerURL" : "https://custom.issuer.com",
        "groupsClaim" : "groups",
        "groupsPrefix" : "-",
        "signingAlgs" : ["RS256"],
        "usernamePrefix" : "-",
        "usernameClaim" : "sub",
        "requiredClaims" : ["first-claim=value", "second-claim=value"],
        "encodedJwksArray": ""
      }
    ]
  }
}
```

...

}

âf

Updating Kyma Runtime with the oidc List

You can update the OIDC configuration from a single oidc object to an oidc list, but updating from an oidc list to a single oidc object is not supported.

When updating your OIDC configuration, you have the following options:

If you are updating your oidc list, you must explicitly define values for all its elements, because they do not inherit default values.

The update operation overwrites the OIDC configuration values provided in JSON for the oidc list, so if you provide OIDC parameters with empty values, they are considered valid and replace the existing settings.

If you provide an empty oidc list, it clears the OIDC configuration for the instance.

Example

Updating an instance with a list of oidc objects.

Suppose your current OIDC configuration is the following:

âE

ã

[

{

"clientId": "first-custom-client-id",

"issuerURL": "https://first.custom.issuer.com",

"groupsClaim": "groups",

"groupsPrefix": "-",

"usernameClaim": "sub",

"usernamePrefix": "-",

"signingAlgs": ["RS256"],

"requiredClaims": ["first-claim=value", "second-claim=value"],

"encodedJwksArray": ""

},

{

"clientId": "second-custom-client-id",

"issuerURL": "https://second.custom.issuer.com",

"groupsClaim": "groups",

"groupsPrefix": "-",

"usernameClaim": "sub",

"usernamePrefix": "acme-",

"signingAlgs": ["RS256"],

"requiredClaims": ["example-claim=value"],

"encodedJwksArray": ""

}

]

âf

To update this configuration, send an HTTP PATCH request with the following payload:

âE

ã

{

"service_id": "47c9dcbf-ff30-448e-ab36-d3bad66ba281",

"plan_id": "4deee563-e5ec-4731-b9b1-53b42d855f0c",

"context": {

"globalaccount_id": "{GLOBAL_ACCOUNT_ID}"

},

"parameters": {

"name": "{CLUSTER_NAME}",

"oidc": {

```

"list": [
{
"clientId": "new-client-id",
"issuerURL": "https://new.issuer.com",
"groupsClaim": "groups",
"groupsPrefix": "-",
"signingAlgs": ["RS256"],
"usernameClaim": "sub",
"usernamePrefix": "-",
"requiredClaims": [],
"encodedJwksArray": ""
},
{
"clientId": "updated-client-id",
"issuerURL": "https://updated.issuer.com",
"groupsClaim": "groups",
"groupsPrefix": "-",
"usernameClaim": "sub",
"usernamePrefix": "acme-",
"signingAlgs": ["RS256"],
"encodedJwksArray": ""
}
]
}
}
}

```

You see the configuration with the values provided in the request:

␣

␣

```

[
{
"clientId": "new-client-id",
"issuerURL": "https://new.issuer.com",
"groupsClaim": "groups",
"groupsPrefix": "-",
"usernameClaim": "sub",
"usernamePrefix": "-",
"signingAlgs": ["RS256"],
"requiredClaims": [],
"encodedJwksArray": ""
},
{
"clientId": "updated-client-id",
"issuerURL": "https://updated.issuer.com",
"groupsClaim": "groups",
"groupsPrefix": "-",
"usernameClaim": "sub",
"usernamePrefix": "acme-",
"signingAlgs": ["RS256"],
"requiredClaims": [],
"encodedJwksArray": ""
}
]

```

Configuring the OIDC Property as a Single oidc Object␣

This solution is not recommended, and is only supported for maintaining backward compatibility with existing automations. With this approach you cannot have multiple OIDC configurations for your Kyma runtime. In the SAP BTP cockpit, you can only see the parameters that you configure yourself. Since you must provide values only for `clientId` and `issuerURL` in a single `oidc` object, the visibility of your complete OIDC configuration may be limited.

Provisioning Kyma Runtime with a Single `oidc` Object

When configuring your OIDC setup, you have the following options:

If you do not include an `oidc` object in the provisioning request at all, the default OIDC configuration is applied.

If you leave the parameters of a single `oidc` object empty, it defaults to the predefined values.

Example

See a request with an `oidc` object with empty keys:

```
â€”
{
  "service_id" : "47c9dcbf-ff30-448e-ab36-d3bad66ba281",
  "plan_id" : "4deee563-e5ec-4731-b9b1-53b42d855f0c",
  "context" : {
    "globalaccount_id" : {GLOBAL_ACCOUNT_ID}
  },
  "parameters" : {
    "region": {REGION},
    "name" : {CLUSTER_NAME},
    "oidc" : {
      "clientId" : "",
      "issuerURL" : "",
      "groupsClaim" : "",
      "groupsPrefix" : "",
      "signingAlgs" : [],
      "usernamePrefix" : "",
      "usernameClaim" : "",
      "requiredClaims" : ["-"],
      "encodedJwksArray": "-"
    }
  }
}
```

The result is the default OIDC configuration:

```
â€”
{
  ...
  "oidc" : {
    "clientId" : "12b13a26-d993-4d0c-aa08-5f5852bbdff6",
    "issuerURL" : "https://kyma.accounts.ondemand.com",
    "groupsClaim" : "groups",
    "groupsPrefix" : "-",
    "signingAlgs" : ["RS256"],
    "usernamePrefix" : "-",
    "usernameClaim" : "sub",
    "requiredClaims" : ["-"],
    "encodedJwksArray": "-"
  }
  ...
}
```

Updating Kyma Runtime with a Single `oidc` Object

You can update the OIDC configuration from a single oidc object to an oidc list, but updating from an oidc list to a single oidc object is not supported.

When updating your OIDC configuration, you have the following options:

If you provide no oidc object in the update request, the existing OIDC configuration remains unchanged.

If you update using a single oidc object, empty keys do not change the configuration, and only the provided values are updated.

Example

Updating an instance with a single oidc object.

Suppose your current OIDC configuration is the following:

␣

␣

```
{
  "clientId": "some-client-id",
  "issuerURL": "https://some.issuer.com",
  "groupsClaim": "groups",
  "groupsPrefix": "-",
  "usernameClaim": "sub",
  "usernamePrefix": "-",
  "signingAlgs": ["RS256"],
  "requiredClaims": ["-"],
  "encodedJwksArray": "-"
}
```

To update this configuration, send an HTTP PATCH request with the following payload:

␣

␣

```
{
  "service_id": "47c9dcbf-ff30-448e-ab36-d3bad66ba281",
  "plan_id": "4deee563-e5ec-4731-b9b1-53b42d855f0c",
  "context": {
    "globalaccount_id": "{GLOBAL_ACCOUNT_ID}"
  },
  "parameters": {
    "name": "{CLUSTER_NAME}",
    "oidc": {
      "clientId": "new-client-id",
      "issuerURL": "https://new.issuer.com",
      "groupsClaim": "",
      "groupsPrefix": "",
      "signingAlgs": [],
      "usernamePrefix": "",
      "usernameClaim": "",
      "requiredClaims": ["-"],
      "encodedJwksArray": "-"
    }
  }
}
```

You see the configuration with the values provided in the request, similar to this one:

␣

␣

```
{
  "clientId": "new-client-id",
  "issuerURL": "https://new.issuer.com",
  "groupsClaim": "groups",
  "groupsPrefix": "-",

```

```

"usernameClaim": "sub",
"usernamePrefix": "-",
"signingAlgs": ["RS256"],
"requiredClaims": ["-"],
"encodedJwksArray": "-"
}

```

Configuration with no OIDC Property

You can provision or update your Kyma runtime without providing the OIDC property at all.

Provisioning Kyma Runtime with No OIDC Property

If you do not include an `oidc` parameter in the provisioning request at all, the default OIDC configuration is applied.

Example

See an example of a provisioning request without specifying any `oidc` configuration:

```

{
  "service_id" : "47c9dcbf-ff30-448e-ab36-d3bad66ba281",
  "plan_id" : "4deee563-e5ec-4731-b9b1-53b42d855f0c",
  "context" : {
    "globalaccount_id" : {GLOBAL_ACCOUNT_ID}
  },
  "parameters" : {
    "region": {REGION},
    "name" : {CLUSTER_NAME}
  }
}

```

The resulting configuration has default values like in the following example. You can only view this configuration in your cluster.

```

{
  ...
  "oidc" : {
    "clientId" : "12b13a26-d993-4d0c-aa08-5f5852bbdff6",
    "issuerURL" : "https://kyma.accounts.ondemand.com",
    "groupsClaim" : "groups",
    "groupsPrefix" : "-",
    "signingAlgs" : ["RS256"],
    "usernamePrefix" : "-",
    "usernameClaim" : "sub",
    "requiredClaims" : ["-"],
    "encodedJwksArray": "-"
  }
  ...
}

```

Updating Kyma Runtime with No OIDC Property

If you provide no `oidc` property in the update request, the existing OIDC configuration remains unchanged.

Related Information

OpenID Connect (OIDC)

Configuring a Custom Identity Provider for Kyma

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/deploy-workloads-in-kyma-environment-to-extend-sap-systems>

Title: Deploy Workloads in the Kyma Environment to Extend SAP Systems | SAP Help Portal

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Deploy Workloads in the Kyma Environment to Extend SAP Systems

Access the Kyma environment and start creating extensions for SAP systems.

Prerequisitesà

You have the Application Connector and Serverless modules in your cluster. See [Adding and Deleting a Kyma Module](#).

If you want to expose and secure your workload, you must have the default Istio and API Gateway modules in your cluster.

If you want your workload to react to events, you must have the Eventing module in your cluster.

If you want to use CloudEvents, make sure that your ecosystem supports it.

Contextà

You can extend a given SAP system in the Kyma environment with workloads. There are two kinds of workloads - Functions and microservices:

Functions are simple code snippets that are called when the Function is triggered. They implement the exact business logic you define in the code, and you don't need to worry about containerization and deployment in Kyma runtime. Because Kyma Serverless is taking care of that, using Functions is convenient and fast.

You can also deploy your extension code as a microservice. In this variant, you must contain your microservice code in a Docker image and deploy it in Kyma runtime. You need more Kubernetes experience but get more control over the shape of your extension application, because you can separate the tasks into smaller pieces that interact with each other as loosely coupled, independently deployable units of code.

Whichever workload variant you choose, you can make it part of the SAP system extension. You can do this in two ways. Depending on your solution's needs, choose one or both of them:

Expose your workload's API using API Rule and make it accessible to other components of your ecosystem with REST calls.

Subscribe your workload to events emitted from the SAP systems and extend them.

Note

Most of the linked tutorials contain steps that you can perform both on the Kyma dashboard and in the terminal after downloading the kubeconfig file with the cluster configuration.

Follow these steps to get familiar with workloads and learn how to use them.

Procedureà

Connect the SAP system you want to extend ([Extending SAP Solutions Using Automated Configurations](#)).

Choose a namespace in your cluster and create a simple Function ([Create and Modify an Inline Function](#)) or a microservice.

If you want to expose your workload outside the cluster, read [Expose and Secure Workloads](#).

If you want your workload to react to events, read [Subscribe to Multiple Event Types](#).

To subscribe to events with Kyma, you must create a Subscription custom resource (CR) including the following parameters:

Application name: The name of the external Application connected to Kyma runtime. Typically, it starts with mp-*. This name can be found in the UI under [Integration Applications](#). It must be bound to the namespace.

Event name: The event name depends on your CX solution:

SAP Commerce Cloud - for custom events, check the configuration of your Destination Target for the Kyma runtime

SAP Sales Cloud / SAP Service Cloud - you can check the event specification in [Event Notification Monitoring](#).

Event version: v1 for legacy events or v2 for CloudEvents

Caution

Your workload must be error-free, otherwise event loss could occur. If you're facing problems, check:

[Troubleshooting for the API Gateway Module](#)

[Troubleshooting for the Eventing Module](#)

[Troubleshooting for the Serverless Module](#)

[Related Informationà](#)

API Gateway Module

Eventing Module

Serverless Module

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/development-in-kyma-environment>

Title: *Development in the Kyma Environment* | SAP Help Portal

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Development in the Kyma Environment

Learn more about developing applications in SAP BTP, Kyma runtime.

Overviewà^

With SAP BTP, Kyma runtime, you can build simple Functions, develop, and deploy more complex microservices, or mixtures of those, depending on your use case complexity level. Functions and microservices can act as stand-alone applications or extensions of these SAP systems:

SAP S/4HANA

SAP S/4HANA Cloud

SAP S/4HANA Cloud, public edition

SAP Field Service Management

SAP SuccessFactors

SAP Customer Experience systems:

SAP Commerce Cloud

SAP Cloud for Customer

SAP Marketing Cloud

Language of Choiceà^

With the Kyma runtime, you can create microservices in the language of your choice and run them as containerized applications. Additionally, you can create Functions in Node.js or Python.

Kyma Dashboard and kubectlà^

The Kyma runtime comes with a central administration dashboard (Kyma dashboard), which allows you to add a module, deploy microservices, create Functions, and manage their configurations. You can also use it to connect SAP BTP services to your cluster and manage them using the SAP BTP Operator module, create instances of these services, and use them in your microservices or Functions. For more information, see Kyma Dashboard.

For those who prefer to work with command-line tools, the Kyma runtime also offers the Kubernetes command-line tool, kubectl.

SAP Cloud Application Programming Modelà^

The SAP Cloud Application Programming Model (CAP) is the recommended framework for application and service development in the Kyma runtime. To learn more, see Programming Models.

Using APIà^

The APIs of Kyma modules and third-party modules' may vary from the least stable alpha version through a more stable beta and the most stable vX version (X being an integer). While using the alpha version, you can face issues with module upgrades, because there is no guarantee that the API will not change or won't be removed in the future. For more information, check the Kubernetes API versioning.

Tip

To get the list of the APIs, their kind, and version, run:

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kubectl get crd -o custom-

columns="KIND:.spec.names.kind,GROUP:.spec.group,VERSION:.spec.versions[*].name"

âf

If an API is available in two versions, always use the version that is more stable.

Development Tutorialà^

To help you get started with the development process, go through the following missions that show how to

develop applications in the Kyma runtime.

Build an Application in SAP BTP, Kyma Runtime

Develop a Full-Stack Application in SAP BTP, Kyma Runtime

Deploy a Full-Stack CAP Application in SAP BTP, Kyma Runtime Following SAP BTP Developer's Guide

Related Information^

Extension Samples

Kyma provides a ready-to-use project that contains sample applications. You can use them to build event- and API-based extensions in the Kyma runtime using your favorite technology. These samples are implemented in multiple languages and demonstrate various Kyma features and use case scenarios.

Using SAP BTP Services in the Kyma Environment

With the Kyma runtime, you can extend the SAP systems to build and deploy your own applications.

Deploy Workloads in the Kyma Environment to Extend SAP Systems

Access the Kyma runtime and start creating extensions for SAP systems using Functions.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/enable-and-disable-kyma-module>

Title: *Adding and Deleting a Kyma Module | SAP Help Portal*

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Adding and Deleting a Kyma Module

To use a Kyma module, you must add it first. Use Kyma dashboard or Kyma CLI to do that. If you don't need the module anymore, delete it to save resources.

Context^

Tip

To see which modules you have in your cluster, go to Kyma dashboard and under Cluster Overview, choose Modify Modules.

If you prefer Kyma CLI, run `kyma module list`.

Adding a Kyma Module^

Context

Procedure

Use Kyma dashboard

Use Kyma CLI

Related Information

Kyma Dashboard

Kyma CLI

Deleting a Kyma Module^

Procedure

Use Kyma dashboard

Use Kyma CLI

Related Information

Kyma Dashboard

Kyma CLI

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/enable-consumption-of-sap-s-4hana-cloud-events>

Title: *Enable the Consumption of SAP S/4HANA Cloud Events | SAP Help Portal*

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S/4HANA Cloud in the Cloud Foundry and Kyma Environ...Enable the Consumption of SAP S/4HANA Cloud Events

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Enable the Consumption of SAP S/4HANA Cloud Events

To create event-based extensions for SAP S/4HANA Cloud you need to set up the messaging between the

SAP S/4HANA Cloud system and the SAP Event Mesh.

Prerequisites

You have registered an SAP S/4HANA Cloud tenant in the global account in SAP BTP. See [Register an SAP S/4HANA Cloud System in a Global Account in SAP BTP](#).

Context

To consume SAP S/4HANA Cloud events, you need to configure the connectivity between SAP Event Mesh and the SAP S/4HANA Cloud tenant. To do so, you need to create and configure service instances for both SAP S/4HANA Cloud Extensibility service and SAP Event Mesh.

When you create a service instance in a subaccount, an event bus for this subaccount is created. All service instances in a subaccount point to the same event bus and are nothing else but clients providing access credentials for the event bus. You configure the clients using JSON descriptors. Each client has its own name which can be configured with the `emname` parameter in the JSON descriptor file. Thus, the client is identified with a human readable name instead of a GUID. The name of each client must be unique. As all of the clients point to the same event bus, they must have different namespaces in which to publish events. Each namespace must be unique per subaccount. Each client should publish only in its own unique namespace.

Communication flow

The SAP S/4HANA Cloud Extensibility service instance is the client which creates the SAP S/4HANA Cloud integration and provides the credentials for the SAP S/4HANA Cloud system so that the SAP S/4HANA Cloud system can send events to that event bus. The purpose of this client is to send messages to a topic. It sends messages to the namespace which was configured while creating the service instance of S/4HANA Cloud Extensibility service. See [Define SAP S/4HANA Cloud Extensibility Service Descriptor JSON/YAML File](#)

The SAP Event Mesh service instance is the client which receives the events. This service instance must have a different name and a different namespace than the ones configured for the SAP S/4HANA Cloud extensibility service instance. Since the purpose of this client is to receive messages from a different client the rules have to take care of that. So, for the SAP Event Mesh service instance, you need to configure the subscribe filter to the SAP S/4HANA Cloud Extensibility service instance. See [Define SAP Event Mesh Service Descriptor JSON/YAML File](#).

After both instances are created and configured, you can create the topic-to-queue subscription. To do so, you first create a queue in the client that refers to the SAP Event Mesh service instance and then subscribe the topic of the namespace of the SAP S/4HANA Cloud Extensibility service to that queue. See [Manage Queues in the SAP Event Mesh documentation](#)

The service instance of SAP Event Mesh is the client which consumes the message and should be bound to the extension application.

Procedure

To configure the connectivity between SAP Event Mesh and the SAP S/4HANA Cloud tenant so that SAP S/4HANA Cloud events can be produced and then consumed by applications running on SAP BTP, you need to perform the following tasks:

Add the required quotas to your subaccount as follows:

Assign the messaging SAP S/4HANA Cloud Extensibility service plan to the subaccount in SAP BTP you want to pair with the SAP S/4HANA Cloud tenant. See [Configure the Entitlements for the SAP S/4HANA Cloud Extensibility Service](#).

Note

You can use the service plan on regions where the SAP Event Mesh service is available.

Note

The messaging service plan enables the Enterprise Eventing Integration (SAP_COM_0092) communication scenario which allows you to consume SAP S/4HANA Cloud events and create event-based extensions.

This service plan has been renamed from `sap_com_0092`.

Assign the required number of SAP Event Mesh instances to the subaccount in SAP BTP that you want to pair with the SAP S/4HANA Cloud tenant. See [Configure the Entitlements for SAP Event Mesh](#).

Configure the connectivity between SAP Event Mesh and the SAP S/4HANA Cloud tenant, and then configure event topics for the channel inside SAP S/4HANA Cloud tenant. See [Set Up the Connectivity Between Event Mesh and the SAP S/4HANA Cloud Tenant](#).

Enable the SAP Event Mesh for your subaccount in SAP BTP, create a queue that is specific to your application, and then subscribe this queue to the channel topic that the SAP S/4HANA Cloud tenant uses to

produce events. See [Enable the SAP Event Mesh Service for Your Subaccount in SAP BTP](#).

Related Information

What Is SAP Event Mesh?

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/enabling-side-by-side-extensibility>

Title: [Setting Up System Landscape for Kyma | SAP Help Portal](#)

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Setting Up System Landscape for Kyma

Use

Note

The content in this section is not relevant for China (Shanghai) and Government Cloud (US) regions.

You want to extend the functionality on top of your existing SAP solution because your specific business needs require it. You also want to use the SAP BTP Kyma environment. You can build an extension application that contains your business logic and UI changes and deploy it in SAP BTP Kyma runtime. For the users of your SAP solution, this change will be seamless, it will look as if it's part of the original functionality. At the same time, you can use the extension capabilities of SAP BTP to implement the additional workflows or modules. This is called side-by-side extensibility with Kyma.

Procedure

To enable the side-by-side extensibility with Kyma for your SAP solution, you have to:

Create an SAP BTP, Kyma environment instance in your subaccount in SAP BTP.

Add the Application Connector module on your Kyma runtime.

Connect the corresponding SAP system with the global account in SAP BTP. Use the automated configurations for the following SAP solutions:

Extending SAP S/4HANA Cloud in the Cloud Foundry and Kyma Environment

Extending SAP Marketing Cloud in the Cloud Foundry and Kyma Environment

Extending SAP SuccessFactors in the Cloud Foundry and Kyma Environment

Extending SAP Customer Experience Products in the Kyma Environment

Create a formation of type Side-by-Side Extensibility with Kyma where you can include one or more SAP systems depending on your use case.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/eventing-module>

Title: [Eventing Module | SAP Help Portal](#)

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Eventing Module

Use the Eventing module to set up event-driven communication between applications in your Kyma cluster using a publish-subscribe model.

What is Eventing?

The Eventing module enables event-driven communication between applications in your Kyma cluster. One application publishes an event (the publisher), and other applications (subscribers) subscribe to receive it. This decouples your services, as publishers and subscribers don't need to know about each other. They can communicate asynchronously and evolve independently.

Features

The Eventing module provides the following features:

Publish-subscribe (pub/sub) messaging: Decouples applications so you can build resilient and scalable event-driven systems.

Flexible backend support: Use the in-cluster NATS backend (see NATS Module) or configure SAP Event Mesh (see SAP Event Mesh) for enterprise messaging.

Standardized event format: All events follow the CloudEvents specification, ensuring a consistent and portable format.

Automatic legacy event conversion: Converts older, non-standard Kyma event formats into valid CloudEvents automatically. For details, see [Event Name Cleanup](#).

At-least-once delivery: Ensures that each event is delivered at least one time when you use the NATS backend, preventing message loss during temporary failures.

Declarative subscriptions: Manage event subscriptions with a simple Subscription custom resource (CR).

Built-in observability: Exposes key health and performance metrics in Prometheus format (see [Eventing Metrics](#)). Use the Telemetry module to collect, process, and forward these metrics to your observability backend.

Scope
The Eventing module focuses on in-cluster, asynchronous communication using the CloudEvents standard. It uses only one stream, with file storage, for all the events.

For cross-cluster or hybrid scenarios, configure SAP Event Mesh as the backend.

Architecture

The Eventing module uses the Eventing Manager, a Kubernetes operator, to manage the components that process and deliver events within your cluster.

The architecture involves the following flows:

Subscription: The Eventing Manager watches for Subscription custom resources and configures the eventing infrastructure.

Delivery: The Eventing Publisher Proxy receives events, and the configured backend (NATS or SAP Event Mesh) delivers them.

You create a Subscription CR. In the resource manifest, you specify the type of event you want to receive and the sink where to send it.

The Eventing Manager, continuously watching for Subscription resources, detects your Subscription CR.

Based on the backend configured in the Eventing CR, the Eventing Manager connects to the backend (NATS or SAP Event Mesh) and creates a subscription mechanism (a consumer or webhook).

A publisher app sends an event to the Eventing Publisher Proxy.

The Eventing Publisher Proxy forwards the event in CloudEvents format to your eventing backend.

The eventing backend processes the event and pushes it to the Eventing Manager.

The Eventing Manager receives the event, looks up the sink from the Subscription CR, and sends the event to your subscriber app. Your application can now process the event payload.

Eventing Manager

The Eventing Manager is the module's controller. It watches for Subscription custom resources and configures the underlying eventing infrastructure.

When you create or update a Subscription, the Eventing Manager performs the following tasks:

Configures the selected eventing backend to manage event streams and consumers for subscriptions.

Ensures that events are routed from the correct publisher to the specified subscriber (the sink).

Creates and manages Kubernetes resources, such as ConfigMap, Services, and DestinationRules, adapting them to the desired state.

Eventing Publisher Proxy

The Eventing Publisher Proxy provides a single, stable endpoint for your applications to publish events using a standard HTTP POST request (endpoint: /publish for CloudEvents and <application_name>/v1/events for legacy events). This simplifies integration, as you can use common tools like curl or any standard HTTP client.

The proxy performs the following tasks:

Receives inbound events from your applications.

Converts events from legacy formats into the standard CloudEvents format.

Forwards validated CloudEvents to the configured eventing backend for delivery.

API/Custom Resource Definitions

You configure the Eventing module by creating and applying Kubernetes Custom Resource Definitions (CRD), which extend the Kubernetes API with custom additions:

To understand and configure the module's global settings, see the [Eventing CRD](#).

To create a subscriber, define a Subscription CRD. You cannot delete the Eventing module as long as Subscription CRs exist.

Authorization

To assign access permissions to the Eventing module resources, use the following aggregated ClusterRoles:

- kyma-eventing-view - Grants read-only access to all Eventing module resources and their status.
- kyma-eventing-edit - Grants full access to all Eventing module resources and read-only access to their status.

Resource Consumption

To learn more about the resources used by the Eventing module, see [Kyma Modules' Sizing](#).

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/extending-sap-customer-experience-products-in-kyma-environment>

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Getting Started in the Kyma Environment

As an administrator, you must perform several steps to set up a fully operational Kyma environment to which you can connect the chosen SAP solutions.

Prerequisites

To perform administrative and development tasks, you need a global account and one or several subaccounts for which you can provision the Kyma environment. For details, see [Creating Kyma Instances](#).

The subaccount must have the following entitlements assigned:

Kyma environment (added by the subaccount manager)

Service Manager (added by default). However, if you removed the entitlement, you must add it again in your subaccount

For details, see [Configure Entitlements and Quotas for Subaccounts](#).

Procedure

Set up a Kubernetes cluster with the project "Kyma" to connect and extend SAP systems: [Creating Kyma Instances](#)

Add the modules you'd like to use: [Kyma Modules](#).

Assign the roles to users to allow the administrators to manage Kyma and the developers to create Functions:

[Assign Roles in the Kyma Environment](#)

Caution

Assign the roles before the users start using the Kyma dashboard. Not granting the roles results in an error.

Use the Kyma environment to integrate external systems: [Register an SAP Customer Experience System](#)

You have the following options:

Integrate a CX system and the Kyma environment, so that the Functions you develop can use the system's API and receive business events.

Integrate an SAP S/4 HANA Cloud system to use the services it provides to extend your applications.

After you've integrated external systems, the developers can start working on their Functions.

Extend SAP S/4HANA Cloud by developing event-driven extensions and applications: [Extending SAP S/4HANA Cloud in the Cloud Foundry and Kyma Environment](#)

Group several solutions into one formation to meet the requirements of your business scenario: [Including SAP Systems in a Formation](#)

Results

You have set up the Kyma environment and connected it as required.

Next Steps

After the administrator has set up the Kyma environment, the developers can access it through the Kyma dashboard. After logging on, developers can start creating extensions for the SAP systems either from the Kyma dashboard or from the terminal after downloading the kubeconfig file with the cluster configuration. For details, see [Development in the Kyma Environment](#).

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/getting-started-with-trial-account-in-kyma-environment>

Title: [Getting Started with a Trial Account in the Kyma Environment | SAP Help Portal](#)

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[Getting Started with a Trial Account in the Kyma Environment](#)

Quickly get started with a trial account in the Kyma environment.

Procedure

Get a global account.

Sign up for a free trial account. See [Get a Free Trial Account](#). For more information about the scope of our trial offering, see [Trial Accounts and Free Tier](#).

For more information on available plans, including free tier, see [Available Plans in the Kyma Environment](#).

Set up your subaccount.

When you register for a trial account, a subaccount is created for you.

If you want to create more subaccounts to break down your account model and structure it according to your development scenario, see [Create a Subaccount](#). But first it's important you understand how to navigate to your accounts using the SAP BTP cockpit.

You can also download and use the SAP BTP command line interface (btp CLI) to create new subaccounts.

See [Download and Start Using the btp CLI Client and Working with Global Accounts, Directories, and](#)

Subaccounts Using the btp CLI.

Create your Kyma environment. For details, see [Creating Kyma Instances and Enable Trial Kyma Environment](#).

Configure your Kyma environment.

Now that you've set up your account model, it's time to think about member management. You can add members at different levels:

For Kyma-specific role assignment, see [Assign Roles in the Kyma Environment](#).

To learn more about how user roles are assigned in SAP BTP, have a look at [User and Member Management](#).

Develop your applications and extend SAP solutions.

Check out [Development in the Kyma Environment](#) to learn more.

You can also provide extensions for the SAP Commerce Cloud, SAP Cloud for Customer, and SAP Field Service Management systems. For more information, read [Extending SAP Solutions](#).

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/ingress-and-egress-ip-addresses>

Title: [Ingress and Egress IP Addresses | SAP Help Portal](#)

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Ingress and Egress IP Addresses

Learn how to check ingress and egress IP addresses for your Kyma runtime cluster. These addresses are needed to integrate with partner systems, enforce IP-based access controls, connect to external databases, and more.

Egress IP Addressà^

How to Check the Egress IP

You can see the IP addresses for NAT Gateway that handles the egress traffic as values of the `cloud.natGatewayIps` parameter in the `kyma-info` ConfigMap in your Kyma instance.

To check the NAT IPs, use either Kyma dashboard or `kubectl`:

In Kyma dashboard, go to the Cluster Overview section.

With `kubectl`, run:

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```
kubectl --namespace kyma-system get configmap kyma-info -o json | jq -r '.data["cloud.natGatewayIps"]'
```

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Egress IP Stability

The NAT Gateway IP address is assigned when you create the Kyma runtime, and remains unchanged throughout the cluster's lifecycle.

Ingress IP Addressà^

How to Check the Ingress IP

You can see the IP addresses for Istio Ingress Gateway that, by default, handles the ingress traffic. To get the LoadBalancer IP for `istio-ingressgateway`, run:

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```
LOAD_BALANCER_ADDRESS=$(kubectl get services --namespace istio-system istio-ingressgateway --output jsonpath='{.status.loadBalancer.ingress[0].ip}')
```

```
if [[ -z $LOAD_BALANCER_ADDRESS ]]; then
```

```
LOAD_BALANCER_ADDRESS=$(kubectl get services --namespace istio-system istio-ingressgateway --output jsonpath='{.status.loadBalancer.ingress[0].hostname}')
```

```
fi
```

```
echo "The load balancer address is ${LOAD_BALANCER_ADDRESS}"
```

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When the Ingress IP Is Assigned

When you create a Kyma runtime cluster and add the Istio module, the module deploys the `istio-ingressgateway` Service of type `LoadBalancer`. Then, Gardener creates a cloud `LoadBalancer`. When it is ready, its IP address appears in the `istio-ingressgateway` Service.

When you also enable the API Gateway module, it creates a default Gateway custom resource named kyma-gateway. This Gateway configures the istio-ingressgateway as the cluster's ingress traffic entry point. If the Istio module isn't enabled, the cluster doesn't have the istio-ingressgateway Service, so there's no ingress LoadBalancer IP. If the Istio module is enabled but the API Gateway module isn't, the istio-ingressgateway exists, but there's no default Gateway resource that routes external traffic through it. In that case, you must manually configure your own Gateway and related routing resources.

Ingress IP Stability

The external ingress IP of the istio-ingressgateway LoadBalancer Service changes in these cases:

You delete the istio-ingressgateway LoadBalancer Service and recreate it.

You migrate from Elastic LoadBalancer to Network LoadBalancer.

You remove the Istio module (which deletes the istio-ingressgateway Service) and then add it again, which creates a new LoadBalancer.

In other cases, the ingress IP remains the same.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/istio-module>

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Istio Module

Use the Istio module to manage and configure the Istio service mesh.

What Is Istio?à^

Istio is an open-source service mesh that provides a uniform way to manage, connect, and secure microservices. It helps to manage traffic, enhance security capabilities, and provide telemetry data for understanding service behavior. See the open-source Istio documentation.

The Istio module installs and manages Istio in your Kyma cluster. By default, the Istio module is automatically added once you create a Kyma runtime instance.

Featuresà^

The Istio module offers the following features:

Management of Istio installation and upgrades: The module installs Istio and simplifies the process of managing its installation, reducing the complexity and time required for maintenance.

Default Istio configuration: You can quickly have Istio installed with default settings.

Fine-tuning capabilities: You can optimize Istio and fine-tune its settings according to specific performance or operational requirements.

Synchronization of the data plane with the Istio control plane: This ensures that changes you make to the control plane are consistently reflected in the data plane, ensuring that the network operates consistently and reliably without any discrepancies.

Support for the X-Forwarded-For (XFF) header: You can configure the XFF header to manage and track the source of incoming requests.

Architectureà^

Istio Operator

Within the Istio module, Istio Operator handles the management and configuration of the Istio service mesh. It contains one controller, referred to as Istio Controller.

Istio Controller

Istio Controller manages Istio installation as defined in the Istio custom resource (CR). Istio Controller is responsible for:

Installing, upgrading, and uninstalling Istio

Restarting workloads that have Istio sidecar proxy injected to ensure that these workloads use the correct version of Istio

API/Custom Resource Definitionsà^

The `istios.operator.kyma-project.io CustomResourceDefinition (CRD)` describes the Istio CR that Istio Controller uses to manage the installation of Istio. See Istio Custom Resource.

Resource Consumptionà^

To learn more about the resources used by the Istio module, see Kyma Modules' Sizing.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/keda-module>

Title: Keda Module | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Kyma ModulesKeda Module

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à

Keda Module

Learn more about the Keda module. Use it to install and manage the KEDA autoscaler in your Kubernetes cluster.

What Is KEDA?à^

Kubernetes-based Event Driven Autoscaler (KEDA) is an autoscaler that allows you to easily scale your Kubernetes-based resources. You can scale your applications based on the data of your choice.

KEDA supports a great number of scalers that help you manage your deployments. For the complete list, see the KEDA Scalers documentation.

For more information about KEDA features, see the KEDA documentation.

Featuresà^

With the Keda module, you can have a custom event-driven autoscaling for Kubernetes workloads.

Architectureà^

User configures the Keda custom resource (CR). With the CR you can determine the way the Keda module runs.

Keda Manager watches the Keda CR.

Keda Manager manages the KEDA workloads.

To learn more about the KEDA architecture, see the KEDA architecture diagram.

Keda Manager

Keda Manager helps you to install and manage KEDA in your cluster. It manages the lifecycle of KEDA based on the dedicated Keda CR.

API/Custom Resource Definitionsà^

For the Keda CR conditions, check Keda Custom Resource Conditions.

To learn more about the KEDA CR, see KEDA Custom Resources.

Authorizationà^

To assign access permissions to the Keda module resources, use the following aggregated ClusterRoles:

kyma-keda-view

kyma-keda-edit

Resource Consumptionà^

To learn more about the resources used by the Keda module, see Keda Module's Sizing.

Keda Module Demo Applicationsà^

To learn how to scale the Kubernetes workloads using the KEDA API based on a simple CPU consumption case, and how the Keda module can complement other Kyma components, see Demo Applications.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/kyma-dashboard>

Title: Kyma Dashboard | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Kyma EnvironmentKyma Dashboard

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à

Kyma Dashboard

Use Kyma dashboard to access various features and functionalities of SAP BTP, Kyma runtime.

Kyma dashboard is a web-based UI utilizing the React library and the SAP Fiori Horizon design system. Kyma uses Kyma dashboard as a central administration dashboard, which provides a graphical overview of your cluster and all its resources.

To access Kyma dashboard, enable Kyma in your subaccount. The link to the dashboard will be visible in the

Kyma Environment section.

Featuresà

You can use Kyma dashboard to perform the following actions:

Add, manage, and delete Kyma modules.

Manage your namespaces.

Manage your Kubernetes and Kyma resources.

Generate access tokens based on service accounts and Kubernetes RoleBindings and ClusterRoleBindings, which regulate access using Role-based access control (RBAC).

Have a quick look at the basic cluster condition.

Related Informationà

Creating Kyma Instances

Adding and Deleting a Kyma Module

Setting Your Module to the Unmanaged and Managed State

Using RBAC Authorization

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/kyma-env-setting-up-your-trial-account>

Title: *Setting Up Your Trial Account | SAP Help Portal*

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Getting Started with a Trial Account in the Kyma EnvironmentSetting Up Your Trial Account

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à

Setting Up Your Trial Account

Your global trial account is set up automatically, so that you can start using it right away. However, if one or more of the automatic steps fail, you can also finalize the setup manually by following the steps below.

Create Your Trial Subaccountà

The first thing that is needed in the setup of your trial account is the creation of a subaccount. If this step was successful in the SAP BTP cockpit, you can directly skip to the next section.

Procedure

Navigate into your global account by choosing Enter Your Trial Account.

Choose New Subaccount.

Configure your trial account:

Field"—ç W@

Display Name

Note

The value for this parameter is a sample one, you can provide a name of your choice.

trial

Description

Optional

Provider

Desired infrastructure provider

Region

Desired region

Subdomain

<your_id>trial

Example: P0123456789trial

Enable beta features

Optional

Enables the use of beta services and applications.

Custom Properties

Optional

You can use custom properties to identify and organize the subaccounts in your global account. For example, you can filter subaccounts by custom property in some cockpit pages.

Results

You have successfully set up your trial subaccount.

Add Entitlements

Once you have a subaccount (whether it was created automatically or you followed the steps described above), you need entitlements to get your trial up and running.

Procedure

Navigate to your subaccount by choosing its tile.

Choose Entitlements from the left-hand side navigation.

Choose Configure Entitlements to enter edit mode.

Choose Add Service Plans and select all the service plans available for your subaccount.

Note

To select a service plan, choose a service from the left and tick all the service plans that appear on the right.

Do that for all services.

Once you've added all the service plans, you see them all in a table. Before you choose Save, for all the plans with numerical quota, choose ∞ to increase the amount to the maximum (until the icon becomes disabled).

Finally, choose Save to save all your changes and exit edit mode.

Enable Trial Kyma Environment

After you've successfully added the entitlements, you can enable your trial Kyma environment and start developing.

Procedure

Navigate to your subaccount.

Create your Kyma environment. For details, see Creating Kyma Instances.

Provide the name of your cluster and, optionally, a description.

Wait until the cluster is provisioned and you see the link to the Kyma dashboard.

Access the Kyma dashboard. For further information on the development options, see Development in the Kyma Environment.

Results

Your trial Kyma environment is ready to use.

Disable Expired Trial Kyma Environment

After your trial Kyma environment has expired, you must disable it to remove all the data connected to the cluster from SAP systems.

Procedure

Navigate to your subaccount.

In the Kyma Environment section of your subaccount overview, click Delete Kyma Instance.

Note

On deletion of the expired cluster, we attempt to delete the Service Instances that you created with the cluster.

If we cannot do that, you have to remove the Service Instances yourself before you disable such a Kyma Environment.

Extend Trial Kyma Environment

You cannot extend your trial Kyma environment. After your trial Kyma environment has expired, you must disable it on your subaccount and enable it again to create a new cluster.

Procedure

Navigate to your subaccount.

Disable your expired Kyma environment.

After your Kyma environment has been successfully disabled, create a new Kyma environment.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/kyma-environment-backup>

Title: Kyma Environment Backup | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformAdministration and OperationsAdministration and Operations in the Kyma EnvironmentKyma Environment Backup
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Kyma Environment Backup

The user load in a Kyma cluster typically consists of various Kubernetes objects and volumes.

Objects hold the configuration of user's application applied to the API server using Kubernetes resources

Volumes hold the actual data that a user may store in a persistent way

Kyma does not provide backup, neither for objects nor volumes, that can replace the configuration or data from a specific point in time.

Object Backup for Kubernetes Configuration

Kyma environment relies on the managed Kubernetes cluster for periodic backups of Kubernetes objects. It means that it is possible to automatically restore Kubernetes objects in case of inconsistencies or data corruption. However, Kyma environment does not provide backup that can replace the configuration from a specific point in time. Automatic backup doesn't include Kubernetes volumes.

For example, project "Gardener" uses etcd as the Kubernetes' backing store for all cluster data. This means that all Kubernetes objects are stored on etcd. Gardener uses periodic jobs to take major and minor snapshots of the etcd database. A major snapshot (including all the resources) takes place every day, and each minor snapshot (including only the changes in between) takes place every 5 minutes. If the etcd database experiences any problems, Gardener automatically restores the Kubernetes cluster using the latest snapshot.

Tip

Follow GitOps principles to reapply the desired state of your Kubernetes objects to any Kubernetes cluster at any time.

GitOps means managing your Kubernetes objects using Git.

A Git repository becomes a single source of truth where you declare the desired state of your Kubernetes objects.

Automation tooling of your choice pulls the desired state from that single source and applies the declared configuration to your cluster.

For more information, see OpenGitOps.

Volume Backup for Customer Data

Your customer data isn't backed up automatically. If you're using Kubernetes volumes to store data, we recommend that you use Kubernetes VolumeSnapshots to back up and recover your data. The backup and recovery using Kubernetes VolumeSnapshots are supported for AWS, Azure, and Google Cloud.

Related Information

Creating On-Demand Volume Snapshots

Change Storage Size in Kyma

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/kyma-environment>

Title: Kyma Environment | SAP Help Portal

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Kyma Environment

SAP BTP, Kyma runtime provides a fully managed cloud-native Kubernetes application runtime based on the open-source project "Kyma". Based on modular building blocks, Kyma runtime includes all the necessary capabilities to simplify the development and to run enterprise-grade cloud-native applications.

Kyma as a Managed Service

Kyma environment permits a native consumption of the Multi-Cloud Foundation Services (What Is the Multi-Cloud Foundation?) and smooth consumption of SAP and non-SAP applications. It also supports out-of-the-box CAP, SAP Cloud SDK, application router, and HTML5 deployer.

Kyma implements a dedicated application runtime to deploy highly scalable, robust, and secure containerized microservices.

Note

Kyma as a managed service automatically checks all Kyma-managed resources. Any unexpected modifications are discarded, and the resource is reverted to the original state.

Every Kyma environment consists of:

One or more Kubernetes clusters based on project "Gardener" on a cloud provider and region (data center) of your choice. To find out the available regions and providers, see Regions for the Kyma Environment.

A set of Kyma modules picked by a user in the release channel of their choice installed on the provisioned cluster.

Integration

Kyma environment runs on one or more Kubernetes clusters created in a specific subaccount. The configuration of the Kyma environment enables you to connect it to a multitude of SAP systems. This way, you can build various formations that aggregate the SAP systems and environment according to your business use cases.

SAP systems connected to a Kyma environment expose APIs and events. To extend the existing logic of these SAP services, you can build serverless applications called "Functions", and trigger them to react to particular events or calls to your application's API. You can also use the Kyma environment to deploy microservices or even build full-stack applications.

Related Information[^]

Getting Started in the Kyma Environment

Development in the Kyma Environment

Administration and Operations in the Kyma Environment

Security in the Kyma Environment

Discovery Center

What's New for SAP Business Technology Platform

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/kyma-module-releases>

Title: Kyma Module Releases | SAP Help Portal

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Kyma Module Releases

For each of your Kyma modules, you can choose if you want the regular release, or an earlier preview so you can test new features and provide early feedback.

Kyma Release Channels[^]

Kyma is released in a modular approach. To learn more about the available Kyma modules, read: Kyma Modules.

Kyma modules are deployed in the following release channels:

Regular channel is the default release channel.

Fast channel provides more frequent releases. It offers early previews of all new features and changes before they are promoted to the regular channel. It also allows you to test and provide feedback on the new features sooner.

According to the Kyma modules` release cycle, we first release a new module's major or minor version in the fast channel. After approximately two weeks, we promote the release to the regular channel.

Note

In case of important functionality fixes or critical vulnerabilities identified by our security organization, the timeline doesn't apply, as we provide hotfixes between regular releases.

You can use one or both release channels in your Kyma cluster, but you can define only one release channel per module. For example, you can mix different modules from the regular and fast channels in your development cluster, but you cannot deploy the same module in the regular and fast versions in one cluster.

Tip

You can upgrade module versions, but you cannot downgrade them. To test the upstream versions, you can switch a module or an entire cluster from the regular channel to the fast one.

To return to the regular channel, you must wait until the version you are using in the fast channel is promoted to the regular channel. Once the versions in both the fast and regular channels are the same, you can switch back to regular. Alternatively, you can delete and add your module from the regular channel.

To find out which module version is running in your cluster, go to Kyma dashboard.

Release Notes[^]

A release of a new module's version is announced with a release note in What's New for SAP Business Technology Platform for both, the fast and regular channels:

On the day of the release in the fast channel, a release note is published with the Preview label.

After approximately two weeks, the module version becomes available in the regular channel and the Preview label is removed.

For details on the Kyma release schedule, see 3459911 .

Note

For any changes that require user action, we notify users at least three months in advance in the “What’s New for SAP Business Technology Platform”. You get information on the required or recommended actions, as well as links to more detailed guidance, including details on the opt-in solutions and migration timelines.

To get notifications on such changes and other updates, subscribe to What’s New for SAP Business Technology Platform.

Related Information

Kyma Functionalities

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/kyma-modules-sizing>

Title: Kyma Modules' Sizing | SAP Help Portal

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Kyma Modules' Sizing

The following tables show the resource consumption, such as memory, CPU, and storage, of Kyma modules.

Istio

Istio Module's Sizing

Infrastructure size

Memory

CPU

Small cluster minimum

352 Mi

0.08

Large cluster minimum

1984 Mi

0.61

API Gateway

API Gateway Module's Sizing

Infrastructure size

Memory

CPU

Small cluster minimum

212 Mi

0.03

Large cluster minimum

416 Mi

0.34

SAP BTP Operator

SAP BTP Operator Module's Sizing

Infrastructure size

Memory

CPU

Minimum

53 Mi

0.02

Maximum

1000 Mi

2

Application Connector

Application Connector Module's Sizing

Infrastructure size

Memory

CPU

Minimum

64-128 Mi

0.2-0.5

Eventingà^

Eventing Manager's Sizing

Infrastructure size

Memory

CPU

Minimum

128 Mi

0.01

Maximum

512 Mi

0.5

Eventing Publisher Proxy's Sizing

Infrastructure size

Memory

CPU

Max Publisher instances

Low EPS (up to 200 Events per second)

64-512 Mi

0.04-0.5

2 (second instance for high availability)

High EPS (up to 3000 Events per second)

64-512 Mi

0.04-0.5

3

Kedaà^

Keda Module's Sizing

Infrastructure size

Memory

CPU

Minimum

692 Mi

0.62

Maximum

928 Mi

1.8

NATSà^

By default, the NATS module creates a NATS-cluster with three nodes for data safety reasons. The values are given per NATS-cluster node:

NATS Module's Sizing

Infrastructure size

Memory

CPU

Storage

Minimum

64 Mi

0.04

1000 Mi

Maximum

1000 Mi

0.5

1000 Mi

Serverlessà^

Serverless Module's Sizing

Infrastructure size

Memory

CPU

Storage

Minimum

400 Mi

0.03

Maximum

2300 Mi

1.7

Up to 20000 Mi (for PVC for internal Docker registry)

Telemetryà^

Resource consumption depends on the APIs (pipelines) you are using. For details, see Telemetry Pipeline API.

Using the Telemetry module without activating any pipelines results in the Telemetry Manager running with the following footprint:

Infrastructure size

Memory

CPU

Minimum

5 Mi

0.01

Maximum

100 Mi

0.1

Activation of the first LogPipeline causes the deployment of the log agent running an instance per node:

Infrastructure size

Memory

CPU

Minimum per node

50 Mi

0.1

Maximum per node

1000 Mi

1

Activation of the first TracePipeline causes the deployment of the trace gateway running with two replicas:

Infrastructure size

Memory

CPU

Minimum

2*32 Mi

2*0.2

Maximum

2*2000 Mi

2*1.2

Activation of the first MetricPipeline causes the deployment of the metric gateway running with two replicas and the metric agent running per node:

Infrastructure size

Memory

CPU

Minimum

2*30 Mi + 50 Mi per node

2*0.01 + 0.01 per node

Maximum

2*1000 Mi + 1200 Mi per node

2*1.00 + 1 per node

Related Informationà

Sizing

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/kyma-modules>

Title: Kyma Modules | SAP Help Portal

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à

Kyma Modules

With Kyma's modular approach, you can install just the modules you need, instead of a predefined set of components.

Pick the modules that best meet your business needs. You can add or delete modules on demand and configure them, thus determining the costs generated by the infrastructure that your modules use. To learn how, see [Adding and Deleting a Kyma Module](#).

You can decide which modules are needed for your workload, although some features may still require the presence of other modules or related capabilities. For example, the API Gateway module depends on Istio capabilities, but it doesn't require the Istio module from Kyma. Instead, the API Gateway module needs Istio custom resources, such as Gateway and VirtualService, to be present in the cluster. If you want to use your own Istio installation, the API Gateway module will use it as soon as the required resources are available.

Tip

A release of a new module's version is announced with a release note in [What's New for SAP Business Technology Platform](#) for both, the fast and regular channels:

On the day of the release in the fast channel, a release note is published with the Preview label.

When the module version becomes available in the regular channel (after approximately two weeks), the Preview label is removed.

For details, see [Kyma Module Releases](#).

To find out which module version is running in your cluster, go to [Kyma dashboard](#).

Default Kyma Modulesà

When you create Kyma runtime in SAP BTP cockpit, it is provisioned with the default modules added. The default modules are not mandatory. If you don't need them, you can delete them in [Kyma dashboard](#).

Default Kyma Modules

Module

Technical Name

Purpose

Documentation

Istio

istio

Istio is a service mesh with Kyma-specific configuration.

Istio Module

API Gateway

api-gateway

API Gateway provides functionalities that allow you to expose and secure APIs.

API Gateway Module

SAP BTP Operator

btp-operator

Within the SAP BTP Operator module, BTP Manager installs the SAP BTP service operator that allows you to consume SAP BTP services from your Kubernetes cluster using Kubernetes-native tools.

SAP BTP Operator Module

What's New for SAP BTP Operator

GitHub repository: BTP Manager

Optional Kyma Modules

When you create Kyma runtime in SAP BTP cockpit, the following modules are not added by default, but you can choose to add and delete them anytime in Kyma dashboard.

Optional Kyma Modules

Module

Technical Name

Purpose

Documentation

Application Connector

application-connector

Application Connector allows you to connect with external solutions. No matter if you want to integrate an on-premise or a cloud system, the integration process doesn't change, which allows you to avoid any configuration or network-related problems.

Application Connector Module

GitHub repository: Application Connector Manager

Cloud Manager

cloud-manager

Cloud Manager brings cloud provider products and resources into the Kyma cluster in a secure way.

Cloud Manager Module

GitHub repository: Cloud Manager

Eventing

eventing

Eventing provides functionality to publish and subscribe to CloudEvents.

At the moment, the SAP Event Mesh default plan and NATS are supported. If you choose NATS, add the NATS module.

Eventing Module

GitHub repository: Eventing

Keda

keda

The Keda module comes with Keda Manager, an extension to Kyma that allows you to install KEDA (Kubernetes Event-Driven Autoscaler).

Keda Module

NATS

nats

NATS deploys a NATS cluster within the Kyma cluster. You can use it as a backend for Kyma Eventing.

NATS Module

GitHub repository: NATS

Registry Cache

registry-cache

With the Registry Cache module, you can add a caching layer for container image registries, reducing outbound traffic to upstream registries and improving image pull performance.

Registry Cache Module

GitHub repository: Registry Cache

Serverless

serverless

With the Serverless module, you can define simple code snippets (Functions) with minimal implementation effort.

Serverless Module

Configuring Serverless

Deploy Workloads in the Kyma Environment to Extend SAP Systems

GitHub repository: Serverless

Telemetry

telemetry

The Telemetry module collects application logs and distributed traces for your application, and dispatches them to your preferred backends.

Telemetry Module

What's New for Telemetry

GitHub repository: Telemetry

Other SAP Modulesà^

These modules are developed and maintained by SAP teams outside of Kyma. To get help or request a feature, contact the module provider directly.

Other SAP Modules

Module

Technical Name

Purpose

Documentation

Transparent Proxy

transparent-proxy

Use the transparent proxy for Kubernetes to connect workloads in a Kubernetes cluster to Internet and on-premise applications.

Transparent Proxy in the Kyma Environment

Connectivity Proxy

connectivity-proxy

Use the connectivity proxy for Kubernetes to connect workloads in a Kubernetes cluster to on-premise systems, exposed via the Cloud Connector

On-Premise Connectivity in the Kyma Environment

Community Modulesà^

Community modules are modules provided by the Kyma community. Use them to enhance your Kyma experience with additional features and capabilities. You can install community modules alongside managed modules. Unlike managed modules, community modules aren't automatically updated or maintained. For more information, see kyma-project.io: Community Modules.

Caution

Community modules aren't subject to the Service Level Agreement (SLA).

Related Informationà^

Adding and Deleting a Kyma Module

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/kyma-runtime-with-dual-stack-support>

Title: *Kyma Runtime with Dual-Stack Support | SAP Help Portal*

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à

Kyma Runtime with Dual-Stack Support

Learn how to apply dual-stack support in Kyma instances.

Contextà^

SAP BTP, Kyma runtime can support both Internet Protocol (IP) versions, IPv4 and IPv6, simultaneously (dual stack), when deployed on Google Cloud and Amazon Web Services IaaS providers. Compared to IPv4, IPv6 offers a significantly greater number of unique IP addresses, which enhances network flexibility. To retrieve support for both Internet protocols, create your Kyma runtime with dual-stack support enabled.

The dual-stack feature is only available for newly created Kyma instances. Existing instances are limited to IPv4.

Caution

The Kyma Istio module does not support dual-stack mode. Any traffic sent or received through the IPv6 network is not part of the Istio service mesh. Consequently, the Istio module does not protect IPv6 traffic, either by transparent encryption or by offering additional security mechanisms (for example, authentication).

Procedureà^

Create a service instance of type LoadBalancer in your Kyma cluster. In the service manifest, add the

ipFamilyPolicy field with the value RequireDualStack, and add the value IPv6 in the ipFamilies field.

âE

ã

apiVersion: v1

kind: Service

...

spec:

ipFamilyPolicy: RequireDualStack

ipFamilies:

- IPv6

- IPv4

...

In your Kyma cluster with dual-stack support, each deployed Pod automatically receives two IP addresses: IPv4 and IPv6.

Related Informationâ

Creating an IPv4/IPv6 (dual-stack) Ingress

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/kyma-telemetry-module>

Title: Telemetry Module | SAP Help Portal

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à

Telemetry Module

Use the Telemetry module to collect telemetry signals (logs, traces, and metrics) from your applications and send them to your preferred observability backend.

What is Telemetry?â

With telemetry signals, you can understand the behavior and health of your applications and infrastructure. The Telemetry module provides a standardized way to collect these signals and send them to your observability backend, where you can analyze them and troubleshoot issues.

The Telemetry module processes three types of signals:

Logs: Time-stamped records of events that happen over time.

Traces: The path of a request as it travels through your application's components.

Metrics: Aggregated numerical data about the performance or state of a component over time.

Telemetry signals flow through the following stages:

You instrument your application so that its components expose telemetry signals.

The signals are collected and enriched with infrastructural metadata.

You send the enriched signals to your preferred observability backend.

The backend stores your data, where you can analyze and visualize it.

The Telemetry module focuses on the collection, processing, and shipment stages of the observability workflow.

It offers a vendor-neutral approach based on OpenTelemetry and doesn't force you into a specific backend. This means you can integrate with your existing observability platforms or choose from a wide range of available backends that best suit your operational needs.

Tip

Build your first telemetry pipeline with the hands-on lesson [Collecting Application Logs and Shipping them to SAP Cloud Logging](#).

Featuresâ

To support telemetry for your applications, the Telemetry module provides the following features:

Consistent Telemetry Pipeline API: Use a streamlined set of APIs based on the OTel Collector to collect, filter, and ship your logs, metrics, and traces (see [Telemetry Pipeline API](#)). You define a pipeline for each signal type to control how the data is processed and where it's sent. For details, see [Collecting Logs](#), [Collecting Traces](#), and [Collecting Metrics](#).

Flexible Backend Integration: The Telemetry module is optimized for integration with SAP BTP observability services, such as SAP Cloud Logging. You can also send data to any backend that supports the

OpenTelemetry protocol (OTLP), giving you the freedom to choose your preferred solution (see [Integrate With Your OTLP Backend](#)).

Recommendation

For production deployments, we recommend using a central telemetry solution located outside your cluster. For an example, see [Integrate With SAP Cloud Logging](#).

For testing or development, in-cluster solutions may be suitable. For examples such as Dynatrace (or to learn how to collect data from applications based on the OpenTelemetry Demo App), see [Integration Guides](#).

Seamless Istio Integration: The Telemetry module seamlessly integrates with the Istio module when both are present in your cluster. For details, see [Istio Integration](#).

Custom Filtering and Processing: You can drop any unnecessary Telemetry data right when it's collected (see [Filter Logs](#), [Filter Traces](#), and [Filter Metrics](#)). For advanced data manipulation, such as redacting sensitive information or creating complex filters, you can use the OpenTelemetry Transformation Language (see [Transform and Filter with OTTL](#)).

Automatic Data Enrichment: The Telemetry module adds resource attributes as metadata, following OTel semantic conventions. This makes your data more consistent, meaningful, and ready for analysis in your observability backend. For details, see [Automatic Data Enrichment](#).

Instrumentation Guidance: To generate telemetry data, you must instrument your code. Based on Open Telemetry (OTel), you get community samples on how to instrument your code using the Open Telemetry SDKs in most programming languages.

Simplified Permission Management: To manage access to Telemetry resources, you can use the module's aggregated ClusterRoles (see [Authorization](#)).

Scope

The Telemetry module focuses only on the signals of application logs, distributed traces, and metrics. Other kinds of signals are not considered. Also, audit logs are not in scope.

Supported integration scenarios are neutral to the vendor of the target system.

Architecture

The Telemetry module is built around a central controller, Telemetry Manager, which dynamically configures and deploys data collection components based on your pipeline resources.

To understand how the core components interact, see [Telemetry Architecture](#).

To learn how this model applies to each signal type, see:

[Logs Architecture](#)

[Traces Architecture](#)

[Metrics Architecture](#)

API/Custom Resource Definitions

You configure the Telemetry module and its pipelines by creating and applying Kubernetes CustomResourceDefinitions (CRDs), which extend the Kubernetes API with custom additions.

To understand and configure the module's global settings, refer to the [Telemetry CRD](#).

To define how to collect, process, and ship a specific signal, use the pipeline CRDs:

[LogPipeline CRD](#)

[TracePipeline CRD](#)

[MetricPipeline CRD](#)

Authorization

To assign access permissions to the Telemetry module resources, use the following aggregated ClusterRoles:

[kyma-telemetry-view](#) to grant read-only access

[kyma-telemetry-edit](#) to grant write access

Resource Consumption

To learn more about the resources used by the Telemetry module, see [Kyma Modules' Sizing](#).

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/managing-kyma-runtime-using-provisioning-service-api>

Title: *Managing Kyma Runtime Using the Provisioning Service API | SAP Help Portal*

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformAdministration and OperationsAdministration and Operations in the Kyma EnvironmentManaging Kyma Runtime Using the Provisioning Service API

SAP Business Technology PlatformEnglish

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Managing Kyma Runtime Using the Provisioning Service API

The SAP Cloud Management service for SAP BTP (technical name: cis) provides the Provisioning Service API to create and manage available environments. Use the Provisioning Service API to automatically manage and access SAP BTP, Kyma runtime.

Prerequisitesà

Your subaccount must have entitlements for SAP BTP, Kyma runtime and the SAP Cloud Management service for SAP BTP. See [Managing Entitlements and Quotas Using the Cockpit](#).

Command line interface (CLI) tools:

kubectl configured to communicate with your Kyma instance (see [Access a Kyma Instance Using kubectl](#))

jq

curl

SAP BTP command line interface (btp CLI). See [Download and Start Using the btp CLI Client](#).

Contextà

To manage a Kyma instance automatically, create a Kyma service binding. The Kyma service binding enables getting a Kyma kubeconfig, which in turn allows for accessing a Kyma cluster, deploying applications, running tests, and deleting the resources in a fully automated way.

Procedureà

Provision the SAP Cloud Management service instance with the local plan and create a binding to get the credentials for the Provisioning Service API. To do that, you can use one of the following methods:

SAP BTP cockpit, as described in [Getting an Access Token for SAP Cloud Management Service APIs](#).

btp CLI and follow these steps:

Set the CIS_INSTANCE_NAME environment variable with the name of the SAP Cloud Management service instance.

âE

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```
export CIS_INSTANCE_NAME={CIS_INSTANCE_NAME}
```

Provision the SAP Cloud Management service instance with the Client Credentials grant type passed as parameters.

âE

ã

```
btp create services/instance --offering-name cis --plan-name local --name ${CIS_INSTANCE_NAME} --  
parameters {"grantType":"clientCredentials"}
```

âf

Create a binding for the instance.

âE

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```
btp create services/binding --name ${CIS_INSTANCE_NAME}-binding --instance-name  
${CIS_INSTANCE_NAME}
```

âf

Set the CLIENT_ID, CLIENT_SECRET, UAA_URL, and PROVISIONING_SERVICE_URL environment variables using the credentials from the binding stored in the clientid, clientsecret, url, and provisioning_service_url fields. Use the btp CLI to get the credentials.

âE

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```
export CLIENT_ID=$(btp --format json get services/binding --name ${CIS_INSTANCE_NAME}-binding | jq -r  
'credentials.uaa.clientid')
```

```
export CLIENT_SECRET=$(btp --format json get services/binding --name ${CIS_INSTANCE_NAME}-binding |  
jq -r 'credentials.uaa.clientsecret')
```

```
export UAA_URL=$(btp --format json get services/binding --name ${CIS_INSTANCE_NAME}-binding | jq -r  
'credentials.uaa.url')
```

```
export PROVISIONING_SERVICE_URL=$(btp --format json get services/binding --name  
${CIS_INSTANCE_NAME}-binding | jq -r 'credentials.endpoints.provisioning_service_url')
```

àf

Get the access token for the Provisioning Service API using the client credentials.

âE

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```
TOKEN=$(curl -s -X POST "${UAA_URL}/oauth/token" -H "Content-Type: application/x-www-form-urlencoded" -u "${CLIENT_ID}:${CLIENT_SECRET}" --data-urlencode "grant_type=client_credentials" | jq -r '.access_token')
```

àf

Check if Kyma runtime is available for provisioning.

âE

ã

```
curl -s "$PROVISIONING_SERVICE_URL/provisioning/v1/availableEnvironments" -H "accept: application/json" -H "Authorization: bearer $TOKEN" | jq
```

àf

Set the ENVIRONMENT_TYPE and SERVICE_NAME environment variables to const values kyma and kymaruntime, and provide values for the NAME, REGION, PLAN, and USER_ID environment variables.

âE

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```
export ENVIRONMENT_TYPE="kyma"
export SERVICE_NAME="kymaruntime"
export NAME={RUNTIME_NAME}
export REGION={CLUSTER_REGION}
export PLAN={KYMA_RUNTIME_PLAN_NAME}
export USER_ID={USER_ID}
```

Provision the Kyma runtime instance and save the instance ID in the INSTANCE_ID environment variable.

âE

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```
INSTANCE_ID=$(curl -s -X POST "$PROVISIONING_SERVICE_URL/provisioning/v1/environments" -H "accept: application/json" -H "Authorization: bearer $TOKEN" -H "Content-Type: application/json" -d '{"environmentType":"'${ENVIRONMENT_TYPE}',"parameters":{"name":"'${NAME}',"region":"'${REGION}',"planName":"'${PLAN}',"serviceName":"'${SERVICE_NAME}',"user":"'${USER_ID}'"}}' | jq -r '.id')
```

àf

Optional: Set the EXPIRATION_SECONDS environment variable to the number of seconds (from 600 to 7200) after which a binding expires.

âE

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```
export EXPIRATION_SECONDS={EXPIRATION_SECONDS}
```

After the provisioning is completed, create the binding and save the binding ID in the BINDING_ID environment variable.

âE

ã

```
[ -z "$EXPIRATION_SECONDS" ] && \
```

```
BINDING_ID=$(curl -s -D - -X PUT "$PROVISIONING_SERVICE_URL/provisioning/v1/environments/$INSTANCE_ID/bindings" -H "accept: application/json" -H "Authorization: bearer $TOKEN" -H "Content-Type: application/json" -d '{"parameters":{"expiration_seconds":600}}' -o /dev/null | sed -n 's/^.*location: //p' | sed 's/\r$//g') || \
```

```
BINDING_ID=$(curl -s -D - -X PUT "$PROVISIONING_SERVICE_URL/provisioning/v1/environments/$INSTANCE_ID/bindings" -H "accept: application/json" -H "Authorization: bearer $TOKEN" -H "Content-Type: application/json" -d '{"parameters":{"expiration_seconds":$EXPIRATION_SECONDS}}' -o /dev/null | sed -n 's/^.*location: //p' | sed 's/\r$//g')
```

àf

Note

You can have a maximum of 10 non-expired bindings. If you try to create more, you get the message stating that you've reached the maximum number of non-expired bindings.

Get the binding credentials and save them in a kubeconfig file.

âE

ã

```
curl -s -X GET "$PROVISIONING_SERVICE_URL/provisioning/v1/environments/$INSTANCE_ID/bindings/$BINDING_ID" -H "accept: application/json" -H "Authorization: bearer $TOKEN" | jq -r '.credentials.kubeconfig' > kubeconfig.yaml
```

àf

To access the cluster through kubectl, set the KUBECONFIG environment variable to the path of the kubeconfig file.

âE

ã

```
export KUBECONFIG=kubeconfig.yaml
```

Verify the connection to the cluster. Run a kubectl command to get Pods.

âE

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```
kubectl get pods
```

kubectl should return the list of Pods in the default namespace running in the cluster, which means that the cluster is accessible.

Optional: To view the details of the binding you have created, list all bindings for the instance.

âE

ã

```
curl -s "$PROVISIONING_SERVICE_URL/provisioning/v1/environments/$INSTANCE_ID/bindings" -H "accept: application/json" -H "Authorization: bearer $TOKEN"
```

àf

Optional: For extra security, revoke the credentials by deleting the binding sooner than it is set to expire in the EXPIRATION_SECONDS environment variable.

âE

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```
curl -s -X DELETE "$PROVISIONING_SERVICE_URL/provisioning/v1/environments/$INSTANCE_ID/bindings/$BINDING_ID" -H "accept: application/json" -H "Authorization: bearer $TOKEN"
```

àf

Try to access the cluster using kubectl. The connection should be refused, which means that the binding was successfully deleted and credentials revoked.

âE

ã

```
kubectl get pods
```

Note

If you skip this step, the binding is automatically deleted after the maximum allowed expiration time (7200 seconds) passes.

Next Stepsà^

To deprovision Kyma runtime, run:

âE

ã

```
curl -s -X DELETE "$PROVISIONING_SERVICE_URL/provisioning/v1/environments/$INSTANCE_ID" -H "accept: application/json" -H "Authorization: bearer $TOKEN"
```

àf

Note

You can delete the runtime independently of the bindings. Existing bindings do not block the runtime deprovisioning.

Related Informationà^

Account Administration Using APIs of the SAP Cloud Management Service

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/nats-module>

Title: NATS Module | SAP Help Portal

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ModulesNATS Module

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NATS Module

Learn more about the NATS Module. Use it to manage and configure the message-oriented middleware, NATS.

What Is NATS?à

NATS is an infrastructure that enables the exchange of data in the form of messages. One feature of NATS is JetStream. JetStream is a distributed persistence system that provides more functionalities and higher qualities of service on top of Core NATS.

The NATS module includes the NATS Manager, which manages the lifecycle of a NATS JetStream deployment. It observes the state of the NATS cluster and reconciles it with the desired state.

For more information about NATS and NATS JetStream, see the official NATS documentation.

The Eventing Module can use NATS as a backend to process events and send them to subscribers.

Caution

Reaching the NATS storage size limits may cause the Eventing module to stop receiving events or delay them.

For more information on the troubleshooting, see NATS Backend Storage Is Full.

Featuresà

Automated NATS JetStream Deployment: Deploys a production-ready NATS JetStream cluster without manual setup.

Persistent Messaging: Use file-based storage to ensure messages are retained even if a Pod restarts. Memory-based storage is available for higher throughput scenarios.

Declarative Configuration: Manage your NATS cluster configuration, including cluster size and storage options, through a simple Kubernetes custom resource.

Configurable Resource Allocation: Define specific CPU and memory requests and limits for the NATS Pods to fit your cluster's capacity.

Seamless integration with the Eventing module.

High Availability

For high availability, set up NATS servers across different availability zones for uninterrupted operation and uptime. NATS Manager deploys the NATS servers in the availability zones where your Kubernetes cluster has Nodes. If the Kubernetes cluster has Nodes distributed across at least three availability zones, NATS Manager automatically distributes the NATS servers across these availability zones. If the Kubernetes cluster doesn't have Nodes distributed across at least three availability zones, high availability is compromised.

Architectureà

The NATS module uses a Kubernetes operator-based architecture.

You create a NATS CR.

The NATS Manager starts the Controller, which creates, watches, and reconciles the relevant resources.

The Controller reacts to changes in the NATS CR to adapt the resources to the desired state.

The Controller creates or deletes the NATS server.

NATS Manager

The NATS Manager is responsible for starting the Controller which creates, watches, and reconciles the relevant resources:

ConfigMaps

Secrets

Services

StatefulSets

DestinationRules

API/Custom Resource Definitionsà

The `nats.operator.kyma-project.io CustomResourceDefinition (CRD)` describes the NATS CR that NATS Manager uses to manage the module. See NATS Custom Resource.

Authorizationà

To assign access permissions to the NATS module resources, use the following aggregated ClusterRoles:

`kyma-nats-view` - Grants read-only access to NATS CRs and their status.

`kyma-nats-edit` - Grants full access to NATS CRs and read-only access to their status.

Resource Consumptionà

To learn more about the resources used by the NATS module, see NATS.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/preconfigured-credentials-and-access>

Title: Preconfigured Credentials and Access | SAP Help Portal

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Preconfigured Credentials and Access

When you create SAP BTP, Kyma runtime, all necessary resources for consuming SAP BTP services are created, and the basic cluster access is configured.

Credentialsà^

When you create a Kyma instance in the SAP BTP cockpit, the following events happen in your subaccount:

An SAP Service Manager service instance with the service-operator-access plan is created.

An SAP Service Manager service binding with access credentials for the SAP BTP Operator is created.

The credentials from the service binding are passed on to the Kyma service instance in the creation process.

The sap-btp-manager Secret is created and managed in the kyma-system namespace.

By default, the SAP BTP Operator module is installed together with the following:

The sap-btp-manager Secret.

The sap-btp-service-operator Secret with the access credentials for the SAP BTP service operator. You can view the credentials in the kyma-system namespace.

The sap-btp-operator-config ConfigMap.

The sap-btp-manager Secret provides the following credentials:

clientid

clientsecret

cluster_id

sm_url

tokenurl

Note

If you modify or delete the sap-btp-manager Secret, it is modified back to its previous settings or regenerated within 24 hours.

To prevent your changes from being reverted, label the Secret with kyma-project.io/skip-reconciliation: "true".

For more information, see Customizing the Default Credentials and Access.

When the SAP BTP Operator module is added to your cluster by default, the sap-btp-manager Secret populates the SAP BTP service operator's resources as shown in the following diagram:

The cluster ID represents and identifies a Kyma service instance created in a particular subaccount. In Kyma dashboard, you can view the cluster ID in the following resources:

The sap-btp-manager Secret.

The sap-btp-service-operator Secret.

The sap-btp-operator-config ConfigMap.

Cluster Accessà^

By default, SAP BTP Operator has cluster-wide permissions. The following parameters manage cluster access:

Cluster Access Parameters

Parameter

Description

CLUSTER_ID

Generated when Kyma runtime is created.

MANAGEMENT_NAMESPACE

Indicates the namespace for the Secrets with credentials to communicate with the SAP Service Manager. By default, set to kyma-system.

RELEASE_NAMESPACE

Indicates the namespace for the sap-btp-service-operator and sap-btp-operator-clusterid Secrets. By default, set to kyma-system.

ALLOW_CLUSTER_ACCESS

You can use every namespace for your operations. The parameter is always set to true. If you change it to false, your setting is automatically reverted, and SAP BTP Operator cluster-wide permissions remain unchanged.

Default Credentials and Kyma Runtime Deletion

The preconfigured credentials described in the Credentials section may affect the deletion of your Kyma cluster. If any non-deleted service instances in your Kyma cluster use the credentials from the SAP Service Manager resources created automatically, you can't delete the cluster. In this case, the existing service instances block the cluster's deletion. Before you attempt to delete the cluster from the SAP BTP cockpit, delete your service instances and bindings in Kyma dashboard. See [Deleting Service Bindings and Service Instances](#).

However, if all existing service instances in your Kyma cluster use your custom SAP Service Manager credentials, the non-deleted service instances do not block the cluster's deletion. See [Customizing the Default Credentials and Access](#).

If you have not deleted service instances and bindings connected to your expired free tier service, you can still find the service binding credentials in the SAP Service Manager instance details in the SAP BTP cockpit. Use them to delete the leftover service instances and bindings.

Related Information

[Customizing the Default Credentials and Access](#)

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/provisioning-and-update-parameters-in-kyma-environment>

Title: [Provisioning and Updating Parameters in the Kyma Environment | SAP Help Portal](#)

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[Provisioning and Updating Parameters in the Kyma Environment](#)

When creating a Kyma cluster, you can configure various parameters to adjust it to your specific needs.

Overview

To configure the cluster parameters, you can use your preferred interface, the SAP BTP cockpit, or the SAP BTP command line interface (btp CLI).

To check which parameters are available for configuration in a particular plan, see [Available Plans in the Kyma Environment](#).

Remember

The parameters marked with an asterisk "*" are mandatory.

Access Control List

Additional Volume Size

Additional Worker Node Pools

Administrators

Audit Log Access

Auto Scaler Max

Auto Scaler Min

Cluster Name*

Colocate Control Plane

Machine Type

Modules

Networking

OpenID Connect (OIDC)

Region*

Related Information

[Creating Kyma Instances](#)

[Available Plans in the Kyma Environment](#)

[Regions for the Kyma Environment](#)

[Kyma Modules](#)

[Account Administration Using the SAP BTP Command Line Interface \(btp CLI\)](#)

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/regions-for-kyma-environment>

Title: Regions for the Kyma Environment | SAP Help Portal

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Regions for the Kyma Environment

To work with the Kyma environment, you need to specify the region for both your subaccount and the cluster.

Note

In Kyma, the ingress and egress IP addresses are independent of the subaccount region. For more information, see Ingress and Egress IP Addresses.

Subaccount Regionsà^

The table lists the regions you can choose from when creating a subaccount.

Subaccount Regions for Kyma

Global Account Type

Region

IaaS Provider

Technical Key

Region Name

Plan ID

Enterprise account

Partner Test, Demo, and Development account

Trial account

ap21

Microsoft Azure

cf-ap21

Singapore

azure

azure_lite

trial

build-runtime-azure

Enterprise account

Partner Test, Demo, and Development account

us20

Microsoft Azure

cf-us20

US West (WA)

azure

azure_lite

build-runtime-azure

Enterprise account

Partner Test, Demo, and Development account

jp20

Microsoft Azure

cf-jp20

Japan (Tokyo)

azure

azure_lite

build-runtime-azure

Enterprise account

Partner Test, Demo, and Development account

us21

Microsoft Azure

cf-us21
US East (VA)
azure
azure_lite
build-runtime-azure
Enterprise account
Partner Test, Demo, and Development account
eu20
Microsoft Azure
cf-eu20
Europe (Netherlands)
azure
azure_lite
build-runtime-azure
Enterprise account
Partner Test, Demo, and Development account
eu22
Microsoft Azure
cf-eu22
Europe (Frankfurt)
azure
azure_lite
Enterprise account
Partner Test, Demo, and Development account
uk20
Microsoft Azure
cf-uk20
UK South (London)
azure
azure_lite
Enterprise account
Partner Test, Demo, and Development account
ap20
Microsoft Azure
cf-ap20
Australia (Sydney)
azure
azure_lite
build-runtime-azure
Enterprise account
Partner Test, Demo, and Development account
br20
Microsoft Azure
cf-br20
Brazil (São Paulo)
azure
azure_lite
build-runtime-azure
Enterprise account
Partner Test, Demo, and Development account
ca20
Microsoft Azure
cf-ca20
Canada (Toronto)

azure
azure_lite
Enterprise account
ch20
Microsoft Azure
cf-ch20
Switzerland (Zurich) EU Access
azure
Enterprise account
cn20
Microsoft Azure
cf-cn20
China (North 3) 1
azure
Enterprise account
Trial account
us10
Amazon Web Services
cf-us10
US East (VA)
aws
trial
build-runtime-aws
Enterprise account
Trial account
eu10
Amazon Web Services
cf-eu10
Europe (Frankfurt)
aws
trial
build-runtime-aws
Enterprise account
eu11
Amazon Web Services
cf-eu11
Europe (Frankfurt) EU Access
aws
build-runtime-aws
Enterprise account
eu13
Amazon Web Services
cf-eu13
Europe (Milan)
aws
Enterprise account
br10
Amazon Web Services
cf-br10
Brazil (São Paulo)
aws
build-runtime-aws
Enterprise account
jp10

Amazon Web Services
cf-jp10
Japan (Tokyo)
aws
build-runtime-aws
Enterprise account
ca10
Amazon Web Services
cf-ca10
Canada (Montreal)
aws
build-runtime-aws
Enterprise account
ap12
Amazon Web Services
cf-ap12
South Korea (Seoul)
aws
Enterprise account
ap10
Amazon Web Services
cf-ap10
Australia (Sydney)
aws
build-runtime-aws
Enterprise account
ap11
Amazon Web Services
cf-ap11
Singapore
aws
build-runtime-aws
Enterprise account
us11
Amazon Web Services
cf-us11
US West (Oregon)
aws
Enterprise account
us30
Google Cloud
cf-us30
US Central (IA)
gcp
build-runtime-gcp
Enterprise account
eu30
Google Cloud
cf-eu30
Europe (Frankfurt)
gcp
build-runtime-gcp
Enterprise account
eu31

Google Cloud
cf-eu31
Europe (Netherlands)
gcp
Enterprise account
in30
Google Cloud
cf-in30
India (Mumbai)
gcp
build-runtime-gcp
Enterprise account
jp30
Google Cloud
cf-jp30
Japan (Osaka)
gcp
Enterprise account
jp31
Google Cloud
cf-jp31
Japan (Tokyo)
gcp
Enterprise account
sa30
Google Cloud
cf-sa30
KSA (Dammam) GCP public sector
gcp
Enterprise account
sa31
Google Cloud
cf-sa31
KSA (Dammam)
gcp
Enterprise account
il30
Google Cloud
cf-il30
Israel (Tel Aviv)
gcp
Enterprise account
br30
Google Cloud
cf-br30
Brazil (São Paulo)
gcp
Enterprise account
ap30
Google Cloud
cf-ap30
Australia (Sydney)
gcp
build-runtime-gcp

Enterprise account
ap31
Google Cloud
cf-ap31
Singapore (Jurong West)
gcp
Enterprise account
eu01
SAP Cloud Infrastructure
cf-eu01
Europe (Frankfurt) EU Access
sap-converged-cloud available in the Germany (Frankfurt) (eu-de-2) region
Enterprise account
eu02
SAP Cloud Infrastructure
cf-eu02
Europe (Rot) EU Access
sap-converged-cloud available in the Germany (Rot) (eu-de-1) region
Enterprise account
ap01
SAP Cloud Infrastructure
cf-ap01
Australia (Sydney)
sap-converged-cloud available in the Australia (Sydney) (ap-au-1) region
Enterprise account
ae01
SAP Cloud Infrastructure
cf-ae01
UAE (Dubai)
sap-converged-cloud available in the UAE (Dubai) (ap-ae-1) region
Enterprise account
us01
SAP Cloud Infrastructure
cf-us01
US East (Sterling)
sap-converged-cloud available in the US East (Sterling) (na-us-1) region
Enterprise account
us02
SAP Cloud Infrastructure
cf-us02
US West (Colorado)
sap-converged-cloud available in the US West (Colorado) (na-us-2) region
Enterprise account
jp01
SAP Cloud Infrastructure
cf-jp01
Japan (Tokyo)
sap-converged-cloud available in the Japan (Tokyo) (ap-jp-1) region
Enterprise account
cn40
Alibaba Cloud
cf-cn40
China (Shanghai)
alicloud

Note

1 This region is available only in the BTP region cf-cn20 and is the sole region available within the Microsoft Azure (azure) plan in China.

Cluster Regions¹

When you enable a Kyma environment for a given subaccount, you must select a plan and region where the cluster is going to be created. Note that there is a number of regions available within each plan. They are all listed in the table:

Cluster Regions

IaaS Provider

Plan ID

Region

Region Name

Microsoft Azure

azure

azure_lite

build-runtime-azure

centralus

US Central (IA)

eastus

US East (VA)

westus2

US West (WA)

northeurope

North EU (Ireland)

uksouth

UK South (London)

germanywestcentral

Europe (Frankfurt)

switzerlandnorth

Switzerland (Zurich)

westeurope

Europe (Netherlands)

southeastasia

Singapore

australiaeast

Australia (Sydney)

japaneast

Japan (Tokyo)

brazilsouth

Brazil (São Paulo)

canadacentral

Canada (Toronto)

chinanorth3 1

China (North 3) 1

Amazon Web Services

aws

build-runtime-aws

eu-central-1

Europe (Frankfurt)

eu-west-2

Europe (London)

eu-south-1

Europe (Milan)

ca-central-1

Canada (Montreal)
sa-east-1
Brazil (São Paulo)
us-east-1
US East (VA)
ap-northeast-1
Japan (Tokyo)
ap-northeast-2
South Korea (Seoul)
ap-south-1
India (Mumbai)
ap-southeast-1
Singapore
ap-southeast-2
Australia (Sydney)
us-west-2
US West (Oregon)
trial
eu-central-1
Europe (Frankfurt)
us-east-1
US East (VA)
ap-southeast-1
Singapore
Google Cloud
gcp
build-runtime-gcp
europe-west3
Europe (Frankfurt)
europe-west4
Europe (Netherlands)
us-central1
US Central (IA)
asia-south1
India (Mumbai)
asia-northeast2
Japan (Osaka)
me-central2
KSA (Dammam)
me-west1
Israel (Tel Aviv)
australia-southeast1
Australia (Sydney)
southamerica-east1
Brazil (São Paulo)
asia-northeast1
Japan (Tokyo)
asia-southeast1
Singapore (Jurong West)
us-west1
North America (Oregon)
us-east4
North America (Virginia)
SAP Cloud Infrastructure

sap-converged-cloud

eu-de-1

Germany (Rot)

eu-de-2

Germany (Frankfurt)

na-us-1

US East (Sterling)

na-us-2

US West (Colorado)

ap-au-1

Australia (Sydney)

ap-jp-1

Japan (Tokyo)

ap-ae-1

UAE (Dubai)

Alibaba Cloud

alicloud

cn-shanghai

China (Shanghai)

Load Balancersà^

Depending on the IaaS Provider, the following Load Balancers are provisioned by default:

Default Load Balancers

IaaS Provider

Default Load Balancer

Microsoft Azure

Standard

Amazon Web Services

Classic Network Load Balancer

Google Cloud

External passthrough Network Load Balancer

For more details on the Load Balancers and their features, check out the official documentation of the respective IaaS provider.

Related Informationà^

Creating Kyma Instances

Available Plans in the Kyma Environment

Provisioning and Updating Parameters in the Kyma Environment

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/registry-cache-module>

Title: Registry Cache Module | SAP Help Portal

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ModulesRegistry Cache Module

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Registry Cache Module

The Registry Cache module adds a caching layer for container image registries in SAP BTP, Kyma runtime instances.

What Is Registry Cache?à^

The Registry Cache module adds a caching layer for container image registries in SAP BTP, Kyma runtime instances. It reduces outbound traffic to upstream registries, improving image pull performance. With Registry Cache, you can also cache images from private registries. To do so, provide credentials for the caching layer to use when authenticating against those registries.

The Registry Cache feature is built on top of the Gardener's registry cache extension. See [Configuring the Registry Cache Extension](#).

Featuresà^

The Registry Cache module provides the following features:

Caches container images from upstream registries to reduce outbound network traffic.

Supports private registries using credential Secrets referenced in RegistryCacheConfig.

Configurable cache volume size and storage class per upstream registry.

Configurable garbage collection Time to Live (TTL); you can disable garbage collection.

Proxy support for HTTP and HTTPS connections used by the cache.

TLS-enabled HTTP server for the Registry Cache endpoint.

Architecture

The Registry Cache module consists of two main runtime components: the RegistryCache controller and the RegistryCacheConfig admission webhook. Both run in the same Registry Cache Manager process.

RegistryCache controller — reconciles RegistryCache custom resources (CRs) and drives status transitions.

Webhook Server — TLS server on port 9443 that validates RegistryCacheConfig resources on create and update.

Certificate Manager — watches TLS certificate files and rotates the CA bundle in

ValidatingWebhookConfiguration on renewal.

API / Custom Resource Definitions

The Registry Cache module defines the following custom resources:

Custom Resource Definitions

CRD

Scope

Description

RegistryCache

Namespaced

Module CR managed by the lifecycle infrastructure. Tracks the installation health of the Registry Cache module.

See RegistryCache Custom Resource.

RegistryCacheConfig

Namespaced

User-created CR that configures a caching layer for a specific upstream container image registry.

See RegistryCacheConfig Custom Resource.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/resilient-applications-in-kyma-environment>

Title: Develop Resilient Applications in the Kyma Runtime | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology

PlatformDevelopmentDevelopment in the Kyma EnvironmentDevelop Resilient Applications in the Kyma

Runtime

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Develop Resilient Applications in the Kyma Runtime

All SAP BTP, Kyma runtime production plans ensure high availability. To benefit from the high-availability setup, make sure the architecture and deployment of your application comply with resiliency best practices. Use the following guidelines for Kubernetes and microservice-based applications to develop a stable application. Read further to find some Istio-related resiliency tips.

Resilient Applications Deployed on Kubernetes

To develop resilient applications deployed on Kubernetes, use:

Kubernetes built-in health checks. For more information, read how to Configure Liveness, Readiness and Startup Probes.

Deployment across multiple availability zones for high availability. For more details, see the blog post.

The Keda Module or Kubernetes autoscaling to automatically adjust the number of replicas based on resource usage.

Rolling updates to minimize downtime during application updates. If you don't provide any configuration, rolling updates are by default enabled in your SAP BTP, Kyma runtime.

BTP backing services, such as PostgreSQL or Object Store on SAP BTP. If you have to store some application data on SAP BTP, Kyma runtime, use persistent storage to ensure data is not lost in the event of a pod failure.

Resource quotas to prevent resource starvation and ensure fair resource allocation.

The Telemetry Module to configure a logging and monitoring solution to identify and troubleshoot issues in real time.

Resilient Microservice-Based Applications[^]

The Istio Module and Istio traffic management features help you achieve inbuilt microservices resiliency.

Moreover, to develop resilient microservice-based applications, use:

Istio's circuit breaker and retry features to prevent service overloads and recover from service failures.

Distributed tracing with Istio's observability features to monitor the flow of requests across microservices and identify performance and error issues by shipping logs to SAP Cloud Logging (for details, see Integrate With SAP Cloud Logging).

The Eventing module for asynchronous communication patterns, such as messaging queues, to decouple services and reduce dependencies. See how to Configure SAP Event Mesh for Kyma Eventing.

Health check endpoints as a part of your microservice. For more details, read Kubernetes API health endpoints.

A centralized logging and monitoring solution to identify issues across multiple microservices and quickly troubleshoot them.

A clear and well-defined boundary in your microservice's design to minimize the impact of failures and enable easy replacement or scaling of individual services.

Istio's service mesh to implement mutual TLS and authorization policies to secure communication between microservices. For more details, read the blog post.

Network resiliency using Istio's traffic management features to test how your microservices respond to various failure scenarios, such as slow or failing services.

The Telemetry Module to ship your Istio's access logs and metrics to your logging backend to monitor traffic patterns and quickly troubleshoot issues.

Related Information[^]

Improve resiliency of your applications deployed on Kyma runtime by using multiple availability zones

Developing enterprise-grade applications with mutual TLS easy with SAP BTP, Kyma runtime and BTP destinations

SAP BTP Kyma API Rules with destinations and a managed approuter

Protect your Kyma workloads from eavesdropping with JWT access strategy and BTP destinations

Protect your Kyma workloads from eavesdropping with introspection and BTP destinations

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/restore-configuration-settings-after-instance-refresh>

Title: Restore Configuration Settings After an Instance Refresh | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Extending SAP SuccessFactors in the Cloud Foundry and Kyma Enviro...Restore Configuration Settings After an Instance Refresh

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Restore Configuration Settings After an Instance Refresh

After an instance refresh, you must restore some of your extension integration artifacts and configuration settings for the SAP SuccessFactors target company.

Prerequisites[^]

You have a dedicated SAP SuccessFactors company instance.

You have a registered SAP SuccessFactors system in your global account in SAP BTP.

You have performed the instance refresh with the Instance Refresh tool or your cloud operators have performed an instance refresh manually.

Context[^]

In SAP SuccessFactors, instance refresh is a procedure in which the data and the settings of a source company instance are copied into another company instance. During the instance refresh the data of the target company instance is deleted and replaced by the data of the source company instance. Therefore, after an instance refresh, if the target company has been integrated with a global account in SAP BTP, the extension integration configuration settings and artifacts in the target company are overwritten by the configuration settings and data coming from the source company.

You can trigger the instance refresh automatically with the Instance Refresh tool or your cloud operators can trigger it manually.

If you have performed an instance refresh with the Instance Refresh tool, the OAuth clients, and the Assertion Consumer Services (ACS) created by the cloud platform are recreated. In Provisioning, the Extension Management Configuration page displays the integration details for the target system.

Note

The OAuth clients are assigned new IDs. This does not affect the configuration settings.

However, if you have the following artifacts, you must reconfigure them in the SAP SuccessFactors target company instance:

- Custom home page tiles

- Permission roles

- Permission groups

Note

After a manual instance refresh, no artifacts are reconfigured. If you want to restore the integration settings for the target company instance, you need to recreate the SAP SuccessFactors Extensibility service instance.

Procedure

To restore the configuration settings after an instance refresh, proceed as follows:

After an automated instance refresh:

If you have configured custom home page tiles in the target company instance, reconfigure them. See Custom Home Page Tiles.

If you have configured permission roles and permission groups, you need to reconfigure them. See What Are Role-Based Permissions?.

After a manual instance refresh:

For SAP SuccessFactors, Second Half 2021 Release or later, proceed as follows:

Recreate the SAP SuccessFactors Extensibility service instance. See Create a Service Instance to Consume the SAP SuccessFactors HXM Suite OData API.

If you have configured permission roles and permission groups, you need to reconfigure them. See Custom Home Page Tiles.

Reconfigure the permission roles and permission groups. See What Are Role-Based Permissions?.

Results

You have restored the extension integration artifacts and configuration setting of the target company instance.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/runtime-monitoring-agent-admission-policies>

Title: Admission Policies | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology PlatformAdministration and OperationsAdministration and Operations in the Kyma EnvironmentAdmission Policies

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Admission Policies

Learn about the validating admission policies deployed in SAP BTP, Kyma runtime clusters, why they block specific actions, and what you can do if you need an exception. These policies are enforced at admission time and protect cluster stability, security, and operational correctness.

Namespaces Delete Protection

This policy prevents the deletion of Kyma-managed namespaces unless the request comes from an allowed user, group, or service account.

The following namespaces host critical platform components and are protected from deletion:

- istio-system

- kyma-system

- sap-transp-proxy-system

- ztis-agent-system

- docker-registry

- registry-proxy

An accidental or unauthorized deletion of these namespaces can cause cluster-wide outages, break service mesh functionality, and interfere with platform lifecycle operations.

When you try to delete these namespaces, you receive an API error. The validating admission policy returns a rejection with the message that you cannot delete the Kyma-managed namespaces. This action requires an

allowed user, group, or service account.

Verify that you're targeting the correct namespace and use application namespaces instead of the Kyma-managed system namespaces.

Kyma System Protection (Workload Deletion in kyma-system)^

This policy prevents the deletion of workloads (Deployments, StatefulSets, DaemonSets, Jobs, CronJobs) inside the kyma-system namespace unless the request comes from an allowed user, group, or service account.

The kyma-system namespace runs managed Kyma components. Deleting workloads there can break the control plane, integrations, and user-facing services. This policy prevents accidental disruption and enforces that only trusted controllers or processes make such changes.

When you try to delete a workload in kyma-system, you see a rejection at the API admission step with the message that you cannot delete workloads in the kyma-system namespace. This action requires an allowed user, group, or service account.

When you delete workloads, verify that you're targeting the correct namespace. For changes to customer workloads, use their application namespaces rather than kyma-system.

Kube-System Istio Injection Prevention^

This policy prevents adding the label istio-injection=enabled to the kube-system namespace.

Enabling Istio sidecar injection in kube-system can unintentionally inject proxies into critical system components. These components can be incompatible with sidecars, which could lead to degraded performance, bootstrap failures, or networking issues for platform services.

The label add or patch requests that attempt to set istio-injection=enabled on the kube-system namespace are rejected with the message that this label cannot be added here.

Double-check why you need injection in kube-system. Commonly, you can isolate your components by enabling sidecar injection in the specific application namespaces instead.

Dynakube Kyma Exclusion^

This policy validates that Dynatrace Dynakube custom resources include namespace-exclusion rules so that Kyma-managed namespaces (for example, kyma-system, istio-system, sap-transp-proxy-system, ztis-agent-system, or namespaces with label operator.kyma-project.io/managed-by=kyma) are excluded from automatic instrumentation.

If the required exclusions are missing, the admission policy denies the resource.

Automatic instrumentation across Kyma-managed infrastructure may interfere with Kyma's lifecycle management, generate noisy metrics or logs, or cause unexpected side effects in platform-managed namespaces. Explicit exclusions ensure that monitoring is applied where safe and desired without interfering with platform-managed components.

The policy checks the following fields depending on your Dynakube API version and configuration:

spec.metadataEnrichment.namespaceSelector.matchExpressions (if metadataEnrichment is enabled)

spec.oneAgent.cloudNativeFullStack.namespaceSelector.matchExpressions (if cloudNativeFullStack is configured)

spec.oneAgent.applicationMonitoring.namespaceSelector.matchExpressions (if applicationMonitoring is configured)

spec.namespaceSelector.matchExpressions (for Dynakube API version v1beta1)

If your Dynakube setup is missing required exclusions, an error message tells you to which field you must add them.

For configuration examples, see [Kyma: Dynatrace Setup](#) and [Dynatrace: DynaKube parameters for Dynatrace Operator](#).

Enforce Kyma Workload Traffic^

This policy prevents the use of network policies with default-deny-all rules in the Kyma-managed namespaces. Such network policies block essential communication between Kyma components within a cluster, potentially leading to system failures.

If you attempt to create or update a network policy that violates this rule, the action is blocked, and you receive a warning. In such a case, review your network policy configuration and ensure it includes specific allow rules instead of denying all traffic. For applications that require strict network isolation, consider using your own namespaces instead of the Kyma-managed namespaces.

Kyma Module Label Protection^

This policy prevents you from setting labels with the kyma-project.io/ prefix on any resource in Kyma-managed

namespaces.

The kyma-project.io/ label prefix is reserved exclusively for Kyma module operators to identify and manage their resources. When your resource in a protected namespace carries a kyma-project.io/ label, Kyma module operators may misidentify it as a module operator-managed resource. This misidentification causes the operator to enter an error state.

The following namespaces are protected:

kyma-system

istio-system

docker-registry

registry-proxy

sap-trans-proxy-system

ztis-agent-system

When you create or update a resource with a kyma-project.io/ label in one of these namespaces, you receive an API error message stating that setting reserved labels isn't allowed.

If you need to categorize or identify your resources, use your own label prefix instead of the reserved kyma-project.io/ prefix.

If you're deploying workloads that require such a label, deploy them into your own application namespaces rather than into Kyma-managed system namespaces.

Kyma Storage Protection[^]

This policy prevents you from deleting PersistentVolumeClaims (PVCs) in the kyma-system namespace and PersistentVolumes (PVs) that are bound to a kyma-system PVC, unless the request comes from an allowed user, group, or service account.

The kyma-system namespace may use persistent storage for platform components. Deleting a PVC or its bound PV can result in permanent data loss for those components. This policy ensures that only trusted users, groups, or service accounts can remove storage resources tied to platform operations.

When you try to delete a PVC in kyma-system, or a PV that is bound to a PVC in kyma-system, you receive an API error. The error message states that this deletion isn't allowed and requires an allowed user, group, or service account.

Verify that you're targeting the correct resource. If you need to remove storage in kyma-system, contact your platform administrator to perform the operation with an allowed user, group, or service account.

Kyverno Namespace Protection[^]

This policy validates that Kyverno ClusterPolicies and policies that affect workloads, such as Pods, Deployments, StatefulSets, DaemonSets, Jobs, CronJobs, ReplicaSets, include an exclude block that covers Kyma-managed namespaces.

Kyverno policies without explicit exclusions can inadvertently match resources in Kyma-managed namespaces, such as kyma-system and istio-system. This can cause mutations or validations to interfere with platform-managed components and lead to instability or failed reconciliation of Kyma modules.

When you create or update a Kyverno policy that affects workloads but is missing the required exclusions, you receive a warning. The operation isn't blocked, but you should address the warning to avoid unintended impact on Kyma components.

Add an exclude block to each rule that affects workloads. Use one of the following approaches:

Label-based exclusion (recommended): Exclude namespaces with the selector operator.kyma-project.io/managed-by NotIn [kyma]. This label automatically covers all current and future Kyma-managed namespaces.

Explicit namespace exclusion: List both kyma-system and istio-system in the exclude block's namespaces field.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/serverless-module>

Title: Serverless Module | SAP Help Portal

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SAP Business Technology PlatformEnglish

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Serverless Module

Learn more about the Serverless module. Use it to run lightweight Functions in a cost-efficient and scalable way using JavaScript and Node.js.

What Is Serverless?â

The Kyma Serverless module is a "code-as-a-service" tool designed to accelerate the process of prototyping applications by removing the technical overhead necessary to obtain the first testable versions. It allows developers to focus on the coding and easily run it in the context of the SAP BTP without dealing with Kubernetes orchestration or containerization complexities. The fact that no Kubernetes knowledge is required to use it makes Serverless a useful tool for beginners who want to experiment with loosely coupled microservice architecture on Kubernetes or for business consultants who need to provide extension logic for an existing Kubernetes-based system with minimal overhead.

With the Serverless module, you can build, run, and manage serverless applications in Kubernetes. These applications are called Functions, and they are based on the Function custom resource (CR) objects. They contain simple code snippets that implement a specific business logic. For example, you can define that you want to use a Function as a proxy that saves all incoming event details to an external database.

A Function can be triggered by other workloads in the cluster (in-cluster events) or business events coming from external sources. You can subscribe to them using a Subscription CR. You can expose your Function to an external endpoint (HTTPS). With an APIRule CR, you can define who can reach the endpoint and what operations they can perform on it.

Featuresâ

The Serverless module provides the following features:

- Quick deployments following a Function approach

- Loosely coupled microservice approach

- Event-driven programming model

- Reduced implementation and operation effort

- Function management with Kubernetes resources like Deployments, Services, and HorizontalPodAutoscalers

Architectureâ

- User applies the Serverless CR.

- Serverless Operator watches the Serverless CR.

- Serverless Operator reconciles the Serverless workloads.

- Serverless Operator

When you add the Serverless module, Serverless Operator takes care of installation and configuration of Serverless in your cluster. It manages the Serverless lifecycle based on the dedicated Serverless CR.

API/Custom Resource Definitionsâ

The API of the Serverless module is based on Kubernetes CustomResourceDefinitions (CRDs), which extend the Kubernetes API with custom additions. To inspect the specification of the Serverless module API, see:

- Function CRD

- Serverless CRD

Security Considerationsâ

To learn how to avoid any potential threats while using Functions, see Function Security.

Authorizationâ

To assign access permissions to the Serverless module resources, use the following aggregated ClusterRoles:

- kyma-serverless-view

- kyma-serverless-edit

- kyma-functions-view

- kyma-functions-edit

Resource Consumptionâ

To learn more about the resources used by the Serverless module, see Serverless.

Related Informationâ

- Function Security

- kyma-project.io: Serverless module tutorials

- kyma-project.io: Serverless module technical reference

- Troubleshooting for the Serverless Module

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/service-plans-and-metering-for-kyma-runtime>

Title: Service Plans and Metering for Kyma Runtime | SAP Help Portal

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EnvironmentService Plans and Metering for Kyma Runtime

Service Plans and Metering for Kyma Runtime

This page explains the relationship between the service plans of the SAP Discovery Center and the service plans of the SAP BTP cockpit and provides information to help you understand how the service is billed.

Serviceà^

Overview

The following diagram shows how the service plans listed in the SAP Discovery Center correspond to the plans you choose in the SAP BTP cockpit. For more information about the commercial model, see [What is the Consumption-Based Commercial Model?](#)

Note that there's no subscription-based commercial model for SAP BTP, Kyma runtime.

SAP BTP Cockpit: Service Plans

Name

Service Plan (Discovery Center)

Description

free

Free

Subscribe to the 30-day free plan provided on Amazon Web Services. This plan uses Kyma on a limited-size cluster (4 CPU—16 GB RAM). You can use the free plan only once in a global account for up to 30 days. The upgrade to the paid plan is not yet supported. Only best-effort support is available for free tier service plans, and these are not subject to SLAs. The services you plan to use must be available in the same region as the subaccount for the Kyma runtime.

aws

Standard

Select Amazon Web Services as the cloud provider where your Kyma cluster is deployed.

gcp

Standard

Select Google Cloud as the cloud provider where your Kyma cluster is deployed.

azure

Standard

Select Microsoft Azure as the cloud provider where your Kyma cluster is deployed.

Metricsà^

The usage metric for SAP BTP, Kyma runtime is a Capacity Unit (CU) per hour.

Metric

Definition

Capacity Unit

Number of units consumed by the usage of the services as outlined in the solution-specific product supplement. For SAP BTP, Kyma runtime, CUs are calculated for nodes and for storage.

Backward Calculationà^

Tip

If you use the Cloud Manager module, also see [Calculation with the Cloud Manager Module](#).

Formula

The formula is a mapping between consumed nodes/storage and the respective CUs. The details are provided in the following tables.

However, as Kyma supports flexible scaling, cost per hour may vary because it depends on the size of the nodes that were used.

Underlying Metrics: Nodes

The nodes calculation takes the number of virtual CPUs (vCPUs) and provides the number of CUs per hour. To better understand the relationship between vCPUs and nodes, note that one node is sized with 2 vCPUs.

The Calculation for vCPU table presents data for general-purpose machine types.

Additionally, the calculation is used for compute-intensive machine types.

Calculation for vCPU

Number of vCPUs

Number of CUs per Hour

2

$0.24 * 1.5$

4

$0.24 * 2$

8

$0.24 * 4$

16

$0.24 * 8$

32

$0.24 * 16$

...

...

Underlying Metrics: Storage

SAP BTP, Kyma runtime is provisioned with 32 GB of storage dedicated to operational data at no additional cost. However, certain Kyma modules may require extra storage. In such cases, they create additional volumes, which incur charges for the extra storage used.

You can calculate CUs for the following storage types:

PersistentVolumes: Storage is provided in blocks of 32 GB. If you use 33 GB, you're charged for 64 GB (2 blocks of 32 GB).

VolumeSnapshotContents: You pay based on the actual amount of storage used.

Additional volume size: Each worker node in a Kyma cluster comes with a default disk volume. The size depends on the machine type. This default volume stores Docker images and container logs for operational purposes and isn't charged separately. If the default volume size isn't sufficient (for example, when running many containers that produce large log volumes or pulling many large Docker images), you can request additional volume size beyond the default. You configure this using service instance parameters. Any storage beyond the default volume size is an additional node volume and is billed accordingly. For example, if you use 4 nodes of 10 GB additional volume, totaling 40 GB, you're charged for 64 GB (2 blocks of 32 GB).

Calculation for Storage

Amount of Storage

Number of CUs per Hour

32 GB

0.02

Examples

Your consumption report for nodes shows 1800 capacity units charged for 4 vCPU nodes for 30 days.

This means that you were running roughly $1800 / (0.24 * 2) = 3750$ hours worth of 4 vCPU nodes. Given that 30 days have 720 hours, this means you were running $3750 / 720 = 5.21$ nodes cluster for the 30 days.

Due to Kyma's automatic scaling, 5.21 does not mean actual size. It means that during some hours in the 30 days, the cluster size was 4 nodes. At other times, there was more load, so the cluster autoscaled to 5 or 6 nodes, based on your configurations. So your average cluster size for the 30 days was 5.21 nodes.

Your consumption report for storage shows 200 capacity units charged for storage (32 GB block) for 30 days.

This means that you used $200 / 0.02 = 10000$ hours of 32 GB storage blocks. Given that 30 days have 720 hours, this means that you have used $10000 / 720 = 13.89$ blocks of 32 GB over the 30 days. So, you used $13.89 * 32 = 444.44$ GB of storage continuously during the 30 days.

This implies that as you started using Kyma, you deployed more applications that used storage of various sizes, such as 4 GB or 8 GB.

Calculation with the Cloud Manager Module

Calculation with the Cloud Manager Module

Using the Cloud Manager module and enabling Redis or NFS storage or both, introduces additional costs.

The cost of NFS storage introduces factor 3 to the calculation. Note that NFS storage is provided in blocks of 32 GB, so if you use 33 GB, you're charged for 64 GB (2 blocks of 32 GB).

Calculation for NFS Storage

Amount of NFS Storage

Number of CUs per Hour

32 GB

0.06 (= 0.02 * 3)

Using Redis adds costs depending on the tier you choose.

Note

Even though the actual available size (GiB) might vary depending on the cloud provider, the tiers for Redis are standardized across cloud providers. Because of that, also the pricing is standardized across all available cloud providers.

Calculation for Redis Tiers

Tier

Approx. Available Size (GiB)

Cost in Storage Size per Hour

Number of CUs per Hour

S1

1

0.25277778

0.1027777778

S2

3

0.50555556

0.2055555556

S3

6

1.32361111

0.5375000000

S4

12

2.65972222

1.0805555556

S5

24

5.26805556

2.1402777778

S6

48

10.54305556

4.2833333333

S7

101

21.08333333

8.5652777778

S8

202

42.15694444

17.1263888889

P1

5

2.64305556

1.0736111111

P2

12

5.31666667

2.1597222222

P3

24

10.53611111
4.280555556
P4
48
21.08333333
8.565277778
P5
101
42.17083333
17.131944444
P6
200
84.31111111
34.251388889

Examples for Cloud Manager

Your consumption report for the NFS storage shows 200 capacity units charged for storage (32 GB block) for 30 days.

This means that you used $200 / 0.06 = 3333.33$ hours of 32 GB storage blocks. Given that 30 days have 720 hours, this means that you have used $3333.33/720 = 4.63$ blocks of 32 GB over the 30 days. So, you used $4.63 * 32 = 148.15$ GB of NFS storage continuously during the 30 days.

Your consumption report for Redis shows 200 capacity units for storage for 30 days.

This means that you used $200 / 0.1027777778 = 1945.946$ hours of the Redis S1 tier. Given that 30 days have 720 hours, this means that you have used $1945.946/720 = 2.703$ tier S1 Redis instances continuously during the 30 days.

Calculatorà

Find the Kyma price calculator at <https://kyma-project.github.io/price-calculator/> and use it to estimate the costs for your SAP BTP, Kyma runtime.

Choose the base configuration for the size and minimum number virtual machines (VM) you need.

Optionally, add more storage to the calculation.

If needed, adjust the conversion rate.

Note that the result is an estimate, and the consumption report may vary depending on the actual hours and size of the Kyma cluster (nodes and storage) that was running in 30 days.

Supplemental Terms And Conditionsà

For more information, see SAP Business Technology Platform Service Description Guide, section “SAP BTP, KYMA RUNTIME”.

Glossaryà

Commercial Information Glossary

Service Specificsà

As soon as you instantiate a Kyma cluster, the basic cost for the empty cluster incurs.

Cost may then vary depending on the actual nodes and storage you consume.

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/setting-your-module-to-unmanaged-state>

Title: *Setting Your Module to the Unmanaged and Managed State | SAP Help Portal*

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Setting Your Module to the Unmanaged and Managed State

In some cases, for example, for testing, you may need to modify your module beyond what is supported by its configuration. By default, when a module is in the managed state, Kyma Control Plane governs its Kubernetes resources, reverting any manual changes during the next reconciliation loop. To modify Kubernetes objects directly without them being reverted, you must set the module to the unmanaged state. In this state, reconciliation is disabled, ensuring your manual changes are preserved.

Caution

Unmanaged modules aren't subject to the Service Level Agreement (SLA). Additionally, setting a module back to the managed state does not guarantee its version is correctly updated.

Setting Your Module to the Unmanaged State

Setting Your Module to the Managed State

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/taints-configuration>

Title: Taints Configuration | SAP Help Portal

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Taints Configuration

Learn how to use taints to effectively schedule your workloads to nodes in additional worker node pools.

The Additional Worker Node Pools (additionalWorkerNodePools) parameter supports an optional Taints list.

Use it to control which workloads are scheduled to nodes in a given worker pool.

Note

Taints alone dictate scheduling restrictions on nodes, but these restrictions take effect only when combined with tolerations. While you configure taints on nodes, you set tolerations on your workloads (Pods) to allow them to run on tainted nodes. For more information on configuring tolerations, see Taints and Tolerations.

Taints Parameters

The Taints (taints) parameter includes three nested parameters: Key*, Value, and Effect*.

Remember

The parameters marked with an asterisk "*" are mandatory.

Taints Nested Parameters

Nested Parameter

Description

Allowed Input

Key*

btp CLI parameter: key

Identifies the taint.

String

Value

btp CLI parameter: value

Provides additional context.

String

Effect*

btp CLI parameter: effect

Defines what happens to workloads that do not tolerate the taint.

NoSchedule (excludes certain types of Pods from running on a specific set of nodes)

PreferNoSchedule (discourages scheduling Pods on certain nodes, but allows it as a last resort, such as if no other nodes are available, to maintain availability or resource utilization)

NoExecute (prevents any new workloads that cannot tolerate the taint from being scheduled onto a given node and evicts any existing Pods that already violate the taint and do not have the required toleration)

For each taint within a single worker node pool, the combination of the key and effect parameters must be unique. However, you can use the same key value with different effect values.

Provisioning a Cluster with Taints

To provision a cluster with taints in an additional worker node pool, use a configuration similar to the following in your request:

âE

ã

{

"additionalWorkerNodePools": [

{

```

...
"taints": [
{
"key": "dedicated",
"value": "gpu",
"effect": "NoSchedule"
},
{
"key": "dedicated",
"value": "gpu",
"effect": "NoExecute"
}
]
}
]
}

```

Updating Taints

To update or overwrite the existing taints for your worker node pool, you must explicitly provide a complete list of desired taints, for example:

```

{
"additionalWorkerNodePools": [
{
...
"taints": [
{
"key": "dedicated",
"value": "gpu",
"effect": "NoSchedule"
}
]
}
]
}

```

Removing Taints

To remove existing taints for a worker node pool, either omit the taints field in the update request or set it to an empty list ([]), for example:

```

{
"additionalWorkerNodePools": [
{
...
"taints": []
}
]
}

```

Related Information

Assigning Workloads to Worker Node Pools

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/tracking-kubeconfig-and-cluster-associations-in-kyma>

Title: Tracking Kubeconfig and Cluster Associations in Kyma | SAP Help Portal

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Tracking Kubeconfig and Cluster Associations in Kyma

You can identify which kubeconfig is associated with a particular Kyma cluster by the value of the Cluster Name parameter.

The Cluster Name* (technical name: name) parameter specifies the name of your cluster. Its value is also used in the kubeconfig of a Kyma instance as a context name and a user name (if OIDC authentication is configured). With the {CLUSTER_NAME} value used in this way, you can track the kubeconfig - cluster associations.

If multiple OIDC configurations are provided, user entries are generated following this incremental naming convention:

{CLUSTER_NAME}

{CLUSTER_NAME-2}

{CLUSTER_NAME-3}

The pattern continues for additional users. Without any OIDC configuration, the users section is omitted.

To view the kubeconfig file, you can download it from the SAP BTP cockpit. In your subaccount Overview, go to the Kyma Environment section and choose the KubeconfigURL link.

See how the {CLUSTER_NAME} value is used in a typical kubeconfig file:

âE

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apiVersion: v1

kind: Config

current-context: {CLUSTER_NAME}

clusters:

- name: {CLUSTER_NAME}

cluster:

certificate-authority-data: {CERTIFICATE_DATA}

server: {SERVER_URL}

contexts:

- name: {CLUSTER_NAME}

context:

cluster: {CLUSTER_NAME}

user: {CLUSTER_NAME}

users:

- name: {CLUSTER_NAME}

user:

exec:

apiVersion: client.authentication.k8s.io/v1beta1

args:

- get-token

- "--oidc-issuer-url={ISSUER_URL}"

- "--oidc-client-id={CLIENT_ID}"

- "--oidc-extra-scope=email"

- "--oidc-extra-scope=openid"

command: kubectl-oidc_login

installHint: |

kubelogin plugin is required to proceed with authentication

Homebrew (macOS and Linux)

brew install int128/kubelogin/kubelogin

Krew (macOS, Linux, Windows and ARM)

kubectl krew install oidc-login

Chocolatey (Windows)

choco install kubelogin
àf

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/updating-kyma-environment-plans>

Title: *Updating Kyma Environment Plans* | SAP Help Portal

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Updating Kyma Environment Plans

Update standard enterprise Kyma environment plans to build runtime plans.

Prerequisitesà^

Assign entitlements for Kyma runtime with the build-runtime-aws, build-runtime-gcp, or build-runtime-azure plan to your subaccount. For more information, see [Configure Entitlements and Quotas for Subaccounts](#).

Contextà^

You can switch from standard enterprise plans to build runtime plans within the same cloud provider. Options include:

aws !' build-runtime-aws

gcp !' build-runtime-gcp

azure !' build-runtime-azure

Note

You can't switch from build runtime plans to standard ones.

To switch plans, you must update your environment instance.

Procedureà^

In the SAP BTP cockpit, navigate to your subaccount [Instances and Subscriptions](#).

Go to [Environments](#), and from the three-dot menu at the end of the Kyma environment row, select [Update](#).

In the [Update Instance](#) window, choose the new plan, and select [Update Instance](#).

Resultsà^

You have updated your Kyma environment service plan.

Related Informationà^

[What Is SAP Build?](#)

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/using-sap-btp-services-in-kyma-environment>

Title: *Using SAP BTP Services in the Kyma Environment* | SAP Help Portal

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Using SAP BTP Services in the Kyma Environment

With the Kyma environment, you can connect SAP BTP services to your cluster and manage them using the SAP BTP Operator module.

Contextà^

To consume SAP BTP services and use their functionality for building and deploying your own applications, you must create and manage the service instances and bindings. To learn how to do it, see [Creating Service Instances and Service Bindings](#) and [Deleting Service Bindings and Service Instances](#).

For an example of how to use a service, see [Create an SAP S/4HANA Cloud Extensibility Service Instance in the Kyma Environment](#).

Related Informationà^

[SAP BTP Operator Module](#)

[Preconfigured Credentials and Access](#)

[Working with Multiple Subaccounts](#)

[Instance-Level Mapping](#)

[Namespace-Level Mapping](#)

Passing Parameters
Rotating Service Bindings
Formatting Service Binding Secrets
